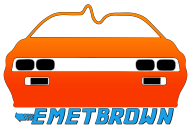

THÈSE PRÉSENTÉE

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Simulations numériques du mouvement brownien dans les environnements complexes.

Sous la direction de **Thomas Salez**

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List of Symbols and abbreviations

Symbols are separated per chapter of occurrence. Symbols may appear in multiple chapters once they have been defined.

Symbol	Description
BM	Brownian Motion
PDF	Probability Density Function
MSD	Mean Squared Displacement
RPY	Rotne-Prager-Yamakawa
RBW	RigidMultiblobsWall library
ABP	Active Brownian Particle
RTP	Run and Tumble Particle
a	Particle radius
D_0	Bulk diffusion coefficient
k_B	Boltzmann's constant
Θ	Temperature
γ	Stokes drag coefficient, friction coefficient
η_0	Bulk dynamic fluid viscosity
d	Dimensionality ($d = 2$ for 2D)
t	Time
m	Particle mass
τ_m	Momentum relaxation time, inertial time
Δt	Time step
t_max	Total time
z	Colloid center to wall distance
h	Colloid apex to wall distance ($z - a$)
$\gamma_\perp$	Effective friction for a normal motion
$\gamma_\parallel$	Effective friction for an in-plane motion
ρ	Density
g	Gravity acceleration
Δm	Buoyant mass
z_i	Ion valency
σ	Surface charge density
l_B	Boltzmann's length

l_{D}	Debye's length
B	Adimensionalised electrostatic repulsion force
U_{grav}	Gravitational potential
U_{elec}	Electrostatic interaction potential
U	Total potential
$D_{\perp}$	Effective diffusion coefficient for a normal diffusion
$D_{\parallel}$	Effective diffusion coefficient for an in-plane diffusion
$D_{i\alpha,j\beta}$	Diffusion matrix component for colloids labeled i and j, in directions α and β
$\bar{v}_{\mathrm{d}}$	Total drift velocity
v_{s}	Spurious drift velocity
v_{d}	Deterministic drift velocity
$\boldsymbol{\mu}^{\mathrm{tt}}$	Translational block of the mobility matrix
$\mathbf{D}^{\mathrm{tt}}$	Translational block of the diffusion matrix
$D_{\perp}^{\mathrm{C,R}}$	Collective (C) and relative (R) pair diffusion coefficients perpendicular ($\perp$) and parallel ($\parallel$) to the pair line of centres
q_i	Degree of freedom representing colloid i
$\phi(q_1)$	Sinusoidal potential acting upon q_1
ΔU	Sinusoidal trap amplitude
λ	Sinusoidal trap spatial period
κ	Inverse stiffness or compliance
ϵ	Adimensionalised compliance
γ	Friction matrix
γ_{ij}	Friction matrix element
$\boldsymbol{\mu}$	Mobility matrix
$\boldsymbol{\mu}_{ij}$	Mobility matrix elements
E	Elastic modulus
k_{eff}	Local trap stiffness around a minimum of $\phi(q_1)$
D_{m}	Microscopic diffusion coefficient of mode q_1
D_2	Microscopic diffusion coefficient of mode q_2
τ_i	Characteristic relaxation time of q_i
T_{sim}	Total simulation time
n_{steps}	Total number of iteration inside the trajectory simulation loop
$\mathcal{C}$	Convergence criterion
$\sigma(x)^2$	Variance of x
$\boldsymbol{\mu}_{\mathrm{e}}$	Dressed mobility
$D^*(\kappa)$	Effective long-time diffusion coefficient
Re	Reynolds number
Pe	Péclet number
W	Channel half-width

H	Channel thickness
τ_{D}	Transverse diffusion time
$D_{x,\mathrm{TA}}$	Taylor-Aris dispersion coefficient
ΔP	Entry-to-exist pressure difference
$\dot{s}$	Shear rate
D_y	Crosswise diffusion coefficient
D_y^{∞}	Crosswise diffusion coefficient away from wall interactions
N	Number of colloids
ϕ	Volume fraction
L_i	Periodic box length in direction $i \in (x, y)$
H_{box}	Virtual simulation box height
D_{T}	Translational diffusion coefficient
D_{R}	Rotational diffusion coefficient
D_{eff}	Active translational diffusion coefficient
ω	Angular velocity
v	Self-propelling, active velocity
τ_{run}	Typical time between tumbles
τ_{p}	Run persistence duration
T	Flow oscillation period
R_{h}	Hydraulic resistance
Pe_{A}	Active Péclet number

Résumé de la thèse

Logic: *Do it for the picture puzzle. Put it all together. Solve the world. One conversation at a time.*

Disco Elysium

Introduction

Ma thèse porte sur l'étude du mouvement brownien dans des environnements complexes. En tant que scientifiques et a fortiori physiciens, nous nous familiarisons avec le mouvement brownien dès nos années lycéennes, tant il est parfaitement connu et prévisible pour l'étude de cas simples. On nous fournit, typiquement, une solution aqueuse dans laquelle se trouvent de petites particules (des colloïdes). Nous les observons au microscope optique et nous nous étonnons, parce que notre culture scientifique n'est pas encore tout à fait développée, du mouvement inarrêtable et imprévisible des colloïdes, alors que le fluide les baignant est parfaitement au repos. Nous apprenons ensuite que ce que nous observons n'est autre qu'une des premières preuves expérimentales de la nature atomique de la matière. Notre instrument est limité dans ses capacités d'observation, et sa limite basse en termes de taille se situe plusieurs ordres de grandeur au-delà de la taille d'une molécule. Nous voyons, en fait, la conséquence de l'existence des atomes, plutôt que les atomes eux-mêmes : c'est la température du fluide qui se traduit en mouvement des molécules à leur échelle. Ces molécules, bien que minuscules devant la taille d'un colloïde, sont animées d'une vitesse élevée, de plusieurs centaines de mètres par seconde en moyenne. En heurtant le colloïde, elles lui transfèrent à chaque impact un peu de leur quantité de mouvement, et comme ces impacts eux-mêmes sont aléatoires tant dans le temps que dans l'espace, le colloïde s'anime d'un mouvement lui-même imprévisible et erratique.

Ces observations ont été largement confirmées et formalisées, et je décris le cas historique du mouvement brownien simple dans le premier chapitre de ce manuscrit. Je précise aussi dans ce chapitre les différents développements théoriques importants pour notre bonne compréhension de mon travail de thèse et y fais référence tout au long du manuscrit. En outre, le mouvement brownien n'est pas qu'un simple cas d'école, fabriqué pour nous faire déduire la réalité moléculaire. Il s'agit, en fait, d'un processus extrêmement com-

mun dans de nombreux systèmes biologiques. Si nous reprenons sa première observation historique dans des grains de pollen, nous notons déjà que la forme des colloïdes elle-même entraîne des modifications aux théories canoniques; cela constitue d'ailleurs une des premières généralisations apportées à la théorie du mouvement brownien. L'on peut aussi se questionner, puisque ce mouvement naît de l'agitation thermique au sein du fluide, de ce qu'il en advient quand le fluide lui-même est perturbé, par exemple en en modifiant les propriétés mêmes (viscosité, température), ou encore en quittant la limite du cas dilué où les colloïdes sont traités comme infiniment éloignés; un régime dense où le fluide déplacé par le mouvement de chaque colloïde affecte directement celui de ses pairs et où un couplage hydrodynamique apparaît. L'on note aussi l'influence de parois rigides à proximité du ou des colloïdes, là encore modifiant le comportement du fluide. Ces thématiques ont été explorées, et en partie résolues, mais leur complexité ne permet pas toujours de comprendre totalement et parfaitement leur superposition. J'explore en partie cette question dans le chapitre 2. Puis vient la question de l'influence de la déformabilité d'une paroi dans un système confiné. Le vivant regorge de membranes déformables telles qu'une paroi cellulaire, une muqueuse ou un tissu mou. Parfois, ces membranes elles-mêmes sont sujettes à ce même mouvement provenant des chocs avec les molécules de fluide. Comment prédire l'influence de leur déformabilité et de leurs fluctuations sur un colloïde voisin ? Peut-on contrôler, améliorer ou diminuer la mobilité d'une molécule à des fins thérapeutiques ? Au contraire, serait-il possible de stopper la propagation d'un pathogène sur de telles membranes, responsable de la dégénérescence parfois incurable de nos capacités ? Ce problème est par essence complexe à résoudre, mais je décris dans le chapitre 3 un modèle élémentaire et théorique pour capturer qualitativement l'effet de la déformabilité sur le mouvement brownien. L'écoulement du fluide lui-même affecte aussi le mouvement des colloïdes, d'abord et évidemment puisque ce dernier se retrouve entraîné dans l'écoulement, mais aussi pour des raisons moins triviales. En outre, je cherche à comprendre la possibilité ou non d'un effet de la densité, même faible, d'une solution de colloïdes dans un écoulement sur la dynamique de ceux-ci dans le chapitre 4. Enfin, nous abordons certes déjà de nombreux cas de figure propres au monde du vivant, mais *quid* du vivant lui-même ? Les colloïdes eux-mêmes peuvent dans certains cas se déplacer par leurs propres moyens en utilisant l'énergie disponible qui les entoure. S'ils sont parfois trop gros pour être visiblement affectés par l'agitation thermique du fluide, nous pouvons alors prédire leur comportement en les traitant par extension comme des particules browniennes effectives. Il est alors possible d'extraire encore de l'information de leurs trajectoires, tout comme nous le ferions pour un colloïde passif de petite taille. J'utilise dans le chapitre 5 ce type de modèle pour étudier le mouvement de micronageurs d'une taille bien supérieure à celle d'une particule brownienne, afin notamment d'étudier les comportements de microalgues dans un écoulement.

Libre accès

L'ensemble des résultats que j'ai produit tout au long de ma thèse s'appuie sur la simulation numérique comme principal outil. J'ai choisi au début de ma thèse d'aborder la recherche par cet angle, car il permet de proprement isoler différents effets pour mieux prédire leurs conséquences. Il ne serait pas possible, expérimentalement, d'observer une sphère parfaite, une température au zéro absolu, un écoulement strictement parabolique ou une eau infiniment pure. Pour moi, il ne s'agit à chaque fois que de quelques lignes de code tout au plus. Naturellement, avec cette capacité vient la question de la réalité de ce que je simule. Je dois toujours garder un œil extrêmement sceptique sur mes résultats, et les comparer aux théories et expériences, quand celles-ci existent. Je vois l'outil numérique comme un fantastique couteau suisse, permettant à la fois l'exploration d'effets inaccessibles expérimentalement à cause des limites actuelles de la technologie, et la validation de théories inaccessibles expérimentalement, alors qu'elles apportent tout de même, sinon quantitativement, au moins conceptuellement, des développements vitaux pour la recherche.

Puisque je me suis formée tout au long de ma thèse grâce aux ressources (forums, publications ou tutoriels) gratuites en libre accès, ainsi qu'aux interactions humaines qui sont deux piliers de la recherche, tous les codes que j'ai développés sont disponibles publiquement sur une page GitHub[1]. J'accorde une immense importance à la science libre, et j'estime que la réussite d'un chercheur ne doit pas se limiter aux fonds qu'on lui accorde, mais à sa curiosité, à sa capacité à poser les bonnes questions, à s'informer et à collaborer pour les résoudre, et à sa pédagogie et son intégrité dans la communication de ses résultats.

[1] https://github.com/Laacherez/thesis/Codes

Introduction

Logic: *Do it for the picture puzzle. Put it all together. Solve the world. One conversation at a time.*

Disco Elysium

As scientists, and physicists in particular, we discover Brownian motion as early as in high school, since it is perfectly well understood and predictable for the study of simple cases. Typically, we are given an aqueous solution containing small particles (colloids). We observe them under an optical microscope and marvel, because our scientific culture is not yet fully developed, at the unstoppable, unpredictable motion of the colloids, even though the fluid surrounding them is perfectly quiescent. We then learn that what we are observing is none other than one of the first experimental proofs of the atomic nature of matter. Our instrument is restained in its observational capabilities, and its lower size limit sits several orders of magnitude beyond the size of a molecule. What we actually see is the consequence of the existence of these atoms, rather than the atoms themselves: the temperature of the fluid translates into the motion of molecules at their own scale. These molecules, though minute compared to the size of a colloid, move at high speeds: several hundred meters per second on average. By striking a colloid, they transfer a bit of their momentum with each impact, and since these impacts are themselves random both in time and direction, the colloid is set into an equally unpredictable, erratic motion.

All these observations have been extensively confirmed and formalised, and I describe the historical case of simple Brownian motion in the first chapter of this manuscript. I also lay out in this chapter the various theoretical developments important for a proper understanding of my doctoral work, and refer back to them throughout the manuscript. Moreover, Brownian motion is not merely a textbook example, constructed to make us infer molecular reality. It is in fact an extremely common process in many biological systems. Going back to its first historical observation in pollen grains, we already note that the shape of the colloids itself leads to modifications of the canonical theories; this in fact constitutes one of the earliest generalisations brought to Brownian motion theory. One may also wonder, since this motion arises from thermal agitation within the fluid, what happens when the fluid itself is disturbed, for example by modifying its properties (viscosity, temperature), or by leaving the dilute limit where colloids are treated as infinitely far apart, entering a dense regime where the fluid displaced by the motion of one colloid

directly affects that of another, giving rise to hydrodynamic coupling. Another natural line of exploration concerns the effect of rigid walls, which likewise modify the behavior of the fluid and therefore of the colloid's motion. These questions have been explored, and partly resolved, over the past century, but their complexity does not always allow for a complete and full understanding of their unification. I explore this in chapter 2 and provide some insights. Then comes the matter of the influence of boundary deformability in confined systems. Living matter is full of deformable membranes: a cell wall, a mucous membrane, a soft tissue. Sometimes these membranes are themselves subject to the same motion arising from collisions with fluid molecules. How can we predict the influence of their deformability and their fluctuations on a neighboring colloid? Could we control, enhance, or reduce the mobility of a molecule for therapeutic purposes? Conversely, might it be possible to halt the spread of a pathogen across such membranes, a pathogen responsible for the sometimes incurable degeneration of certain of our capacities? This problem is inherently complex to solve, but in chapter 3 I describe an elementary theoretical model to qualitatively capture the effect of deformability on Brownian motion. The flow of the fluid itself also affects the motion of colloids, first and most obviously because the colloid gets carried along by the flow, but also for less trivial reasons that I develop in chapter 4. In addition, I seek to understand whether the density, even low, of a colloidal solution in a flow can affect the dynamics of the colloids themselves. Finally, we have already touched on many scenarios specific to the living world, but what about living matter itself? Colloids can, in some cases, move on their own by using the energy available around them. When they are too large to be visibly affected by the thermal agitation of the fluid, we can nonetheless predict their behavior by treating them, by extension, as effective Brownian particles. It is then possible to extract further information from their trajectories, just as we would for a small passive colloid. In chapter 5, I use this type of model to study the motion of microswimmers of a size much larger than that of a Brownian particle, in particular to study the behavior of these microswimmers in a flow.

Open-access

All of the results I produced throughout my thesis rely on numerical simulation as the main tool. I chose, from the start of my thesis, to approach research from this angle, since it permits different effects to be properly isolated in order to better predict their consequences. It would not be possible, experimentally, to observe a perfect sphere, an absolute zero temperature, a strictly parabolic flow, or infinitely pure water. For me, each of these is at most a few lines of code. Naturally, with this capability comes the question of how "real" what I simulate actually is. I must always keep a highly skeptical eye on my results, and compare them against theory and experiment whenever these exist. I see numerical tools as a fantastic Swiss Army knife, allowing both the exploration

of effects that are experimentally inaccessible due to current technological limitations, and the validation of theories that are unrealisable because of the laws of nature, even though such theories still contribute, if not quantitatively then at least conceptually, vital developments for research. Since I trained throughout my thesis using freely accessible resources (forums, publications, tutorials) as well as human interactions, the two pillars of research, all the code I developed is publicly available on dedicated GitHub repository[2]. I place immense importance on open science, and I believe that a researcher's success should not be limited to the funding they are granted, but should rest on their curiosity, their ability to ask the right questions, to inform themselves and collaborate to answer them, and on their pedagogy and integrity in communicating their results.

[2]https://github.com/Laacherez/thesis/Codes

Chapter 1

Brownian motion

This first chapter introduces the notion of Brownian motion from a historical standpoint. I intend to remain broad in this chapter, and provide only the key developments for a simple colloid in an unbounded fluid. Brownian motion is not only relevant in the field of History of Sciences, but also in current research.

1.1 On serendipity

Serendipitous discoveries are common in physics, though they often rely on numerous contributors to be finally formalised. An example I am particularly fond of is the birth of surface physics. Agnes Pockels was a self-taught chemist who, despite not being allowed enrolment at a University, demonstrated a prowess of scientific intuition by noticing during her house duties that the behaviour of water would show modified by the presence of impurities or soap [1, 2]. She then designed a device so as to measure surface tension with good reproducibility. This device is the precursor to the Langmuir-Blodgett through, and her findings and methods were acknowledged in Langmuir's Nobel prize-related publications [3].

Fortunately, History has a tendency to repeat itself: the discovery of Brownian motion also falls under the qualification of "serendipitous". Over the end of the eighteenth century to the middle of the nineteenth century, Robert Brown, a Scottish botanist, made significant contributions to his field through the use of optical microscopy, a state-of-the-art tool at the time [4]. However, he is mostly known amongst physicists for a discovery which decorrelates completely from his field of work. In 1827, he observed *Clarkia pulchella* pollen grains (Fig. 1.1 *a*) for the whole plant, and *b*) for single pollen grains) with a one-lens microscope. Surely, he supposed, much could be revealed on the process of pollinisation by studying pollen structure closely. He noted that the pollen grains were packed with smaller "molecules" (amyloplasts and spherosomes, Fig. 1.1 *c*)), a common word at the time to qualify small granules. To his surprise, once submerged in water, the pollen grains released some of these "molecules". They seemed to be dancing in what he

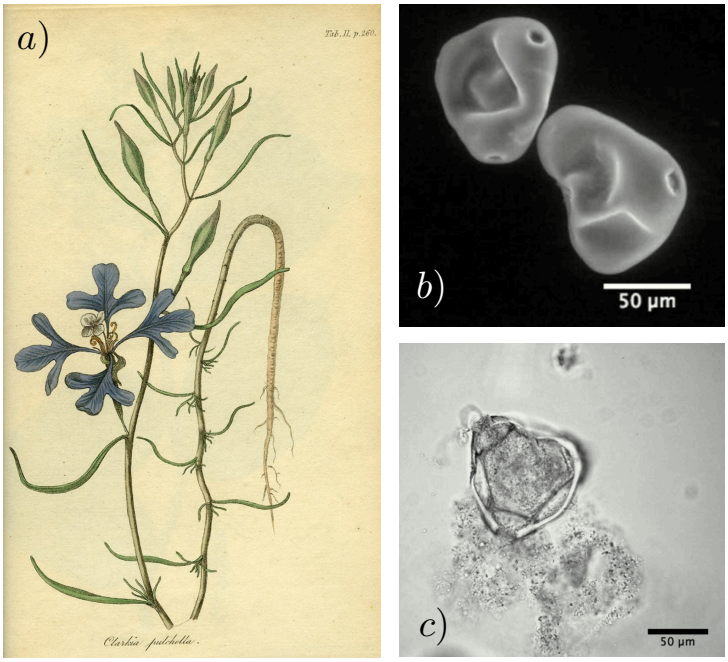

Figure 1.1: *Clarkia pulchella* and its pollen grains. *a*) Painting of a *Clarkia pulchella* specimen's anatomy by Frederick Pursh. *b*) *Clarkia* pollen grain under a microscope. *c*) Bursting grain of pollen. *b*) and *c*) adapted from [4].

called an "oscillatory motion"[1].

Brown then repeated the experiment with other essences, and other parts of the specimens he was working on. The dance was still present, although from different sources than male organelles. He concluded that if the process was alive, it was therefore not related to reproduction; pollen grains were not consciously seeking a mate through their motion. The then-widespread theory of George-Louis Leclerc that living things were built upon "organic molecules" seduced him [5]; perhaps this motion could still owe to a living mechanism. The final blow to this theory of "organic molecules" with an intrinsic voluntary motion was dealt when he repeated the same observations with "inert" mineral dusts of similar size.

The dance of "molecules", though not formalised at the time, came to be known as *"Brownian motion"*. The theory of Brownian motion only really began to develop in the beginning of the twentieth century, through two fundamental contributions: Albert Einstein's formal description of the atomistic nature of matter in 1905, which was undeniably validated experimentally in 1908 by Jean Perrin and rewarded with a Nobel prize. Einstein subscribed to the hypothesis that matter is not a continuum, and is rather

[1]Often in literature and dissemination events, it is improperly said that Brown observed pollen grains moving in water. This is simply not possible, neither it is reported by Brown, as pollen grains are too large to undergo such motion with a sufficient amplitude to be visible with microscopes at the time. We illustrate this quantitatively further down.

made of elementary and indivisible constituents appropriately called "atoms". In the opening of his historical publication on what we now know to be Brownian motion [6], he stated that "bodies of microscopically visible size suspended in liquids must, as a result of thermal molecular motion, perform motions of such magnitude that these motions can be easily detected by a microscope". This meant that the colloids were not moving from a process inherent to themselves, but rather the molecules of fluid had a certain velocity owing to thermal energy. They would then collide with the colloid in a random and disordered fashion, transferring a portion of their momentum with every impact (Fig. 1.2). Under a microscope, the human eye could only perceive the resulting erratic motion of the colloid, which led Brown to his first hypotheses. Einstein started interpreting a suspension of particles in a fluid not as fundamentally distinct entities from the fluid molecules. Rather, they were larger molecules which still obeyed the same physical laws of the molecular-kinetic theory of heat, previously developed by Maxwell and Boltzmann [7, 8]. He connected his developments to ordinary diffusion from Fick's law [9], and quantitatively derived the expression for the diffusion coefficient of a dilute suspension of particles in a fluid:

$$\boxed{D_0 = \frac{k_\mathrm{B}\Theta}{\gamma}}\,, \tag{1.1}$$

where k_B is the Boltzmann constant, Θ the fluid temperature, and γ is the friction coefficient deriving from the Stokes drag ($\gamma = 6\pi\eta_0 a$ for a perfect sphere of radius a in a fluid of viscosity η_0). This expression, known as the Stokes-Einstein relation could finally give an estimate of the size of the particle of interest, which was for some cases impossible to do experimentally due to optical limitations. I provide, in Table 1.1, an estimate of the diffusion coefficient for a whole pollen grain and a single amyloplast, approximated as a spherical particle. Additionally, this expression could as well predict how quickly a particle would explore space with respect to elapsed time t, with the expression of its Mean Squared Displacement (MSD):

$$\boxed{\mathrm{MSD} = 2dD_0t}\,, \tag{1.2}$$

with d a factor owing to the dimensionality of the system ($d = 2$ for a two-dimensional motion). A spherical colloid would thus explore its available space within a simple fluid linearly with respect to elapsed time. In terms of displacement relative to characteristic size, amyloplasts explored their available space much more efficiently that the larger pollen grains. Perrin later confirmed Einstein's predictions by designing a whole experimental program [10]. He prepared suspensions of known particles, measured displacements (Fig. 1.2), varied particle sizes, liquids, viscosity, and other parameters, and obtained consistent values of Avogadro's number. The agreement made it impossible to doubt the kinetic theory of Brownian motion, and thus the existence of atoms.

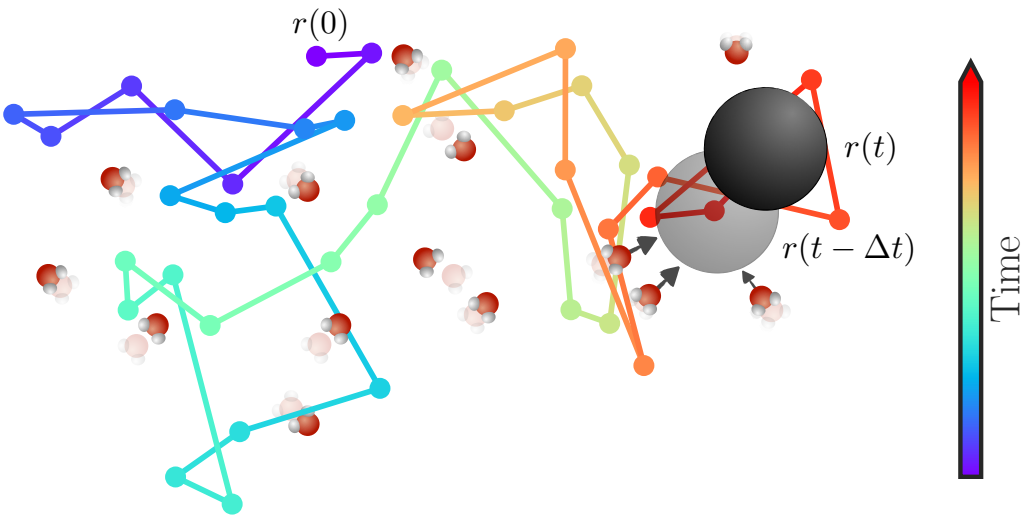

Figure 1.2: Observation of the random and erratic motion of a colloid in a fluid by Jean Perrin (adapted from "Mouvement brownien et réalité moléculaire", 1909). The coarse-grained trajectory of the colloid is represented in a rainbow line. The positions are indicated as r, with $r(0)$ the position at origin, $r(t)$ the position at time t and $r(t - \Delta t)$ the position at one observation time step before t.

Object	Estimated radius a	D_0	MSD at 60 s	$\sqrt{\text{MSD}}$
Clarkia pollen grain	50 µm	4.4×10^{-3} µm^2 s^{-1}	1.05 µm^2	1.0 µm
Amyloplast	3 µm	7.3×10^{-2} µm^2 s^{-1}	17.5 µm^2	4.2 µm

Table 1.1: Order-of-magnitude comparison of Brownian motion for a whole *Clarkia pulchella* pollen grain and for the amyloplasts observed by Brown. The diffusion coefficient is estimated from the Stokes-Einstein relation (Eq. 1.1), using $\Theta = 298$ K and $\eta_0 = 10^{-3}$ Pa.s. The mean-squared displacement corresponds to the two-dimensional projected motion observed under a microscope, as MSD $= 4D_0t$.

1.2 Theoretical developments

This section aims at providing the theoretical tools to extract information from Brownian trajectories, as well as to simulate these trajectories accurately. Most of the information gathered here was based upon David S. Dean's lecture notes on advanced statistical mechanics. First, I present the Langevin equation for the underdamped regime, where inertial effects are measurable at short time lags. I derive the amplitude of the stochastic term in the Langevin equation and deduce the Stokes-Einstein relation (Eq. 1.1). I then express the crossover to the overdamped limit, where the colloid is treated as inertialess. I present the stochastic conventions I use in the Fokker-Planck equation, which I use to derive the diffusion equation.

1.2.1 The underdamped Langevin equation

Some stochastic processes can be described by a Langevin equation [11], which is essentially a Newton's second law with a stochastic forcing owing to the thermal agitation of the fluid molecules. The velocity of a colloid of mass m and friction coefficient γ obeys the following Langevin equation:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -\gamma v + \xi(t) \ , \tag{1.3}$$

where $\xi(t)$ is a stochastic forcing such that $\xi(t) = \sqrt{A}\eta(t)$, and η is a Gaussian white noise, as $\langle \eta(t) \rangle_t = 0$ and $\langle \eta(t)\eta(t') \rangle_t = \delta(t-t')$, where $\langle . \rangle_t$ indicates the temporal average. We aim at deriving an expression for A, the noise strength. We do so, in this development, by expressing $\mathrm{d}(v^2)$ and identifying A using the equipartition of energy.

We rewrite Eq. 1.3 as an Ornstein-Uhlenbeck [12] process:

$$\mathrm{d}v = -\lambda v \mathrm{d}t + \sigma \mathrm{d}W_t \ , \tag{1.4}$$

where $\mathrm{d}W_t$ is an increment of a Wiener process with $\mathrm{d}W_t^2 = \mathrm{d}t$, and with $\lambda = \gamma/m$ and

$\sigma = \sqrt{A}/m$. Taking the square of Eq. 1.4, we get:

$$(\mathrm{d}v)^2 = (\lambda v)^2\ (\mathrm{d}t)^2 - 2\lambda\sigma v \mathrm{d}t\,\mathrm{d}W_t + \sigma^2 \mathrm{d}t\ . \tag{1.5}$$

For $\mathrm{d}t \to 0$, terms of higher order than $\mathcal{O}\mathrm{d}t$ vanish, and we get:

$$\boxed{(\mathrm{d}v)^2 = \sigma^2 \mathrm{d}t\ .} \tag{1.6}$$

Therefore, when differentiating (v^2), the quadratic increment cannot be discarded (see the next section). One obtains:

$$\mathrm{d}(v^2) = 2v\,\mathrm{d}v + (\mathrm{d}v)^2\ . \tag{1.7}$$

Thus,

$$\begin{aligned} \mathrm{d}\left(v^2\right) &= -2v^2\lambda \mathrm{d}t + 2v\sigma \mathrm{d}W_t + \sigma^2 \mathrm{d}t\ , \\ \frac{\mathrm{d}\langle (v^2)\rangle}{\mathrm{d}t} &= -2\langle v^2\rangle\lambda + \sigma^2\ , \\ &= -\frac{2\langle v^2\rangle\gamma}{m} + \frac{A}{m^2}\ , \end{aligned} \tag{1.8}$$

because $\langle \mathrm{d}W_t\rangle = 0$. However, at equilibrium, the variance of the velocity is stationary:

$$\frac{\mathrm{d}\langle (v^2)_{\mathrm{eq}}\rangle}{\mathrm{d}t} = 0\ . \tag{1.9}$$

Thus,

$$\langle v^2\rangle_{\mathrm{eq}} = \frac{A}{2\gamma m}\ . \tag{1.10}$$

Finally, we invoke the theorem on the equipartition of energy:

$$\boxed{\frac{1}{2}m\langle v^2\rangle_{\mathrm{eq}} = \frac{1}{2}k_{\mathrm{B}}\Theta}\ , \tag{1.11}$$

and, comparing to Eq. 1.10, we finally obtain:

$$A = 2\gamma k_{\mathrm{B}}\Theta\ , \tag{1.12}$$

which is the noise amplitude of the process described in Eq. 1.3. Therefore, one has:

$$\langle \xi(t)\xi(t')\rangle = 2\gamma k_{\mathrm{B}}\Theta\delta(t-t')\ . \tag{1.13}$$

1.2.2 Ballistic-to-diffusive crossover and overdamped limit

With the noise strength, we can now identify the time scale over which the velocity loses memory of its initial value. Starting from Eq. 1.4, we first consider the deterministic relaxation of the mean velocity. The formal solution of the Ornstein-Uhlenbeck equation is [13]:

$$v(t) = v(0)\mathrm{e}^{-\lambda t} + \sigma \int_0^t \mathrm{e}^{-\lambda(t-s)}\,\mathrm{d}W_s \ . \tag{1.14}$$

Multiplying by $v(0)$ and averaging gives:

$$\langle v(t)v(0)\rangle = \langle v^2(0)\rangle \mathrm{e}^{-\lambda t} + \sigma \left\langle v(0) \int_0^t \mathrm{e}^{-\lambda(t-s)}\,\mathrm{d}W_s \right\rangle \ . \tag{1.15}$$

The stochastic integral is independent of the initial velocity and has zero mean, so the second term vanishes. At thermal equilibrium, the mean squared velocity v obeys $\langle v^2\rangle_{\mathrm{eq}} = k_{\mathrm{B}}\Theta/m$, and thus:

$$\boxed{\langle v(t)v(0)\rangle = \frac{k_{\mathrm{B}}\Theta}{m} \exp\left(-\frac{t}{\tau_{\mathrm{m}}}\right)}\ , \tag{1.16}$$

where:

$$\tau_{\mathrm{m}} = \frac{1}{\lambda} = \frac{m}{\gamma} \tag{1.17}$$

is the momentum relaxation time. This exponential decay of the velocity autocorrelation controls the ballistic-to-diffusive crossover of the mean-squared displacement. Since the displacement is the time integral of the velocity, as:

$$\Delta x(t) = \int_0^t v(s)\,\mathrm{d}s \ , \tag{1.18}$$

the MSD is [13]:

$$\left\langle \Delta x^2(t) \right\rangle = 2D_0 \left[t - \tau_{\mathrm{m}} \left(1 - \exp\left(-\frac{t}{\tau_{\mathrm{m}}} \right) \right) \right] \ . \tag{1.19}$$

Eq. 1.19 contains two limiting regimes. At short times ($t \ll \tau_{\mathrm{m}}$), the exponential can be expanded, and the MSD behaves as:

$$\left\langle \Delta x^2(t) \right\rangle \simeq \frac{k_{\mathrm{B}}\Theta}{m} t^2 \ . \tag{1.20}$$

The motion is therefore ballistic. At long times lags ($t \gg \tau_{\mathrm{m}}$), the exponential term vanishes and the MSD becomes:

$$\left\langle \Delta x^2(t) \right\rangle \simeq 2D_0 t \ . \tag{1.21}$$

The motion is therefore diffusive. For colloids, τ_{m} is typically much shorter than the experimental observation time. The velocity can therefore be treated as a fast variable and eliminated, leading to the overdamped Langevin equation:

$$\boxed{\gamma\dot{x} = F(x) + \xi(t)} \,. \tag{1.22}$$

Equivalently,

$$\mathrm{d}x = \gamma^{-1}F(x)\,\mathrm{d}t + \sqrt{2D_0}\,\mathrm{d}W_t \,. \tag{1.23}$$

1.2.3 Stochastic conventions

The Langevin equation is a trajectory-level description of Brownian motion. It shows how the position of a colloid evolves under deterministic forces and thermal fluctuations. However, we are sometimes more interested in the probability of finding the colloid at a given position and time, which can be provided by the Fokker-Planck equation [14, 15, 16]. Historically, this equation was introduced by Fokker and Planck in the context of statistical dynamics [17, 18]. Before deriving the Fokker-Planck equation in the next section, we must choose which convention we use to integrate a stochastic differential equation of the form

$$\mathrm{d}X_t = u(X_t)\,\mathrm{d}t + g(X_t)\,\mathrm{d}W_t \,. \tag{1.24}$$

Brownian trajectories are nowhere-differentiable, so $g(X_t)$ inside the stochastic integral must be chosen carefully. A general way to define it is through:

$$\int_t^{t+\Delta t} g(X_s)\,\mathrm{d}W_s \simeq g\left[(1-\alpha)X_t + \alpha X_{t+\Delta t}\right]\Delta W_t \,, \tag{1.25}$$

where $\Delta W_t = W_{t+\Delta t} - W_t$. Different values of α correspond to different stochastic conventions [19]. The Itô convention [20] corresponds to $\alpha = 0$. In this case,

$$\int_t^{t+\Delta t} g(X_s)\,\mathrm{d}W_s \simeq g(X_t)\Delta W_t \,. \tag{1.26}$$

The noise amplitude is therefore evaluated at the beginning of the time step. This makes the integral non-anticipating, as the integrand depends only on information available before the Brownian increment ΔW_t is realised. As a result, the stochastic integral

$$\int_0^t g(X_s)\,\mathrm{d}W_s \tag{1.27}$$

is a martingale. In particular, the expected future increment is zero. This property is quite useful when averaging: stochastic terms of the form $g(X_t)\,\mathrm{d}W_t$ then have zero mean. However, the Itô convention modifies the usual chain rule. Since Brownian increments

scale as:

$$\mathrm{d}W_t \sim \sqrt{\mathrm{d}t} \ , \tag{1.28}$$

one has the Itô rule:

$$(\mathrm{d}W_t)^2 = \mathrm{d}t \ . \tag{1.29}$$

Therefore, terms of second order in $\mathrm{d}W_t$ contribute at order $\mathrm{d}t$ and cannot be neglected. For a smooth function $f(X_t)$, one obtains Itô's formula:

$$\mathrm{d}f(X_t) = f'(X_t)\,\mathrm{d}X_t + \frac{1}{2}f''(X_t)(\mathrm{d}X_t)^2 \ . \tag{1.30}$$

For the Itô stochastic differential equation:

$$\mathrm{d}X_t = u(X_t)\,\mathrm{d}t + g(X_t)\,\mathrm{d}W_t \ , \tag{1.31}$$

this gives:

$$\mathrm{d}f(X_t) = \left[u(X_t)f'(X_t) + \frac{1}{2}g^2(X_t)f''(X_t)\right]\mathrm{d}t + g(X_t)f'(X_t)\,\mathrm{d}W_t \ . \tag{1.32}$$

We keep in mind this second order contribution as we use the Itô convention further down when deriving the Fokker-planck equation. Another common convention is the Stratonovich convention, corresponding to $\alpha = 1/2$:

$$\int_t^{t+\Delta t} g(X_s)\,\mathrm{d}W_s \simeq g\left(\frac{X_t + X_{t+\Delta t}}{2}\right)\Delta W_t \ . \tag{1.33}$$

The noise amplitude is then evaluated at the midpoint of the time step. The Stratonovich convention preserves the usual rules of calculus, which often makes it more natural when stochastic equations are obtained as limits of deterministic equations driven by rapidly fluctuating noise with a finite correlation time. However, the corresponding stochastic integral is no longer non-anticipating, because the midpoint involves $X_{t+\Delta t}$, which already contains information about the increment ΔW_t. For multiplicative noise, the convention changes the drift term that appears in the Itô representation of the dynamics. In this manuscript, we use the Itô convention.

1.2.4 From Langevin dynamics to the Fokker-Planck equation

Let $P(x, t|x_0)$ denote the probability density of finding the particle at position x at time t, knowing that it was initially located at x_0. Its initial condition is therefore:

$$P(x, t = 0|x_0) = \delta(x - x_0) \ . \tag{1.34}$$

The Fokker-Planck equation is the evolution equation obeyed by P. We consider the Itô stochastic differential equation defined in section 1.2.3 on the particle position in time X_t:

$$\mathrm{d}X_t = u(X_t)\,\mathrm{d}t + g(X_t)\,\mathrm{d}W_t \ , \tag{1.35}$$

where $u(X_t)$ is the drift, $g(X_t)$ is the noise amplitude, and W_t is a Wiener process. The corresponding Fokker-Planck equation reads:

$$\frac{\partial P(x,t|x_0)}{\partial t} = -\frac{\partial}{\partial x}\left[u(x)P(x,t|x_0)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left[g^2(x)P(x,t|x_0)\right] \ . \tag{1.36}$$

Equivalently, the density P can be written as a conservation equation:

$$\frac{\partial P}{\partial t} = -\frac{\partial J}{\partial x} \ , \tag{1.37}$$

where $J(x,t)$ is the probability current. For a stochastic process, this current contains both a deterministic contribution, due to the drift, and a diffusive contribution, due to the noise. Since the total probability is conserved, the probability current reads:

$$J(x,t) = u(x)P(x,t|x_0) - \frac{1}{2}\frac{\partial}{\partial x}\left[g^2(x)P(x,t|x_0)\right] \tag{1.38}$$

in this case.

The first term in the current corresponds to the deterministic transport by the drift velocity $u(x)$, while the second term is the spreading induced by the stochastic fluctuations. In the particular case of additive noise, where g is constant, this diffusive contribution reduces to the usual diffusion current. For example, for a Brownian particle in an unbounded, quiescent bulk fluid, the overdamped Langevin equation is:

$$\mathrm{d}x = \sqrt{2D_0}\,\mathrm{d}W_t \ . \tag{1.39}$$

We identify:

$$u(x) = 0, \qquad g(x) = \sqrt{2D_0} \ . \tag{1.40}$$

Substituting into Eq. (1.36) gives:

$$J = -D_0\frac{\partial P}{\partial x} \ , \tag{1.41}$$

which is the Fickian diffusion current [9]. A more complex case involving multiplicative noise is presented in chapter 2.

1.3 Bases of numerical simulations

1.3.1 The Euler scheme

Computational physics by essence do not work with continuous processes. One of the biggest challenges is therefore to choose or sometimes develop discretisation methods to remain accurate physically, but also computationally efficient. Throughout this manuscript, the codes I develop mostly rely on the Euler-Maruyama method [21, 22, 23]. Fortunately, this method has largely been applied to Brownian problems as an extension of the Euler method to stochastic processes, and is exceptionally inexpensive to simulate. This method approximates the solution of a stochastic differential equation by discretising time into small steps Δt. The continuous-time stochastic process is then approximated by a discrete-time process, where the state of the system is updated at each time step according to the deterministic and stochastic components of the dynamics. We approximate the differentiation of a variable X with respect to time t using the Euler-Maruyama method as follows:

$$\frac{\mathrm{d}X}{\mathrm{d}t} \simeq \frac{X_{i+1} - Xi}{\Delta t} \ , \tag{1.42}$$

where X_i is the value of X at time $t_i = i\Delta t$. The update rule for the variable X is then given by:

$$X_{i+1} = X_i + u(X_i)\Delta t + g(X_i)\mathrm{d}W_i \ . \tag{1.43}$$

The Euler-Maruyama remains, however, a first-order approximation: it is by no means accurate at any point, and its recursivity means errors build upon each other. The choice of a small enough time step is thus crucial to ensure initial accuracy and minimal error propagation throughout the simulation. For the sake of safety, I follow a simple protocol in the determination of the time step I use in simulations. First, I identify all time scales in the system. For the aforementioned underdamped Brownian motion, we expect only one: the inertial time scale τ_m. Other processes can have various characteristic times, and in that case, we choose the smallest amongst them. Then, I choose a time lag Δt which satisfies:

$$\Delta t \leq \frac{1}{100}\tau_\mathrm{m} \ , \tag{1.44}$$

where 1/100 is a common safety used to ensure the time step is smaller than the smallest time scale of the system. I illustrate the importance of the time lag duration choice in Fig. 1.3. The inertial regime is particularly affected as it depends on the velocity. The exact, deterministic velocity decays as:

$$v(t + \Delta t) = \exp -\frac{\Delta t}{\tau_\mathrm{m}} v(t) \ , \tag{1.45}$$

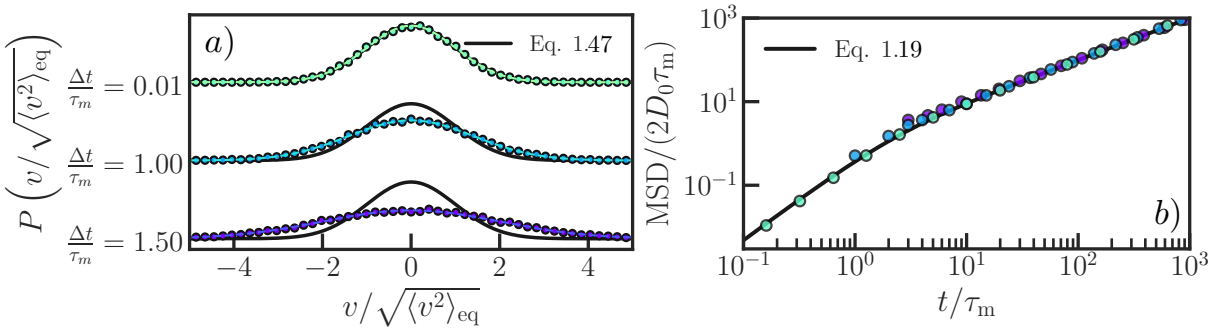

Figure 1.3: Effect of time step choice on numerical results when simulating Eq. 1.3 with an Euler scheme, with purple disks corresponding to simulated data with $\Delta t/\tau_{\rm m} = 1.5$, blue disks with $\Delta t/\tau_{\rm m} = 1.0$, and teal disks with $\Delta t/\tau_{\rm m} = 0.01$, and other parameters in Table 1.2. *a*) PDF of renormalised velocities $v/\sqrt{\langle v^2\rangle_{\rm eq}}$. Black lines show the expected value from Eq. 1.47. Dashed curves show the Gaussian fit to the binned velocity histograms in their respective colours. Curves are offset for readability. *b*) MSD renormalised by $2D_0\tau_m$ as a function of elapsed time for a single trajectory. The black curve indicates the exact prediction from Eq. 1.19.

but Euler discretisation implies:

$$v_{i+1} = \left(1 - \frac{\Delta t}{\tau_{\rm m}}\right) v_i \ . \tag{1.46}$$

Comparatively, for a time step $\Delta t = \tau_{\rm m}$, the exact velocity decay is theoretically predicted with a factor of $\exp(-1) = 0.37$, whereas the Euler method predicts a factor of $\left(1 - \frac{\Delta t}{\tau_{\rm m}}\right) = 0$, a complete erasure of velocity after a single time step. This is readily visible on the first panel of Fig. 1.3, where the probability density function (PDF) of the velocities of a simulated trajectory generated from the discretised Eq. 1.3 is shown for various choices of Δt. In theory, the PDF of velocities should follow the Gaussian distribution of the Maxwell-Boltzmann statistics, as:

$$P(v) = \frac{1}{Z} \exp\left(-\frac{v^2}{2\langle v^2\rangle_{\rm eq}}\right) \ , \tag{1.47}$$

where $1/Z$ is the normalisation constant. The choice of a time step of order $\tau_{\rm m}$ implicates a PDF which does not follow Eq. 1.47. (blue and purple disks on panel *a*) of Fig. 1.3). The choice of a time step $\Delta t = 0.01\tau_{\rm m}$ (teal disks) is, however, small enough to ensure the PDF of velocities follows the expected Maxwell-Boltzmann distribution. Furthermore, since the MSD depends on the velocity variance as:

$$\langle \Delta x^2(t)\rangle = 2\int_0^t (t-s)\langle v(s)v(0)\rangle \mathrm{d}s \ , \tag{1.48}$$

a too large choice of time step entails a wrong ballistic velocity, which immediately invalidates the ballistic and diffusive regimes, though the latter is less noticeably modified in the case of free Brownian motion (b) panel on Fig. 1.3). We can see that the choice of a large time step Δt (purple and blue disks) can lead to significant errors in the simulation results, particularly in the short-time ballistic regime.

1.4 Discretised underdamped Langevin equation

We can discretise Eq. 1.3 using the Euler scheme. We must, however, choose how to treat the noise term $\mathrm{d}W_t/\mathrm{d}t$, which we re-express in its discretised equivalent as $w_i/\Delta t$, where w_i is a random Gaussian variable drawn from a distribution which satisfies $\langle w_i \rangle = 0$ and $\langle w_i w_j \rangle = \Delta t \delta_{ij}$. Under those assumptions, Eq. 1.3 becomes:

$$v_{i+1} = \left(1 - \frac{\Delta t}{\tau_\mathrm{m}}\right) v_i + \sqrt{\frac{2k_\mathrm{B}\Theta}{m\tau_\mathrm{m}}} w_i \ . \tag{1.49}$$

Below, I provide a snippet to simulate easily this equation, including a thermalisation step to draw velocities from equilibrium and avoid altered statistics at the beginning of the simulation, which pollute the ballistic regime. Furthermore, the whole temporal loop is provided to better illustrate the workflow of a simulation for a single trajectory. However, I typically conduct multiple simulations of a same system and stack their statistics together to infer ballistic and diffusive constants from their observables.

Snippet 1.1: Simulation of an underdamped, free bulk Brownian motion

```
import numpy as np

vT2 = kBT / m # Thermal squared velocity
vT = sqrt(vT2)

def simulate_underdamped_euler(dt):
    nsteps = int(np.round(tmax / dt)) # Simulation steps
    t = np.arange(nsteps + 1) * dt    # Time array
    x = np.zeros((ntraj, nsteps + 1)) # Position array
    v = np.zeros((ntraj, nsteps + 1)) # Velocity array

    # v[:, 0] = 0.0                   # Non-thermalised initial velocity
    v[:, 0] = rng.normal(0, vT, size = ntraj) #T Thermalised initial velocity
    noise_prefactor = np.sqrt(2 * vT2 / tau)

    for n in range(nsteps): # v and x array updates from discretised Langevin equations
```

```
            wi = rng.normal(0.0, sqrt(dt), size = ntraj) # Noise, avg=0, std=
    sqrt dt
            v[:, n + 1] = (
                v[:, n] - (dt / tau) * v[:, n] + noise_prefactor * wi
            )
            x[:, n + 1] = x[:, n] + dt * v[:, n]
        return t, x, v
        t, x, v = simulate_underdamped_euler(dt)
```

A typical two-dimensional trajectory generated from Eq. 1.49 is shown in Fig. 1.4.

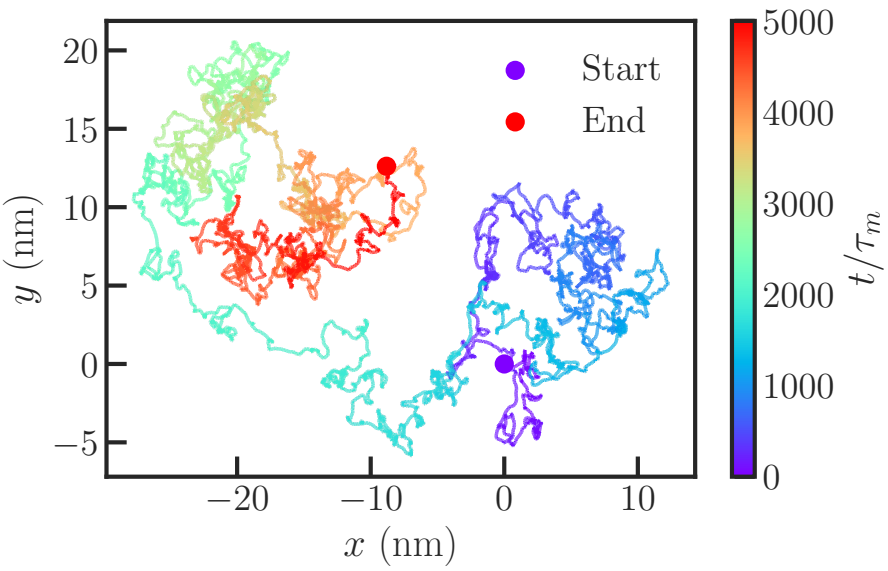

Figure 1.4: Trajectory generated from the two-dimensional equivalent of Eq. 1.49, with parameters given in Table 1.2. Colorbar indicates elapsed time. Start and finish positions are indicated with a purple and a red disk respectively.

1.5 Numerical results

1.5.1 Mean squared displacement and inference of ballistic and diffusive constants

The MSD can be computed from several techniques. The first I present is the most straightforward one: from a generated trajectory, one defines its displacements by simply subtracting the trajectory array to itself, shifted in time, for every possible time lag. We then compute the variance of the resulting displacement arrays at all time lags, and retrieve the MSD. I provide a plot for a simple one-dimensional trajectory in Fig. 1.5.

It is sometimes more convenient to compute the MSD at only certain given time lags

Table 1.2: Physical parameters used in the simulation.

Symbol	Significance	Value	Unit
a	colloid radius	1.0×10^{-6}	m
ρ_f	water density	1000	$\mathrm{kg.\,m^{-3}}$
ρ_p	colloid density	1050	$\mathrm{kg.\,m^{-3}}$
V	colloid volume	4.19×10^{-18}	$\mathrm{m^3}$
$k_\mathrm{B}\Theta$	thermal energy	4.19×10^{-21}	J
η_0	water dynamic viscosity	1.0×10^{-3}	Pa. s
m	colloid mass	4.40×10^{-15}	kg
γ	Stokes drag coefficient	1.88×10^{-8}	$\mathrm{kg.\,s^{-1}}$
τ_m	inertial time scale, Eq. 1.17	2.33×10^{-7}	s
D_0	bulk colloid diffusion coefficient	2.22×10^{-13}	$\mathrm{m^2.\,s^{-1}}$

to reduce the computational cost of data treatment, and this is the technique we use in chapters 4 and 5. However, there exists a way to be even more efficient by not keeping all these arrays in memory, but rather computing the MSD from the variance of the PDF of displacements. In that case, a single histogram per time lag of our choice is created, with pre-computed bins. We then generate a batch of trajectories, compute the displacements at given time lags, fill the displacement histograms and empty the memory. We can then go on with generating more data, up until the displacement PDFs become stable with new iterations. Then, the variance is computed, and directly gives access to the value of the MSD at a time lag. This is the technique we use in chapters 2 and 3 for different reasons. From the MSD, I extract the mean squared velocity (the squared thermal velocity in this case) from the ballistic regime, and the diffusion constant from the diffusive regime. I use log-fits to do so, which avoid giving too much weight to the points corresponding to large time lags, as they are generally less reliable statistically speaking due to less overlap between the trajectory array and the same array, shifted in time. At long times $t \gg \tau_\mathrm{m}$, taking the logarithm of the MSD (Eq. 1.20) gives:

$$\log\left(\Delta x^2\right) = \log(2D_0) + \log(t)\ . \tag{1.50}$$

Thus, we expect the diffusive regime to have a time exponent of 1. We verify numerically that the fit does return these values with a small tolerance of about 5%, and we can then infer from their offsets the value of the diffusion coefficient D_0. We find, for the parameters given in Table 1.2 and the MSD in Fig. 1.5, a diffusion coefficient of $D_{0,\mathrm{fit}} = 2.26 \times 10^{-13}\,\mathrm{m^2\,s^{-1}}$, which is in good agreement with the theoretical value of $D_0 = 2.22 \times 10^{-13}\,\mathrm{m^2\,s^{-1}}$. Arguably, fitting the whole expression from Eq. 1.19 would be more accurate, as it would take into account the whole ballistic-to-diffusive crossover. Since the purpose of this section is to present my typical methods, this is not what I choose to present here. I do not always have access to the full MSD expression beforehand, and I prefer as a rule

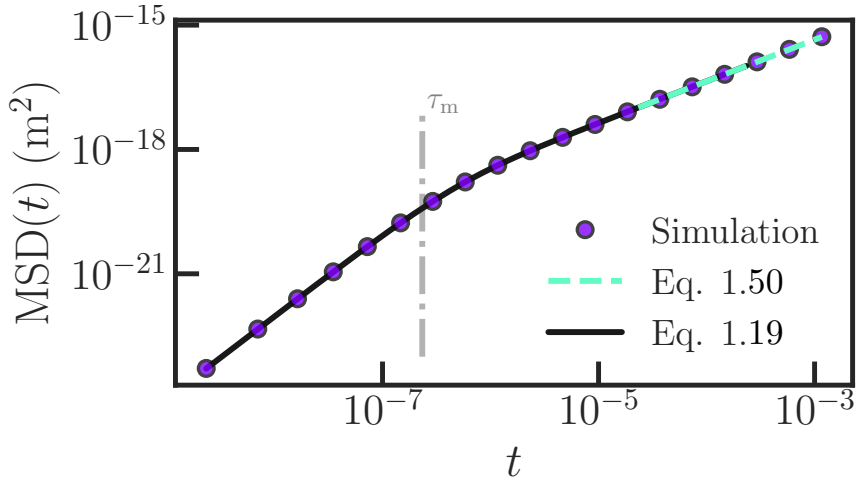

Figure 1.5: Mean squared displacement of an underdamped Langevin simulation. Parameters are given in Table 1.2, with $\Delta t = 0.01\tau_\mathrm{m}$ for 5 stacked trajectory statistics. Purple disks correspond to the MSD computed from the mean of the squared displacement arrays at give times, and the black curve corresponds to the theoretical prediction from Eq. 1.19. The dashed teal line indicates the fit to the diffusive regime (Eq. 1.50) beyond τ_m (grey dash dots) from which we retrieve a diffusion coefficient of $D_{0,\mathrm{fit}} = 2.26 \times 10^{-13}\,\mathrm{m^2\,s^{-1}}$, in good agreement with the theoretical value.

of thumb to keep models and their treatment as general as possible.

1.6 Take-home message

In this introductory chapter, I have presented the history and bases of the theory on Brownian motion. I remained in the most simple framework, for a single colloid in a quiescent and unbounded fluid. I also showed the common methods used to discretise and generate trajectories numerically. I also provided the different means that are usually used to extract dynamical quantities such as the MSD from trajectory data. All these developments will be used in the following chapters. The generation and treatment codes are available on the GitHub page related to this manuscript.

Chapter 2

Hydrodynamic interactions between two colloids near a boundary

2.1 Introduction

The last chapter focused on a very simple, ideal case: a single colloid, far from any boundary. Though we mention Brownian solutes, we consider each colloid as non-interacting with its peers. These assumptions and choices are strong, and very often in biological settings, they do not suffice to describe the dynamics of the system. Nevertheless, confined Brownian motion has been extensively investigated over the last century. Even at long range, hydrodynamic interactions influence the behaviour of colloids diffusing in a fluid [24, 25]. We now know that introducing a single boundary near a colloid effectively increases the dissipation in an inhomogeneous and anisotropic way [26, 27]. The motion of the colloid results in two distinct modes, along the boundary direction, and perpendicular to it [27]. In turn, because the diffusion coefficient depends inversely on the drag coefficient of the colloid (Eq. 1.1), it is hindered by confinement. Moreover, volume fraction also hinders a colloid's diffusion, while collective diffusion of the solute speeds up as osmotic pressure gradients become stronger with higher local concentration [25]. The pair diffusion of colloids in hydrodynamic interaction is thoroughly documented [28, 29]. It is fairly common, in nature, to find highly confined, crowded solutions. For instance, the cell cytoplasm is packed with proteins, RNA, ribosomes, organelles, and confined by a cell membrane. How do crowding and confinement couple hydrodynamically? This coupling is difficult to resolve, even when considering only a pair of colloids in confinement. Indeed, it is permitted only to a certain extent to superimpose the hydrodynamic contributions of a boundary and a pair of colloids. This chapter aims at first decomposing the problem into confined, dilute suspensions (Fig. 2.1 *a*)), then pair diffusion (Fig. 2.1 *b*)). I move to the coupling of these two effects (Fig. 2.1 *c*)), and developments used to model this coupling. I present an experimental study carried out by Dufresne *et al.* [30] exploring the effect on confinement on pair diffusion and recover these results numerically. Finally, I present some prospective numerical works where I go beyond Dufresne's study.

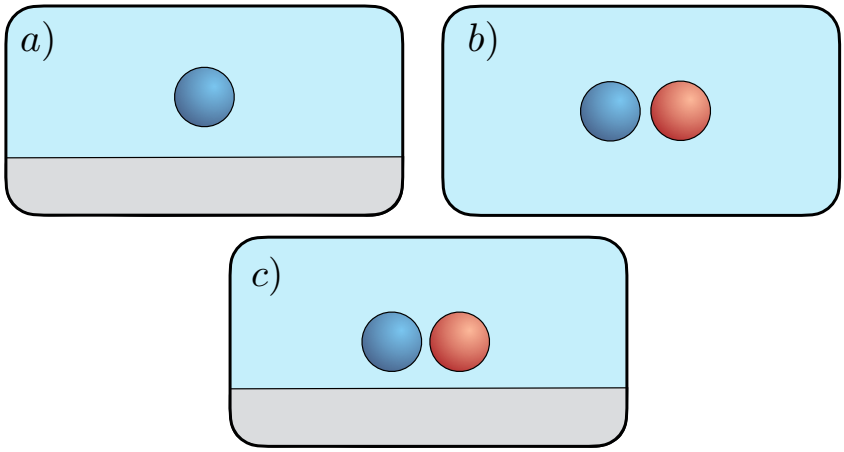

Figure 2.1: Schematic representation of the three cases studied in this chapter. *a*) A single colloid diffusing in a fluid over an infinite, hard boundary. *b*) Two colloids interacting with each other through hydrodynamic interactions in an unbounded fluid. *c*) A combination of the two first cases, where two colloids interact with each other and with an infinite, hard boundary.

2.2 Interactions between a colloid and a boundary

I describe here the interactions that occur in an experimental setup of a negatively-charged colloid in a fluid near a negatively-charged, no-slip boundary. We take the boundary normal to the vertical $z-$ axis, laying at $z = 0$. Over that boundary lays a fluid of density ρ, viscosity η_0, and thermal energy $k_{\mathrm{B}}\Theta$. All parameters for the following derivations are summarised in Table 2.1. In the fluid, we now submerge a negatively-charged colloid of radius a. Due to the no-slip boundary condition, we must modify the effective friction felt by the colloid [27, 26]. This correction brings inhomogeneity along the z-axis and anisotropy, as the x and y components of the friction coefficient differ from the z component.

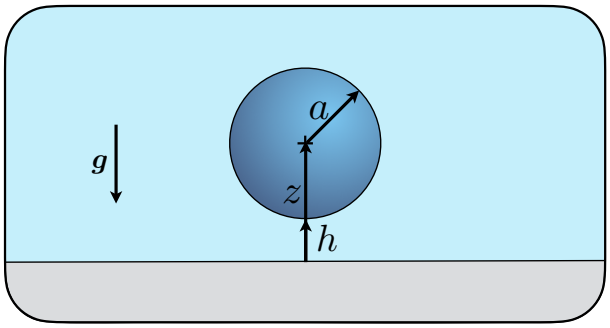

Figure 2.2: Schematic representation of a colloid of radius a at a distance z from an infinite boundary.

Table 2.1: Typical parameters for a silica colloid suspended in water near a silica boundary at room temperature. Values are indicative.

Symbol	Quantity	Typical value
a	Colloid radius	$0.5 - 2$ μm
z	Distance from boundary centre-to-boundary	$a - 20a$ μm
h	Particle-boundary gap	$(z - a)$ μm
ρ	Water density	10^3 kg.m^{-3}
η_0	Water viscosity	10^{-3} Pa.s
Θ	Temperature	298 K
k_B	Boltzmann constant	1.38×10^{-23} J.K^{-1}
$k_\mathrm{B}\Theta$	Thermal energy	4.1×10^{-21} J
$\gamma_\perp(z)$	Effective perpendicular friction	$6\pi\eta_0 a$ to $\gg 6\pi\eta_0 a$
$\gamma_\parallel(z)$	Effective parallel friction	$6\pi\eta_0 a$ to $\gg 6\pi\eta_0 a$
l_B	Boltzmann length	$0.1 - 10$ μm
l_D	Debye length	$1 - 100$ nm

2.2.1 Modified effective friction coefficient

In his seminal work [27], Brenner derived the exact expression for the effective friction felt by a sphere moving perpendicularly to a plane boundary, $\gamma_\perp(h)$ where h is the colloid apex-plane separation, as:

$$\frac{\gamma_\perp(h)}{6\pi\eta_0 a} = \frac{4}{3}\sinh(\alpha)\sum_{n=1}^{\infty}\frac{n(n+1)}{(2n-1)(2n+3)}\left\{\frac{2\sinh\left[(2n+1)\alpha\right]+(2n+1)\sinh(2\alpha)}{4\sinh^2\left[\left(n+\frac{1}{2}\right)\alpha\right]-(2n+1)^2\sinh^2(\alpha)}-1\right\}, \tag{2.1}$$

with $\alpha = \cosh^{-1}(h/a)$. Because Brenner obtained the effective friction coefficient from solving the full Stokes equations with a no-slip boundary condition, it is worth noting that this expression is exact and valid for all values of $h/a > 0$, up until colloid/plane contact. However, we avoid infinite-series computations and prefer the use of Padé approximants [31], which were used by Honig [32], Bevan and Prieve [33] to approximate Brenner's result as:

$$\frac{\gamma_\perp(h)}{6\pi\eta_0 a} \simeq \frac{6h^2 - 9ah + 2a^2}{6h^2 + 2ah}\ . \tag{2.2}$$

The parallel effective friction, on the other hand, was derived by Faxén [26] for $h/a > 0.1$ and is given by the following expression:

$$\frac{\gamma_\parallel(h)}{6\pi\eta_0 a} \simeq \frac{1}{1 - \frac{9}{16}\frac{a}{h} + \frac{1}{8}\left(\frac{a}{h}\right)^3 - \frac{45}{256}\left(\frac{a}{h}\right)^4 - \frac{1}{16}\left(\frac{a}{h}\right)^5}\ . \tag{2.3}$$

Both Eq. 2.2 and Eq. 2.3 show that the self-friction increases as the colloid approaches

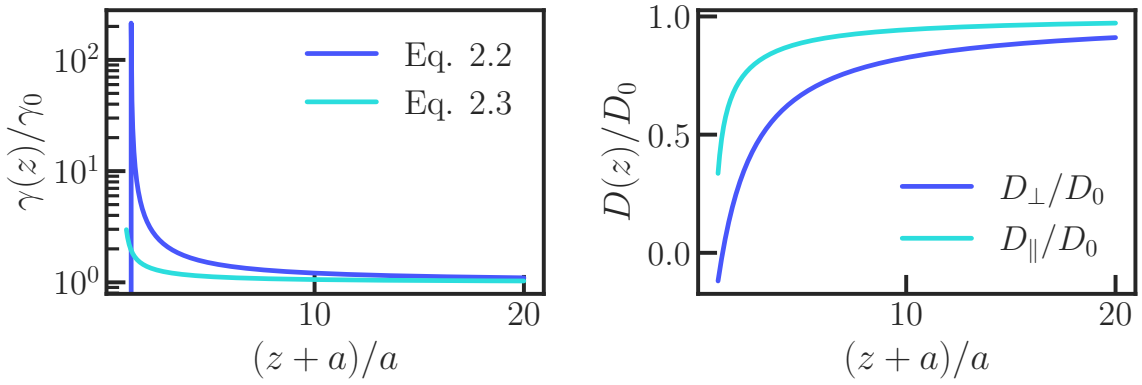

Figure 2.3: Effective friction coefficients and corresponding effective diffusion coefficients for a colloid diffusing over a plane. Left panel: Effective friction coefficients $\gamma_\perp$ (blue line, Eq. 2.2) and $\gamma_\parallel$ (cyan line, Eq. 2.3) normalised by the bulk friction coefficient $6\pi\eta_0 a$ as a function of the distance to the boundary z. Right panel: Effective diffusion coefficients $D_\perp$ (blue) and $D_\parallel$ (cyan) normalised by the bulk diffusion coefficient D_0 as a function of the renormalised colloid centre-to-boundary distance $(z+a)/a$.

the boundary (Left panel of Fig. 2.3). In turn, the effective diffusion coefficients decrease (Right panel of Fig. 2.3). Furthermore, these expressions can be used to simulate the motion of a colloid over a plane up until contact. We note that in the limit of $z/a \to 1$, $\eta_\perp$ captures the divergence close to the boundary. This is an expected behaviour; as the colloid approaches the boundary, the fluid in between is squeezed out and the lubrication forces diverge. However, $\eta_\parallel$ does not capture the logarithmic divergence expected as the surface-to-surface gap tends to zero.

2.2.2 Total potential

In order to write the correct overdamped Langevin equation for a simulation, we recall that the colloid is also subject to a number of forces. We consider a negatively charged colloid in an ionic (water-salt) solution, confined by a negatively-charged and rigid boundary. These assumptions are made to reproduce experimental setups exactly.

Gravitational potential and buoyant mass

We express the gravitational potential acting upon the colloid as:

$$U_{\text{grav}} = gz\Delta m \ , \tag{2.4}$$

$$= \frac{4}{3}\pi a^3(\rho_{\text{colloid}} - \rho_{\text{fluid}})gz \ , \tag{2.5}$$

where Δm is the buoyant mass of the colloid, g is the acceleration of gravity, ρ_{colloid} and ρ_{fluid} are the mass densities of the colloid and the fluid, respectively. We reexpress Eq. 2.5 as:

$$U_{\text{grav}} = \frac{k_{\text{B}}\Theta}{l_{\text{B}}} z \ , \tag{2.6}$$

where l_{B} is Boltzmann's length, defined as $l_{\text{B}} = k_{\text{B}}\Theta/(g\Delta m)$. Boltzmann's length quantifies the distance over which gravitational potential energy is comparable to thermal energy.

Electrostatic potential and double-layer interactions

Since both the boundary and the colloid have negatively charged surfaces, the overlap of their diffuse ionic layers creates a repulsive electrostatic contribution to the colloid-boundary interaction potential. At thermodynamic equilibrium, the local concentration $c_i(\boldsymbol{r})$ of ionic species i in the diffuse layer follows the Boltzmann distribution [34, 35]:

$$c_i(\boldsymbol{r}) = c_i^{\infty} \exp\left[-\frac{z_i e\psi(\boldsymbol{r})}{k_{\text{B}}\Theta}\right] \ , \tag{2.7}$$

where c_i^{∞} is the bulk ion concentration, z_i is the ion valency, e is the elementary charge, and $\psi(\boldsymbol{r})$ is the electrostatic potential. The latter satisfies the Poisson equation:

$$\nabla^2\psi(\boldsymbol{r}) = -\frac{1}{\varepsilon}\sum_i z_i e c_i(\boldsymbol{r}) \ , \tag{2.8}$$

where ε is the permittivity of the fluid. Substituting Eq. (2.7) into Eq. (2.8) yields the Poisson-Boltzmann equation:

$$\nabla^2\psi(\boldsymbol{r}) + \frac{e}{\varepsilon}\sum_i z_i c_i^{\infty} \exp\left[-\frac{z_i e\psi(\boldsymbol{r})}{k_{\text{B}}\Theta}\right] = 0 \ . \tag{2.9}$$

In the Debye-Hückel approximation, which is valid for sufficiently small electrostatic potentials such that $|z_i e\psi| \ll k_{\text{B}}\Theta$, the exponential terms can be linearised, as:

$$\nabla^2\psi(\boldsymbol{r}) + \frac{e}{\varepsilon}\sum_i z_i c_i^{\infty}\left[1 - \frac{z_i e\psi(\boldsymbol{r})}{k_{\text{B}}\Theta}\right] = 0 \ . \tag{2.10}$$

With bulk electroneutrality:

$$\sum_i z_i c_i^{\infty} = 0 \ , \tag{2.11}$$

we get the linearised Poisson-Boltzmann equation:

$$\nabla^2\psi(\boldsymbol{r}) = \kappa^2\psi(\boldsymbol{r}) \tag{2.12}$$

$$= \frac{1}{l_{\mathrm{D}}^2}\psi(\boldsymbol{r}) \ , \tag{2.13}$$

where $\kappa = l_{\mathrm{D}}^{-1}$ is the inverse Debye length:

$$l_{\mathrm{D}} = \sqrt{\frac{\varepsilon k_{\mathrm{B}}\Theta}{e^2\sum_i z_i^2 c_i^\infty}} \ . \tag{2.14}$$

We compute the electrostatic repulsion between a charged sphere and a charged plane. The plane can be obtained as a limiting case of a sphere of infinite radius. We first consider the interaction between two charged spheres [34]. In the Debye-Hückel approximation, the electrostatic potential around a single charged sphere satisfies, in spherical coordinates:

$$\frac{1}{r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r)}{\partial r}\right)\right] = \frac{1}{l_{\mathrm{D}}^2}\psi(r) \ . \tag{2.15}$$

The general solution is:

$$\psi(r) = K_1\frac{\exp(r/l_{\mathrm{D}})}{r} + K_2\frac{\exp(-r/l_{\mathrm{D}})}{r} \ . \tag{2.16}$$

Imposing $\lim_{r\to\infty}\psi(r) = 0$ gives $K_1 = 0$. The remaining constant is fixed by the constant-surface-charge boundary condition at $r = a$, where a is the sphere radius, namely:

$$-\varepsilon\left.\frac{\partial\psi}{\partial r}\right|_{r=a} = \sigma \ , \tag{2.17}$$

where σ is the surface charge density of the sphere. This gives:

$$\psi(r) = \frac{\sigma a^2}{\varepsilon}\frac{\exp(a/l_{\mathrm{D}})}{1+a/l_{\mathrm{D}}}\frac{\exp(-r/l_{\mathrm{D}})}{r} \ . \tag{2.18}$$

Within the linear superposition approximation, and neglecting the van der Waals contribution, the DLVO interaction potential is therefore approximated by its electrostatic contribution. The energy between two charged spheres of radii a_1 and a_2, surface charge densities σ_1 and σ_2, and surface-to-surface separation z can be written as [36]:

$$U_{\mathrm{elec}}(z) = \frac{4\pi}{\varepsilon}\left(\frac{\sigma_1 a_1^2}{1+l_{\mathrm{D}}^{-1}a_1}\right)\left(\frac{\sigma_2 a_2^2}{1+l_{\mathrm{D}}^{-1}a_2}\right)\frac{\exp(-l_{\mathrm{D}}^{-1}z)}{a_1+a_2+z} \ , \tag{2.19}$$

where $a_1 + a_2 + z$ is the centre-to-centre distance between the two spheres. Taking the limit $a_2 \to \infty$ yields the sphere-boundary interaction potential:

$$\frac{U_{\text{elec}}(z)}{k_{\text{B}}\Theta} = B \exp\left(-\frac{z}{l_{\text{D}}}\right) \ , \tag{2.20}$$

with:

$$B = \frac{4\pi}{k_{\text{B}}\Theta\varepsilon}\left(\frac{\sigma a^2}{1 + l_{\text{D}}^{-1}a}\right) l_{\text{D}}\sigma_{\text{boundary}} \ . \tag{2.21}$$

B can be viewed as an effective dimensionless surface charge amplitude.

Equilibrium distribution

The total potential to which the colloid is subjected is expressed as:

$$\frac{U(z)}{k_{\text{B}}\Theta} = B \exp\frac{-z}{l_{\text{D}}} + \frac{z}{l_{\text{B}}} \ , \qquad \forall z > 0 \ . \tag{2.22}$$

The probability to find the colloid at position z follows the Gibbs-Boltzmann equilibrium distribution [7]:

$$P_{\text{eq}} = \frac{1}{Z}\exp\left[-\frac{U(z)}{k_{\text{B}}\Theta}\right] \ , \tag{2.23}$$

where $1/Z$ is the normalising factor.

2.2.3 Spurious drift

In order to reach the correct equilibrium distribution, we must derive from the correct Fokker-Planck equation the corresponding Langevin equation. Indeed, the vertical coordinate Langevin equation acquires multiplicative noise: the noise amplitude along the $z-$coordinate depends on the z position itself. We begin this derivation with the general expression of the Langevin process:

$$\mathrm{d}z = \frac{F(z)}{\gamma_\perp(z)}\mathrm{d}t + \sqrt{2D_\perp(z)}\mathrm{d}W_t \ . \tag{2.24}$$

where $F(z)/\gamma_\perp(z)$ is the total drift. We take the steady-state of Eq. 1.36:

$$\frac{\mathrm{d}}{\mathrm{d}z}\left\{-u(z) + \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}g(z)^2\right\}P_{\text{ss}}(z) = 0 \ , \tag{2.25}$$

with $u(z) = F(z)/\gamma_\perp(z)$. Because $\lim_{z\to\infty} P_{\text{ss}}(z) = 0$ and $\lim_{z\to\infty}(\mathrm{d}P_{\text{ss}}(z)/\mathrm{d}z) = 0$, one has:

$$u(z)P_{\text{ss}}(z) - \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}\left[g(z)^2 P_{\text{ss}}(z)\right] = 0 \ . \tag{2.26}$$

Substituting $A(z) = g^2(z)P_{\rm ss}(z)$ leads to:

$$\frac{\mathrm{d}A(z)}{\mathrm{d}z} - 2\frac{u(z)}{g^2(z)}A(z) = 0 \ , \tag{2.27}$$

which integrates as:

$$\ln\left(A(z)\right) = \ln\left(g^2(z)\right) + \ln\left(P_{\rm ss}\right) \tag{2.28}$$

$$= \int \frac{u(z')}{g^2(z')/2}\mathrm{d}z' \ . \tag{2.29}$$

Thus:

$$P_{\rm ss}(z) = \frac{1}{g^2(z)}\exp\left[2\int_0^z \frac{u(z')}{g^2(z')}\mathrm{d}z'\right] \ . \tag{2.30}$$

With:

$$\ln\left[g^2(z)\right] = \int \frac{2g'(z')g(z')}{g^2(z')}\mathrm{d}z' \ , \tag{2.31}$$

we get:

$$U_{\rm ss}(z) = -k_{\rm B}\Theta\int \frac{u(z') - g(z')g'(z')}{g^2(z')/2}\mathrm{d}z' \ . \tag{2.32}$$

And, recalling that $F_{\rm ss} = -\mathrm{d}U_{\rm ss}/\mathrm{d}z$, we find:

$$\frac{F_{\rm ss}}{\gamma_\perp(z)} = \frac{F(z)}{\gamma_\perp(z)} - g(z)g'(z) \ . \tag{2.33}$$

The mean velocity can be obtained from Eq. 2.24 as:

$$\left\langle\frac{\Delta z}{\Delta t}\right\rangle \equiv \bar{v_{\rm d}} = \frac{F(z)}{\gamma_\perp(z)} \ . \tag{2.34}$$

Then, one has:

$$\bar{v_{\rm d}} = \frac{1}{\gamma_\perp(z)}\frac{\mathrm{d}U_{\rm ss}(z)}{\mathrm{d}z} + \frac{\mathrm{d}D_\perp(z)}{\mathrm{d}z} \ . \tag{2.35}$$

We recover the deterministic drift velocity, as:

$$\boxed{v_{\rm d} = \frac{k_{\rm B}\Theta}{\gamma_\perp(z)}\left[-\frac{1}{l_{\rm D}}B\exp\left(-\frac{z}{l_{\rm D}}\right) + \frac{1}{l_{\rm B}}\right]}, \tag{2.36}$$

whereas the spurious drift $v_{\rm s}$, in the Padé approximation, reads:

$$\boxed{v_{\rm s} = 2D_0 a\frac{2a^2 + 12ah + 21h^2}{(2a^2 + 9ah + h^2)^2}} \ . \tag{2.37}$$

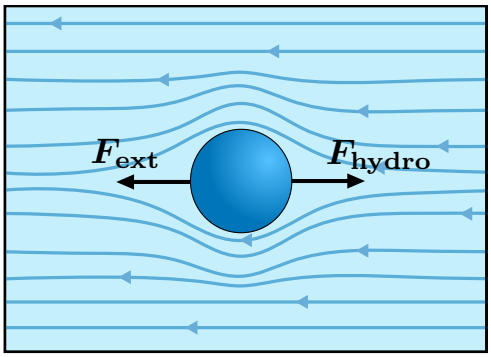

Figure 2.4: Schematic of a colloid in a Stokes flow. The streamlines are represented in dark blue lines. The colloid is pushed with force $\boldsymbol{F}_{\mathbf{ext}}$ and pushes back on the fluid with an equal and opposite force $\boldsymbol{F}_{\mathbf{hydro}}$.

Finally, the overdamped Langevin equations for confined Brownian motion read:

$$\mathrm{d}x = \sqrt{2D_{\parallel}(z)}\mathrm{d}W_x \ , \tag{2.38}$$

$$\mathrm{d}z = \bar{v_{\mathrm{d}}}\mathrm{d}t + \sqrt{2D_{\perp}(z)}\mathrm{d}W_z \ . \tag{2.39}$$

2.3 Colloids in hydrodynamic interaction

Herein, we work in the low-Reynolds-number regime. We define the Reynolds number in more detail in chapter 4. In short, the Reynolds number compares fluid inertia to viscous dissipation. In the present case, $\mathrm{Re} \ll 1$, hence viscous stresses dominate over inertial effects in the fluid. The flow is considered laminar, with a constant profile. The overdamped deterministic force balance reads (Fig. 2.4):

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = 0 \ , \tag{2.40}$$

where m is the mass of the colloid. This means that:

$$\boldsymbol{F}_{\mathbf{ext}} + \boldsymbol{F}_{\mathbf{hydro}} = 0 \ , \tag{2.41}$$

and thus the colloid instantly pushes back with a hydrodynamic force $\boldsymbol{F}_{\mathbf{hydro}}$ on the fluid when submitted to an external force $\boldsymbol{F}_{\mathbf{ext}}$, for example in the case of sedimentation. Stokes equations are linear, and the velocity field generated by one forced particle can be written as a Green function. By evaluating this flow at the position of another particle, we can obtain a hydrodynamic coupling between their velocities. This coupling is encoded in the mobility tensor.

2.3.1 The Stokeslet and the Oseen tensor

The Stokes equations, describing the motion of a fluid in the viscous limit, read [37, 24]:

$$-\nabla p(\boldsymbol{r}) + \eta_0 \nabla^2 \boldsymbol{u}(\boldsymbol{r}) = -\boldsymbol{F}\delta(\boldsymbol{r} - \boldsymbol{r_0}) \ , \tag{2.42}$$

$$\nabla \cdot \boldsymbol{u} = 0 \ , \tag{2.43}$$

where p is the pressure, ∇p is the pressure gradient in the fluid, that is to say, the force density due to pressure differences in the fluid; $\boldsymbol{r}$ is the position to origin, η_0 the dynamic viscosity of the fluid, $\boldsymbol{u}$ the fluid velocity field, and $\eta_0 \nabla^2 \boldsymbol{u}$ is the viscous term; $\boldsymbol{F}\delta(\boldsymbol{r} - \boldsymbol{r_0})$ is the point force at $\boldsymbol{r_0}$; and the second equation denotes fluid incompressibility. Because these equations are linear, we can write the velocity response at $\boldsymbol{r}$ to a point force at $\boldsymbol{r_0}$ as a Green function $\boldsymbol{G}$ [24], as:

$$\boldsymbol{u}(\boldsymbol{r}) = \boldsymbol{G}(\boldsymbol{r} - \boldsymbol{r}_0) \cdot \boldsymbol{F} \ . \tag{2.44}$$

This flow field generated by a point force is known as the Stokeslet. Then, for a point force at origin, the tensor $\boldsymbol{G}(\boldsymbol{r})$ is expressed:

$$\boldsymbol{G}(\boldsymbol{r}) = \frac{1}{8\pi\eta_0 r} \left(\mathbf{I} + \hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\right) \ , \tag{2.45}$$

with $\mathbf{I}$ the identity matrix, and $\hat{\boldsymbol{r}}$ the unit vector relating to $\boldsymbol{r}$. The tensor $\boldsymbol{G}$ is known as the Oseen tensor [38]. It gives the velocity at position $\boldsymbol{r}$ induced by a *point* force applied at another position. Therefore, in the approximation where the particle j exerts a point force $\boldsymbol{F_j}$ on the fluid, the leading-order velocity induced at the position of the particle i is obtained by computing this Green function at $\boldsymbol{r_j} - \boldsymbol{r_i}$ (Fig. 2.5):

$$\boldsymbol{u_i} \simeq \boldsymbol{G}(\boldsymbol{r_j} - \boldsymbol{r_i}) \cdot \boldsymbol{F_j} \ , \qquad i \neq j \ . \tag{2.46}$$

The mobility tensor

We focus here on translational degrees of freedom and neglect torques and rotational dynamics. The velocity field at the position of colloid i generated by the other colloids in the fluid is expressed as:

$$\boldsymbol{u}(\boldsymbol{r}_i) = \sum_{j \neq i} \boldsymbol{G}(\boldsymbol{r}_j - \boldsymbol{r}_i) \cdot \boldsymbol{F}_j \ , \tag{2.47}$$

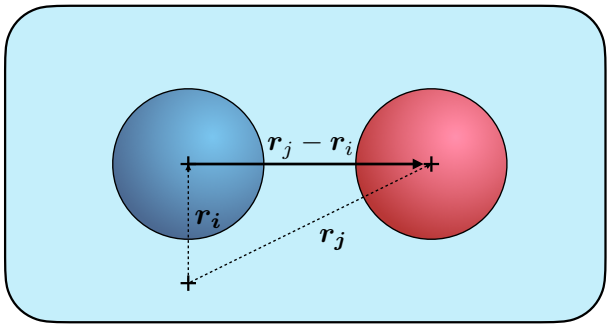

Figure 2.5: Schematic of two colloids interacting in a bulk fluid. Colloids i and j are at positions $\boldsymbol{r_i}$ and $\boldsymbol{r_j}$ from origin respectively, and their separation vector, denoted $\boldsymbol{r_j - r_i}$, is along their line of centres.

in virtue of the superposition principle allowed by the linearity of Stokes equations. We then express the total translational velocity $\boldsymbol{V}_i$ of colloid i as:

$$\boldsymbol{V}_i = \sum_j \boldsymbol{\mu}_{ij}^{tt} \cdot \boldsymbol{F}_j \ , \tag{2.48}$$

where $\boldsymbol{\mu}_{ij}^{tt}$ is the translational mobility block:

$$\boldsymbol{\mu}_{ij}^{tt} = \begin{cases} \dfrac{1}{6\pi_0 a}\mathbf{I} \ , & i = j \ , \\ \boldsymbol{G}(\boldsymbol{r}_j - \boldsymbol{r}_i) \ , & i \neq j \ . \end{cases} \tag{2.49}$$

The diagonal blocks correspond to the self-mobility, and the off-diagonal blocks describe the hydrodynamic coupling between colloids. The grand mobility matrix regroups all mobility blocks:

$$\boldsymbol{\mu}^{tt} = \begin{bmatrix} \boldsymbol{\mu}_{11}^{tt} & \cdots & \boldsymbol{\mu}_{1N}^{tt} \\ \vdots & \ddots & \vdots \\ \boldsymbol{\mu}_{N1}^{tt} & \cdots & \boldsymbol{\mu}_{NN}^{tt} \end{bmatrix} \ . \tag{2.50}$$

The mobility tensor describes the deterministic response of the colloids to applied forces. For Brownian colloids at thermal equilibrium, the same tensor also controls the amplitude and correlations of thermal fluctuations. This connection is given by the fluctuation-dissipation theorem [6, 39]:

$$\mathbf{D}^{tt} = k_\mathrm{B}\Theta\boldsymbol{\mu}^{tt} \ , \tag{2.51}$$

where $\mathbf{D}^{tt}$ is the translational diffusion tensor.

Self- and cross-diffusions

For particles i and j, and Cartesian components $(\alpha, \beta) \in (x, y)$, the diffusion tensor is:

$$2D^{tt}_{i\alpha,j\beta}\Delta t = \langle \Delta r_{i,\alpha}(\Delta t)\Delta r_{j,\beta}(\Delta t)\rangle \ . \tag{2.52}$$

Again, the diagonal blocks ($i = j$) correspond to the self-diffusion of each particle. The off-diagonal blocks ($i \neq j$) give the cross-diffusion between colloids, that is to say, how Brownian displacements of two particles correlate through the fluid. Using the fluctuation-dissipation relation, the Oseen approximation for the translational diffusion blocks becomes:

$$\boldsymbol{D}^{tt}_{ij} = k_{\mathrm{B}}\Theta\boldsymbol{\mu}^{\boldsymbol{tt}}_{\boldsymbol{ij}} = \begin{cases} D_0\mathbf{I} \ , & i = j \ , \\ k_{\mathrm{B}}\Theta\boldsymbol{G}(\boldsymbol{r_j} - \boldsymbol{r_i}) \ , & i \neq j \ . \end{cases} \tag{2.53}$$

The Oseen approximation becomes inaccurate when particles are not well separated. When used to construct a Brownian diffusion tensor, the Oseen approximation may lead to a mobility matrix that is not positive definite at short separations. This is highly problematic because the diffusion tensor must define a positive definite matrix for the Brownian displacements.

2.3.2 Collective and relative modes of motion

For two identical colloids, the diffusion tensor can be diagonalised into collective and relative modes [30]. Along a given direction, either parallel or perpendicular to the line of centres (Fig. 2.5), the two-particle diffusion matrix has the form:

$$\boldsymbol{D} = \begin{pmatrix} D_0 & D_{\mathrm{c}} \\ D_{\mathrm{c}} & D_0 \end{pmatrix} \ , \tag{2.54}$$

where D_{c} is the cross-diffusion coefficient. The eigenvectors are $(1, 1)$ and $(1, -1)$, corresponding to collective and relative motions respectively. The associated self- and cross-diffusion coefficients are therefore:

$$D^{\mathrm{C}} = D_0 + D_{\mathrm{c}} \ , \qquad D^{\mathrm{R}} = D_0 - D_{\mathrm{c}} \ . \tag{2.55}$$

We differentiate the cross-diffusion along (denoted by $D_{\parallel}$) and perpendicular ($D_{\perp}$) to the line of centers of the pair of colloids (Fig. 2.6). Therefore, for the Oseen approximation, the diffusion coefficients in the direction along the line of centres are:

$$\frac{D^{\mathrm{C,R}}_{\parallel}}{2D_0} = 1 \pm \frac{3a}{2r} \ , \tag{2.56}$$

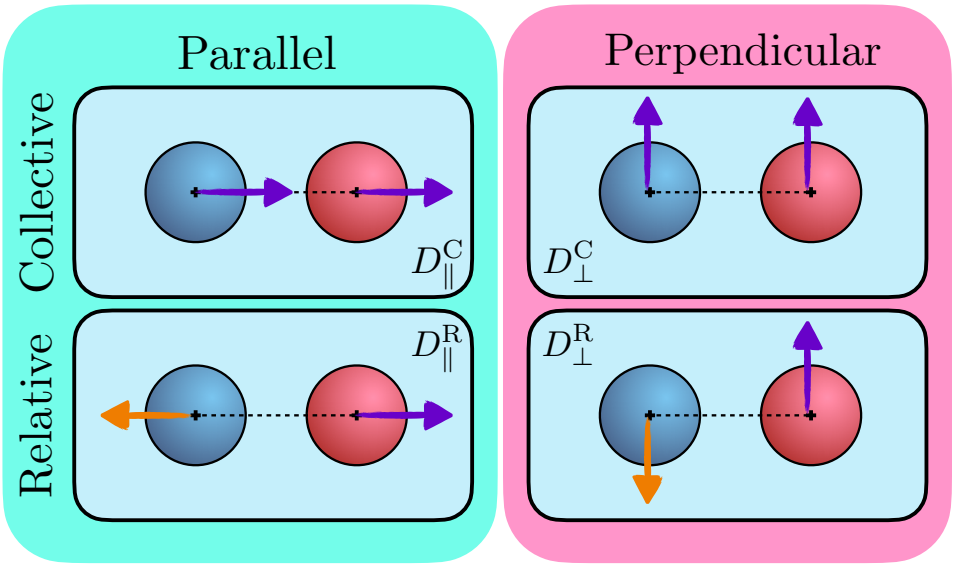

Figure 2.6: Collective and relative diffusion modes. The green quadrant corresponds to the diffusion along the line of centres of the pair of colloid. The pink quadrant corresponds to the perpendicular motion. Top and bottom rows correspond to collective and relative motion, respectively.

and the diffusion coefficients perpendicular to the line of centres read:

$$\frac{D_{\perp}^{\mathrm{C,R}}}{2D_0} = 1 \pm \frac{3a}{4r} \,, \tag{2.57}$$

with + corresponding to the collective mode, and − to the relative mode.

2.3.3 The Rotne-Prager-Yamakawa tensor

To account for finite-size effects, we regularise the Oseen approximation by introducing the Rotne-Prager-Yamakawa (RPY) tensor [40, 41]. This tensor incorporates the finite radius of the colloids at the pairwise level *via* Faxén's first law [26]. For two identical spherical colloids of radius a, separated by $\boldsymbol{r}_{ij}$, the off-diagonal translational RPY mobility for non-overlapping particles reads:

$$\boldsymbol{\mu}_{ij}^{\mathrm{RPY}} = \frac{1}{8\pi\eta_0 r_{ij}} \left[\left(1 + \frac{2a^2}{3r_{ij}^2}\right) \mathbf{I} + \left(1 - \frac{2a^2}{r_{ij}^2}\right) \hat{\boldsymbol{r}}_{ij}\hat{\boldsymbol{r}}_{ij} \right] \,, \qquad i \neq j \,. \tag{2.58}$$

The corresponding self-mobility remains:

$$\boldsymbol{\mu}_{ii}^{\mathrm{RPY}} = \frac{1}{6\pi\eta_0 a}\mathbf{I} \,. \tag{2.59}$$

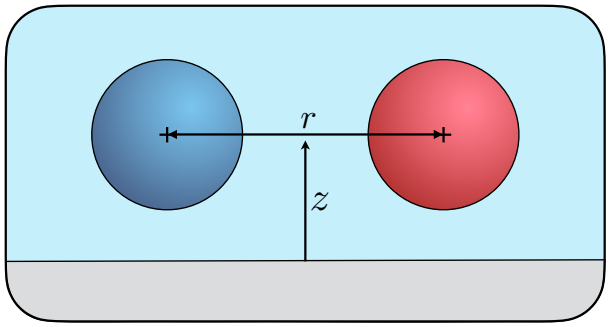

Figure 2.7: Schematic of two colloids interacting near a boundary. The two colloids are separated by a distance r, is along their line of centres. The pair remains in the plane parallel to the boundary, and its height is defined from the centres' height to the boundary as z.

The RPY tensor therefore adds corrections of order $\sim \mathcal{O}(a^2/r_{ij}^2)$ to the Oseen tensor. In the far-field limit, *i.e.* $r_{ij} \gg a$, these corrections vanish and Eq. 2.58 reduces to the Oseen mobility. Again, using the fluctuation-dissipation relation, the RPY diffusion tensor is:

$$\boldsymbol{D}_{ij}^{\mathrm{RPY}} = k_{\mathrm{B}}\Theta\boldsymbol{\mu}_{ij}^{\mathrm{RPY}} \ . \tag{2.60}$$

2.4 Hydrodynamic interactions between two colloids and a boundary

Near a boundary, the pair hydrodynamic coupling must also satisfy the no-slip and no-penetration boundary conditions. This condition proves complex to fulfill, and we present here an experimental study aiming at bridging this gap. We also propose a numerical and experimental study to further investigate the pair-boundary hydrodynamic coupling.

2.4.1 Investigation of the boundary-modified pair hydrodynamic coupling

The hydrodynamic coupling of two Brownian colloids near a planar boundary was studied both experimentally and theoretically by Dufresne *et al.* [30]. In this work, the authors investigate how the presence of a nearby boundary modifies the pair diffusion of two colloids through hydrodynamic interactions by varying the inter-colloid and colloid-boundary distances (Fig. 2.7). Their experimental setup consists of two colloids of radius $a = 0.495 \pm 0.025\,\mu\mathrm{m}$ submerged in a water-salt solution. The fluid layer is $140 \pm 2\,\mu\mathrm{m}$ thick and maintained at $\Theta = 29.00 \pm 0.01\,^{\circ}\mathrm{C}$. Under these conditions, the bulk Stokes-Einstein diffusion coefficient is $D_0 = 0.550 \pm 0.028\,\mu\mathrm{m}^2\,\mathrm{s}^{-1}$. The two colloids are positioned using a single optical tweezer trap. The trap is alternated between the two preset particle

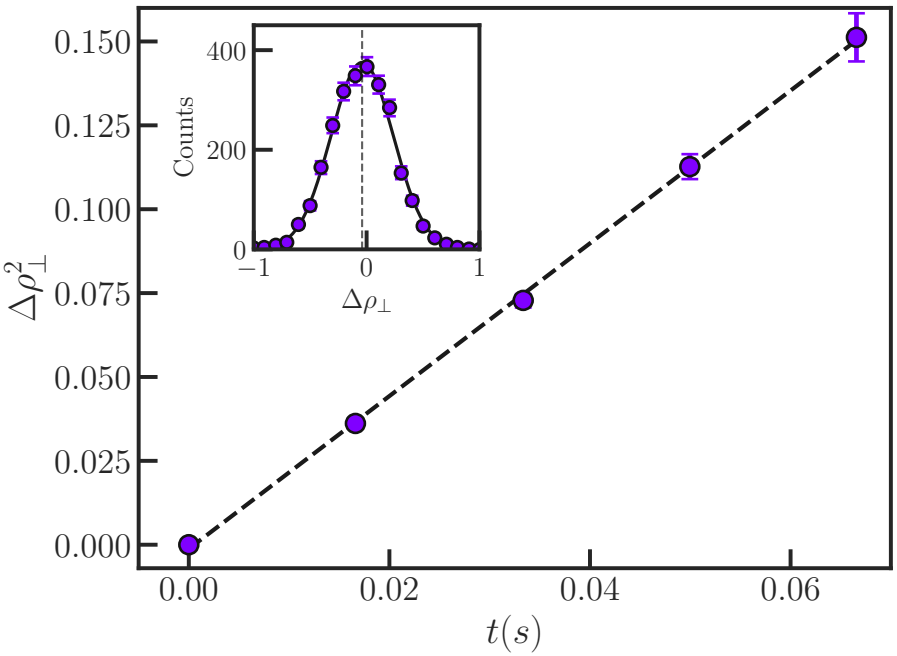

Figure 2.8: Adapted from [30]. Measured pair diffusion for the relative perpendicular mode of motion of two colloids near a planar boundary. The inset shows the histogram of binned displacements for the second time step ($t = 0.018\,\mathrm{s}$), for $z = 25.5 \pm 0.7\,\mu\mathrm{m}$ and $r = 7.00 \pm 0.25\,\mu\mathrm{m}$.

positions (modulable both in height and in pair separation) at a frequency of 200 Hz. The trapping beam is then blocked, the particle diffuse freely and their positions are recorded over five video frames for a total duration of $t = 63$ ms. The short acquisition time limits out-of-plane motion, and we thus assume the line of centres to remain parallel to the plane. The procedure is repeated over a grid of initial configurations (r, z), where r is the centre-to-centre distance between the two colloids, as $2\,\mu\mathrm{m} < r < 10\,\mu\mathrm{m}$, and z is the centre-to-boundary distance, as $1\,\mu\mathrm{m} < z < 30\,\mu\mathrm{m}$. The authors then decompose the pair motion into collective and relative modes (see subsection 2.3.2). For each initial configuration and each lag time, they fill displacement histograms for the collective and relative modes. The MSD corresponding to each mode of motion is computed from the histograms, and the diffusion coefficient is extracted from least square fits to Eq. 1.21. A typical result is shown in Fig. 2.8.

Blake image system

In the presence of a boundary, the Green's function of the Stokes equations must be modified to satisfy the no-slip and no-penetration boundary conditions at the solid boundary. This can be achieved using Blake's image construction [42]. A Stokeslet located at $\boldsymbol{r}_j$

above the boundary is supplemented by an image system located at $\boldsymbol{R}_j = \boldsymbol{r}_j - 2z\boldsymbol{e}_z$, where z is the height of the colloid centre above the boundary. This image system contains an image Stokeslet, a source doublet and a Stokeslet doublet. Taken together, these singularities cancel the fluid velocity at the boundary and yield the boundary-corrected Green's function. For a single colloid diffusing above a boundary, the leading-order corrections to the diffusion coefficients are:

$$\frac{D_z(z)}{D_0} = 1 - \frac{9}{8}\frac{a}{z} + \mathcal{O}\left(\frac{a^3}{z^3}\right) \ , \tag{2.61}$$

$$\frac{D_{xy}(z)}{D_0} = 1 - \frac{9}{16}\frac{a}{z} + \mathcal{O}\left(\frac{a^3}{z^3}\right) \ . \tag{2.62}$$

Here, D_z is the diffusion coefficient normal to the boundary, while D_{xy} is the diffusion coefficient parallel to the boundary. A first approximation consists in superimposing the Oseen pair-diffusion correction with the Blake's single colloid-boundary correction. Dufresne *et al.* write this approximation as:

$$D_\psi^{-1}(r,z) = D_\psi^{-1}(r) + \frac{1}{2}\left[D_{xy}^{-1}(z) - D_0^{-1}\right] \ , \tag{2.63}$$

where ψ denotes one of the collective or relative pair modes. This linear superposition is shown as dashed lines in Fig. 2.9. It captures the qualitative reduction of diffusion near the boundary but fails quantitatively when the colloids are less than approximately 50 radii away from the plane boundary. Each colloid interacts hydrodynamically with its neighbour, with its own image system, and with the image system of its neighbour. Including these contributions gives the pair diffusion coefficients:

$$\frac{D_\perp^{C,R}(r,z)}{D_0} = 1 - \frac{9}{16}\frac{a}{z} \pm \frac{3}{4}\frac{a}{r}\left[1 - \frac{1+\frac{3}{2}\xi}{(1+\xi)^{3/2}}\right] \ , \tag{2.64}$$

$$\frac{D_\parallel^{C,R}(r,z)}{D_0} = 1 - \frac{9}{16}\frac{a}{z} \pm \frac{3}{2}\frac{a}{r}\left[1 - \frac{1+\xi+\frac{3}{2}\xi^2}{(1+\xi)^{5/2}}\right] \ , \tag{2.65}$$

where:

$$\xi = \frac{4z^2}{r^2} \ . \tag{2.66}$$

Again, the positive sign corresponds to collective modes, while the negative sign corresponds to relative modes. These expressions are valid up to corrections of orders $\mathcal{O}\left(\frac{a^3}{r^3}\right)$ and $\mathcal{O}\left(\frac{a^3}{z^3}\right)$. In the far-field limit, where ($r \gg a$) and ($z \gg a$), the authors recover the Oseen predictions for two colloids in an unbounded fluid. At smaller elevations, where ($z \sim a$), they capture the leading-order modification of the pair diffusivities by the

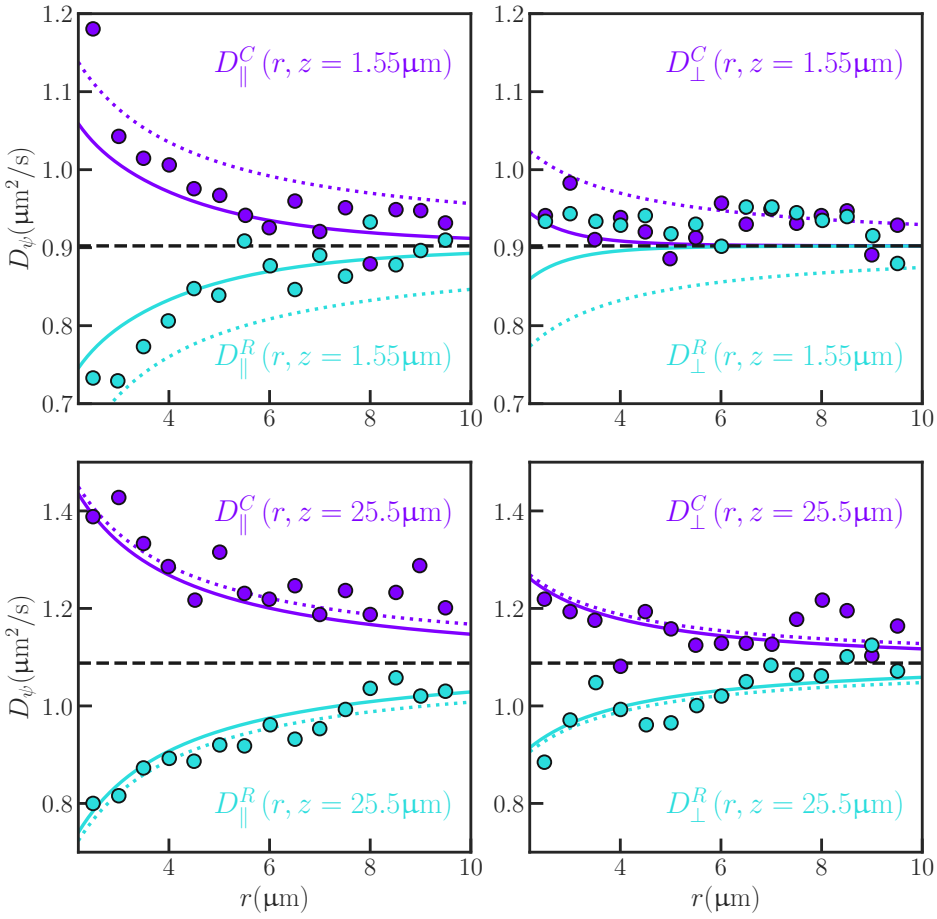

Figure 2.9: Adapted from [30]. Measured collective and relative diffusion coefficients for two colloids near a planar boundary, as functions of the centre-to-centre distance r, for different values of the centre-to-boundary distance z. The dashed lines correspond to the linear superposition approximation (Eqs. 2.63), while the solid lines correspond to the leading-order Stokeslet image-system theory (Eqs. 2.65).

boundary, and show that confinement weakens the hydrodynamic coupling between the particles. This approximation, however, does not resolve the finite-size lubrication effects that appear in the thin gap limit ($r \sim a$ and $z \sim a$). The hydrodynamic description of this regime is particularly delicate. Directly resolving these thin fluid layers using discrete fluid solvers requires a fine resolution and is computationally demanding either in resources or time. We therefore must find a technique which combines both lubrication corrections to the mobility matrix in the thin-gap regime, and high computational efficiency, in order to reproduce the above results and probe at thinner gaps.

The RigidMultiblobsWall library and lubrication corrections

Several numerical approaches have already been developed to account for near-field hydrodynamic interactions. Stokesian dynamics methods combine a multipolar description of the long-range many-body hydrodynamic interactions with explicit pairwise lubrication corrections at short distances [43, 44]. The fluid is treated as a continuum through Stokes equations (Eq. 2.43), and the particle velocities are computed from a mobility problem. Although such methods provide a detailed description of the hydrodynamic interactions, the inclusion of higher moments greatly increases their cost and complexity. Herein, we instead use the minimally resolved, lubrication-corrected Brownian-dynamics method introduced by Delong, Balboa Usabiaga, Donev *et al.* [45] and implemented in the `RigidMultiblobsWall` (RBW) library.[1] This method generates Brownian trajectories of dense colloidal suspensions above a no-slip boundary, but does not resolve the surrounding fluid. Long-ranged particle-particle and particle-boundary hydrodynamic interactions are described using the RPY-Blake mobility. At short separations, the RPY-Blake contributions are subtracted and replaced by more accurate sphere-sphere and sphere-boundary near-field asymptotes. The resulting mobility is written schematically as [46]:

$$\boldsymbol{\mu} = \left[\boldsymbol{\mu}_{\mathrm{RPB}}^{-1} + \boldsymbol{R}_{\mathrm{lub}}^{\mathrm{sup}} - \boldsymbol{R}_{\mathrm{RPB}}^{\mathrm{sup}}\right]^{-1}, \tag{2.67}$$

where $\boldsymbol{\mu}_{\mathrm{RPB}}$ is the RPY-Blake mobility matrix, $\boldsymbol{R}_{\mathrm{lub}}^{\mathrm{sup}}$ is the thin-gap resistance matrix, and $\boldsymbol{R}_{\mathrm{RPB}}^{\mathrm{sup}}$ is the inaccurate RPY-Blake approximation result in the thin-gap, which must be subtracted ("sup" indicates the pairwise-superposed near-field resistance matrices). This construction keeps the computational efficiency of a minimally resolved far-field mobility and preserves the lubrication resistance in the thin-gap regime. The lubrication corrections themselves are divided into regimes depending on the width of the gap between the pair and between the pair and the boundary. The library then handles the whole simulation and returns raw positions at each time step.

[1] `https://github.com/stochastichydrotools/rigidmultiblobsboundary`

2.4.2 Results

It is up to the user to initialise the simulation, and to input all parameters (Table 2.1). The time step selection also requires care (as discussed in chapter 1, subsection 1.3.1). We begin by defining couples of altitude z and pair separation r. Each of these couples sets the initial position of the pair of colloids relative to one another and to the boundary. Then, we let the simulation run for a total duration of $N \times t_{\text{free}} = N \times 83$ ms, where N corresponds to the number of times we reset the pair positions. We use this trick to replicate Dufresne *et al.*'s trapping protocol, while reducing overheads in the simulation by launching only one very long run with resets instead of N short ones. We then segment the trajectories by reset, and compute collective and relative displacement histograms for each simulation time step. We extract from these histograms the collective and relative MSDs, and verify their exponent: the MSD must grow linearly with time so that we can extract the collective and relative diffusion coefficients (Eq. 1.21). As this study was

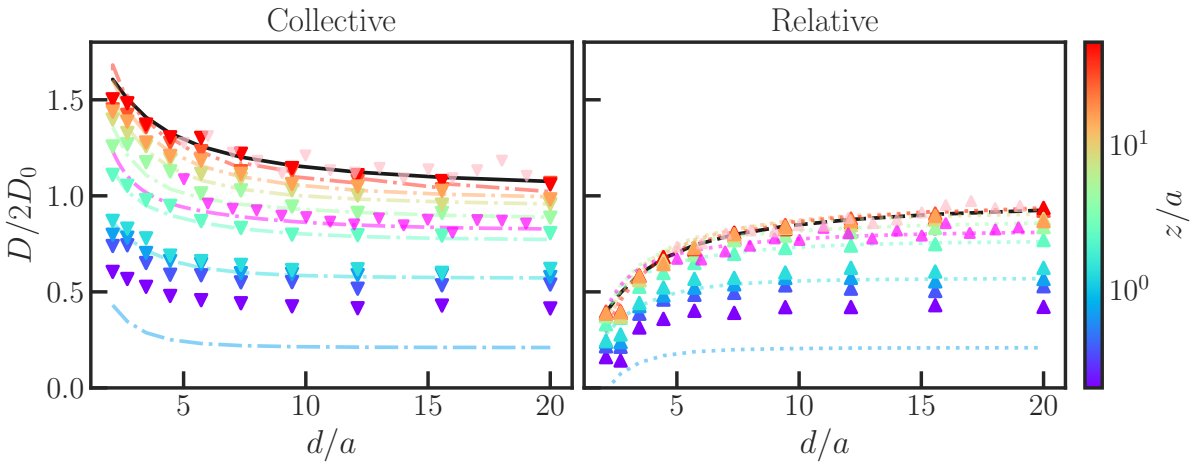

Figure 2.10: Comparison of cross-diffusion coefficients from numerical simulations against experimental results from Dufresne *et al.* [30] along the pair line of centres. Left and right panels correspond to the collective and relative diffusion coefficient, respectively. Pink and magenta markers correspond to Dufresne *et al.* experiments, with $z = 25.5 \pm 0.7$ µm and $z = 1.55 \pm 0.66$ µm, respectively. Rainbow markers correspond to numerical data, and the rainbow colormap indicates the value of the pair-boundary separation distance z. Dash dotted lines and dotted lines correspond to the collective and relative modes of Eqs. 2.65, respectively. The black lines are Oseen's approximations from Eqs. 2.56.

started near the very end of my thesis, the results shown in Fig. 2.10 are preliminary. We will in the near future generate trajectories for many more (r, z) pairs, and explore smaller values of the boundary-pair thin gap. However, we already successfully retrieve the experimental results presented in Dufresne *et al.* with a slightly better agreement with the theoretical values from Eqs. 2.65. We also explore a smaller pair-boundary gap and smaller pair gaps, and show that the theoretical predictions of Eqs. 2.65 hold up to

a ratio d/a of about 2 and a ratio z/a of about 1. Furthermore, we notice a quantitative disagreement for pair separation ratios of about $d/a = 2$ independent of pair-boundary separation, hinting at either a need for a more robust theory on the pair-diffusion at small pair gaps.

2.4.3 Perspectives

Using the RBW library, it seems possible to accurately recover the pair-diffusion of colloids with separations well-below the radius range in altitude and pairwise. This could actually shed light on the behaviour of very dense, confined Brownian solutes. To validate this, experiments are currently being carried our in our group. The experimental protocol was shared to me by Guirec de Tournemire, which I include as follows: we monitor the interaction between of two Brownian particles using *Mie Holography* [47, 48]. This method relies on the holograms produced by interferences between the incident light illuminating a particle, and the light scattered by the latter (see Fig. 2.11*a*)). The interference patterns are extracted as intensity profiles (see Fig. 2.11*b*)) and fitted based on the Lorenz-Mie theory to recover the tri-dimensional position of the particle, for every frame composing the video. The sample consists of two glass slides (thickness $e = 0.13 - 0.17\,\mathrm{mm}$ and refractive index $n_\mathrm{g} = 1.518$) separated by paraffin bands. The space H between the two glass slides, set by the thickness of the paraffin bands, is about $H = 250\,\mu\mathrm{m}$. The Brownian microparticles are made of polystyrene, manufactured by Polyscience. They are negatively surface-charged and have a radius $a_\mathrm{p} = 1.5\,\mu\mathrm{m}$, for a density of $\rho_\mathrm{p} = 1050\,\mathrm{g/cm^3}$. Their motion is recorded through a $\times 100$ oil-immersion Olympus objective (NA = 1.45), configured for total internal reflection microscopy (see chapter 3), using a Basler Camera, with a field of view of 70 µm × 70 µm in the (x, y) plane. Along z, the depth of view is of about 100 µm. The motion of the particles is recorded at a frequency $f = 1/\Delta t = 200\,\mathrm{Hz}$, leaving a lag time of $\Delta t = 5\,\mathrm{ms}$ between two frames.

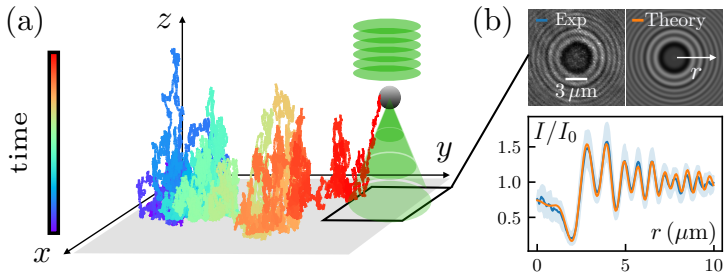

Figure 2.11: Mie Holography method used to probe the motion of a colloid 1.5 μm-radius particles in water, illuminated by a laser at 532 nm at a laser power of 4 mW. *a*) Schematic of a colloid diffusing in three dimensions, illuminated by an incident beam and scattering light. *b*) Resulting intensity profiles from which the three-dimensional tracking is performed. Credits: Guirec de Tournemire.

In order to enforce the hydrodynamic interaction between two particles, we use an optical

trap [49], enabling the translation of a particle from one point to another (see Fig. 2.12). The method consist in finding a diffusing particle, moving it as close as possible to another one, and taking a video with a high frame rate to have as much data as possible, in the region where the particle-particle hydrodynamic interaction is strong. We have already

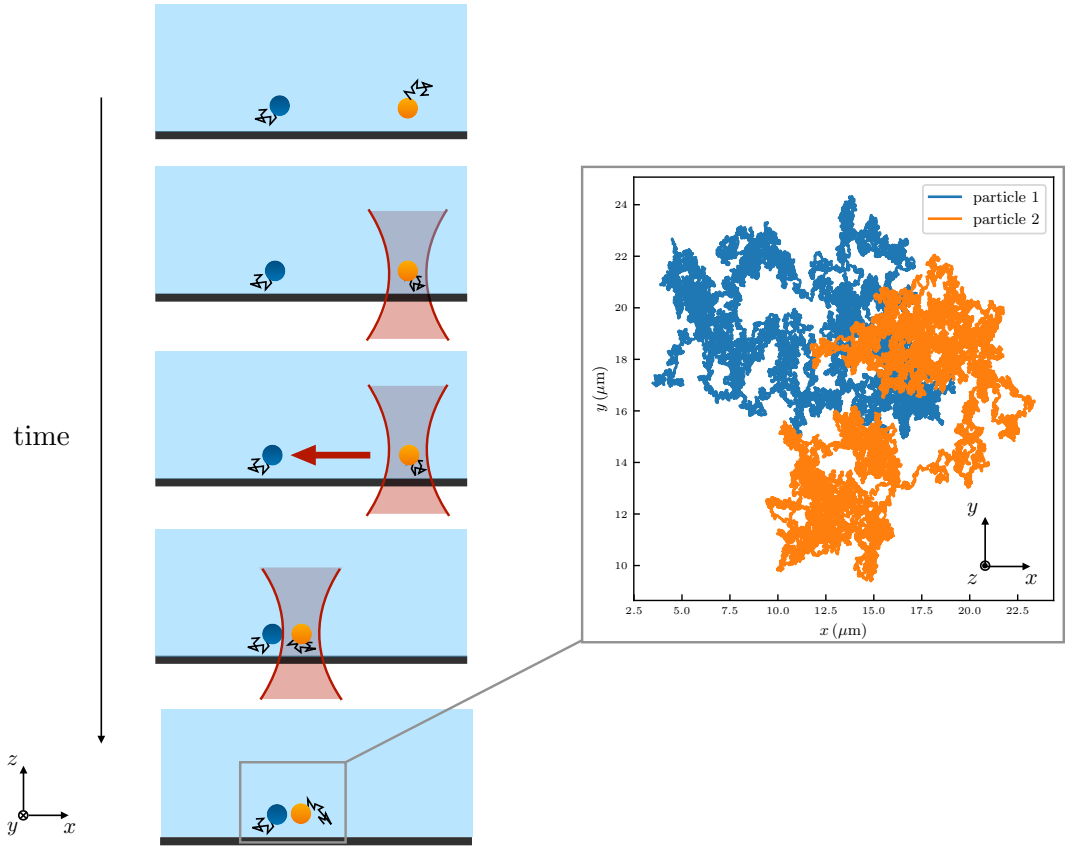

Figure 2.12: Optical trapping used to enforce the colloid-colloid interaction. The trap is set by a laser at 660 nm at a laser power of 50 mW. Credits: Guirec de Tournemire.

gathered sufficient numerical and experimental data to obtain preliminary comparative results, to which I will dedicate the remainder of my thesis.

As a last note, we recall that the RBW library restricts possible studies in the thin-gap limit to spherical colloids. The lubrication corrections in RBW are not implemented for complex shapes. This limitation means that we must use other methods for solving for colloids of different sizes or bodies of different geometries. Activity, however, is permitted within the spherical colloid restriction, and we can thus explore the behaviour of active swimmers (see chapter 5) hydrodynamically interacting in very thin films.

2.5 Take-home message

This chapter was separated in three main sections. First, we described the motion of a colloid in hydrodynamic interaction with a nearby boundary. We noted the emergence of two modes of motion, mediated by the anisotropy of the effective viscosity felt by the colloid. One mode corresponds to the motion perpendicular to the boundary, whereas the second corresponds to the motion in the plane parallel to the boundary. We derived exact and approximate solutions for the expected effective diffusion coefficient and expressed the Langevin equations pertaining to these two modes. We then changed the object of our study to two colloids, diffusing in a bulk fluid. We approximated them as point forces, interacting through a viscous fluid. We derived expressions for their diffusion tensor, as well as expressions for their collective and relative motions along and perpendicular to their line of centres. To account for finite-size effects arising when the colloids are under ten radii of separation distance, we mentioned the Rotne-Prager-Yamakawa tensor, and its limitations: it does not account for lubrication divergence at extremely small separations. We coupled pairwise and boundary hydrodynamic contributions through different means. We followed the first hypothesis of Dufresne *et al.* [30], adding pairwise and boundary contributions, which showed to be satisfactory only for very large separations. We then used the Blake image system to derive more accurate expressions for the diffusion tensor. These expressions showed good agreement with the experimental data from Dufresne *et al.*, but poor agreement with numerical simulations using the RigidMultiblobWalls algorithm. We explained this discrepancy as this algorithm uses asymptotical corrections in the pairwise and boundary sub-radius separation regimes. We finally proposed a numerical and experimental co-study to verify these results, which occupy the end of my doctoral research.

Chapter 3

Brownian Motion in Soft Confinement

3.1 Introduction

In the last chapter, we mentioned the strong influence of confinement on Brownian dynamics. It is common, in microbiological settings for instance, to work with such confined systems. However, strictly rigid confinements are scarce. More often, they can deform under external or active forces, stretch, and fluctuate. Indeed, if we take the example of a cell membrane, which we can describe with a lipid bilayer, the thermal energy $k_{\mathrm{B}}\Theta$ of the system is sufficient to cause long-wavelength fluctuations of the cell membrane, thickness fluctuations, lipid movement within the plane itself, and so forth. A neighbouring fluid's motion is then affected by these shape modifications, which can in turn affect the dynamics of a body evolving in that same bath. The coupling between softness and viscous flows, which we will refer to as the elastohydrodynamic coupling, has been extensively studied in the deterministic framework. We know lift forces active upon a translating body emerge near a deformable boundary at low Reynolds numbers [50]. This has been proposed first theoretically [51, 52, 53, 54, 55] and then experimentally [56, 57, 58, 59, 60]. Studies involving fluid interfaces [61, 62], fluctuating surfaces [63, 64], and biological-inspired membranes [65, 66, 67] have been carried out in that regard, though the field of stochastic elastohydrodynamics is still fairly new. Molecular dynamics simulations have been conducted to investigate the anomalous diffusion of a micelle near a lipid bilayer [68]. Another recent study [69] demonstrated experimentally the emergence of a new, non-conservative force acting upon a Brownian soft colloid near a confining wall. However, we still to this day lack a precise theory, real experiments and numerical simulations that quantitatively encompass the long-time signatures of the coupling between softness, confined flows and fluctuations. I present in this chapter a most simple model, with all the elementary ingredients, which can still qualitatively capture all the effects we are interested in. This model was published in *Communications Physics* in 2025 [70] and the full print and Supplementary Information can be found in appendix of this thesis. In the following sections, I present some theoretical aspects of the model, developed by David S. Dean. I develop the numerical method used to simulate this toy model. Finally, I show how numerics and theory compare.

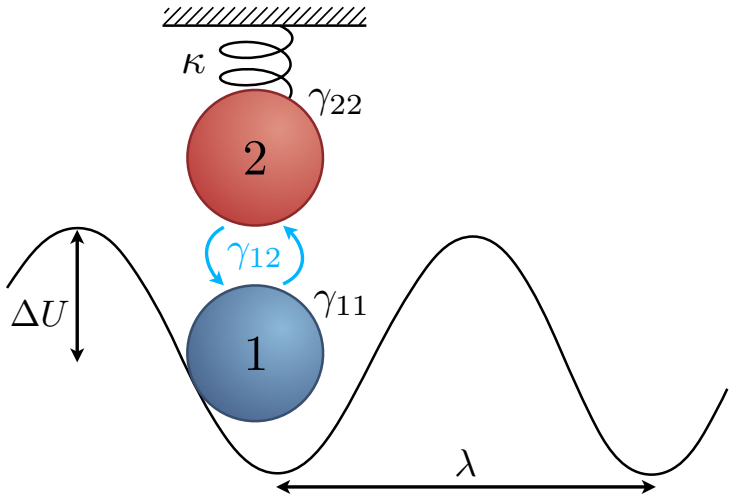

Figure 3.1: Representation of the system. A colloid of interest (blue) is trapped in a periodic potential of amplitude ΔU and spatial period λ. An elastic membrane is modelled locally as a second colloid (red) subject to a harmonic potential of inverse stiffness (*compliance*) κ. This membrane is coupled to the colloid of interest through an indirect, hydrodynamic-like coupling represented by the coefficient γ_{12}, and both the colloid and the membrane feel their own self-frictions, described by the coefficients γ_{11} and γ_{22}, respectively.

3.2 Toy Model for soft Brownian Motion

From here on, we study a purely one-dimensional case of a colloid coupled to an elastic and fluctuating membrane (Fig. 3.1). The colloid is represented by a first degree of freedom, q_1, and is submitted to a periodic trapping potential $\phi(q_1)$ of spatial period λ and amplitude ΔU. We couple to this colloid a minimal version of the membrane. We model the membrane with a colloid that is submitted to a harmonic potential of energy $q_2^2/(2\kappa)$, where κ is the compliance (inverse stiffness), representing the elasticity of the membrane. This second colloid is tracked with the degree of freedom q_2.

Both the colloid and the membrane feel their own friction with the surrounding fluid medium. We refer to this as the self component of the friction matrix and note it as γ_{11} and γ_{22}, respectively. Furthermore, they interact through a hydrodynamic-like interaction γ_{12}. This interaction represents the effect of the fluid that is displaced by the motion of the colloid on the membrane, and *vice-versa*. All together, these terms form the friction matrix γ:

$$\gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix} , \tag{3.1}$$

which can be inverted to give the mobility matrix of the system: $\mu = \gamma^{-1}$. Both matrices must be positive definite, their diagonal terms account for the self-friction and self-mobility respectively. As a consequence, one has $\mu_{11} > 0$ and $\mu_{22} > 0$. We can write the total

energy of the system as:

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa} \ , \tag{3.2}$$

and we notice the absence of Hamiltonian interactions in the system. We work in the overdamped limit and neglect inertial terms. The coupled Langevin equations describing the evolutions of q_1 and q_2 read:

$$\gamma_{11}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{12}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\phi'(q_1) + \eta_1(t) \ , \tag{3.3}$$

$$\gamma_{12}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{22}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\frac{q_2}{\kappa} + \eta_2(t) \ , \tag{3.4}$$

where η_i and η_j (with $j \in \{1, 2\}$) are Gaussian white noises of zero mean and correlation functions $\langle \eta_i(t)\eta_j(t')\rangle = 2k_\mathrm{B}\Theta\delta(t-t')\gamma_{ij}$. The equilibrium probability density function of the system is then given by the Gibbs-Boltzmann distribution, as:

$$P_\mathrm{eq} = \frac{\exp[-\beta H(q_1, q_2)]}{Z} \ , \tag{3.5}$$

where β is the inverse thermal energy as $\beta = (k_\mathrm{B}\Theta)^{-1}$ and $1/Z$ is a normalising factor.

3.3 Theoretical developments

The objective of the following developments is to compute the modification to the long-time dynamics of the colloid diffusing within the periodic trap. The development I choose to present here is not the most rigorous, and corresponds to the stiff ($\kappa \to 0$) limit of the elastic mode q_2. The interested reader can find a more rigorous argument based on multiscale analysis [71, 72] in the appendices of this thesis, alongside a second development for the compliant limit ($\kappa \to \infty$). In the deterministic case (*i.e.* $\Theta = 0$), inverting Eqs. 3.3 and 3.4 yields:

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa}q_2 \ , \tag{3.6}$$

$$\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\mu_{12}\phi'(q_1) - \frac{\mu_{22}}{\kappa}q_2 \ . \tag{3.7}$$

We then integrate to express q_2:

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}}\left(1 + \frac{\kappa}{\mu_{22}}\frac{\mathrm{d}}{\mathrm{d}t}\right)^{-1}\phi'(q_1) \ . \tag{3.8}$$

We define the characteristic relaxation time τ_2 of the harmonic trapping corresponding to the membrane elasticity as $\tau_2 = \kappa/\mu_{22}$. In the small κ limit, we then expand the

right-hand bracket term:

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}}\left[\phi'(q_1) - \frac{\kappa}{\mu_{22}}\phi''(q_1)\frac{\mathrm{d}q_1}{\mathrm{d}t} + O(\tau_2^2)\right] \ , \tag{3.9}$$

Now, substituting this new expression for q_2 in Eq. 3.6, we obtain:

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}}\right)\phi'(q_1) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2}\phi''(q_1(t))\frac{\mathrm{d}q_1}{\mathrm{d}t} \ , \tag{3.10}$$

which we can reexpress as:

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{\mathrm{e}}(q_1)\phi'(q_1) \ , \tag{3.11}$$

where μ_{e} is a dressed mobility with expression:

$$\mu_{\mathrm{e}} = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}\left(1 + \dfrac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2}\right)} \ . \tag{3.12}$$

We finally obtain the dressed mobility:

$$\mu_{\mathrm{e}} = \frac{1}{\gamma_{11}\left(1 + \dfrac{\kappa\phi''(q_1)\gamma_{12}^2}{\gamma_{11}^2}\right)} \ . \tag{3.13}$$

Now, we can avoid reexpressing the noise which should be applied to Eq. 3.11 for the thermal case by working directly with Fokker-Planck formalism. This formalism is introduced in its elementary form at the beginning of my thesis, and developed for more complex cases in chapter 2. Doing such a thing is not rigorous, as I previously mentioned, because the effective stochastic process is not derived. Eliminating fast variables in Eq. 3.11 typically introduces non-trivial corrections. However, since we assume separation of time scales, the effective slow process (q_1 in this case) still permits the writing of a uniquely-defined Fokker-Planck equation. This equation reads:

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\mu_{\mathrm{e}}(q_1)\left[k_{\mathrm{B}}\Theta\frac{\partial P_1(q_1,t)}{\partial q_1} + \phi'(q_1)P_1(q_1,t)\right] \ , \tag{3.14}$$

and $P_1(q_1,t)$ can be expressed in terms of probability flux J as:

$$\frac{\partial P_1(q_1,t)}{\partial t} = -\frac{\partial J}{\partial q_1} \ , \tag{3.15}$$

where:

$$J = -D_{\mathrm{e}}(q_1)\frac{\partial}{\partial q_1}P_1(q_1,t) - \mu_{\mathrm{e}}\phi'(q_1)P_1(q_1,t) \ , \tag{3.16}$$

where $D_{\rm e}(q_1) = \mu_{\rm e}(q_1) k_{\rm B}\Theta$. The above equation can be rewritten as:

$$J = -D_{\rm e}(q_1) \exp\left[-\beta\phi(q_1)\right] \frac{\mathrm{d}}{\mathrm{d}q_1} \left\{\exp\left[\beta\phi(q_1)\right] P_1(q_1,t)\right\} \ . \tag{3.17}$$

Now, we tilt the potential by applying a small force F, which gives a total potential $U(q_1)$, as:

$$U(q_1) = \phi(q_1) - F q_1 \ . \tag{3.18}$$

Then, Eq. 3.17 is rewritten, taking into account this new force, as:

$$\frac{\mathrm{d}}{\mathrm{d}q_1} \left\{\exp[\beta U(q_1)] P(q_1,t)\right\} = -\frac{J}{D_{\rm e}(q_1)} \exp[\beta U(q_1)] \ . \tag{3.19}$$

We can then integrate over a period λ:

$$P_1(q_1) = \frac{J \exp[-\beta U(q_1)]}{1-\exp(-\beta F\lambda)} \int_{q_1}^{q_1+\lambda} \frac{\exp[\beta U(q)]}{D_{\rm e}(q)} \mathrm{d}q \ . \tag{3.20}$$

The reduced density is normalised over one period:

$$\int_0^\lambda P_1(q_1)\mathrm{d}q_1 = 1 \ , \tag{3.21}$$

and thus:

$$J = \frac{1-\exp[-\beta F\lambda]}{\int_0^\lambda \exp[-\beta U(q)]\mathrm{d}q \int_{q_1}^{q_1+\lambda} \frac{\exp[\beta U(q)]}{D_{\rm e}(q)} \mathrm{d}q} \ . \tag{3.22}$$

In the small force limit ($F \to 0$), one has:

$$1-\exp(-\beta F\lambda) \simeq \beta F\lambda \ , \tag{3.23}$$

which, to leading order in F, gives:

$$J = \frac{\beta F\lambda}{\int_0^\lambda \exp[-\beta\phi(q)]\mathrm{d}q \int_0^\lambda \frac{\exp[\beta\phi(q)]}{D_{\rm e}(q)} \mathrm{d}q} \ . \tag{3.24}$$

We note that the drift velocity should read $v = J\lambda$, and that the effective long-time diffusion coefficient is:

$$D^*(\kappa) = \frac{k_{\rm B}\Theta v}{F} = \frac{\lambda^2}{\int_0^\lambda \exp[-\beta\phi(q)]\mathrm{d}q \int_0^\lambda \frac{\exp(\beta\phi(q))}{D_{\rm e}(q)} \mathrm{d}q} \ , \tag{3.25}$$

which is the generalised Lifson-Jackson formula [73]. Finally, we replace $D_{\mathrm{e}}(q_1)$ in the expression, which leads to:

$$D^*(\kappa) = \frac{k_{\mathrm{B}}\Theta\lambda^2}{\gamma_{11}\int_0^\lambda \mathrm{d}q \exp[\beta\phi(q)]\left(1 - \dfrac{\kappa\gamma_{12}^2\phi'^2(q)}{k_{\mathrm{B}}\Theta\gamma_{11}^2}\right)\int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)]} \ . \tag{3.26}$$

And, choosing $\phi(q_1) = \Delta U\psi\left(\dfrac{q_1}{\lambda}\right)$, with $\psi\left(q_1/\lambda\right) = -\cos\left(2\pi q_1/\lambda\right)$ leads to:

$$D^*(\kappa) = \frac{k_{\mathrm{B}}\Theta}{\gamma_{11}\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]\left(1 - \dfrac{\kappa\gamma_{12}^2\Delta U^2\psi'^2(p)}{k_{\mathrm{B}}\Theta\gamma_{11}^2}\right)\int_0^1 \mathrm{d}p \exp[-\beta\Delta U\psi(p)]} \ , \tag{3.27}$$

which is the equation we use to compare with numerical results. In the $\kappa \to 0$ limit, Eq. 3.27 can be approximated as:

$$\boxed{D^*(\kappa \to 0) \simeq D^*(0)(1+\alpha)} \ , \tag{3.28}$$

with:

$$\alpha = \beta\epsilon\frac{\lambda^2\gamma_{12}^2}{\gamma_{11}^2}\frac{\int_0^\lambda \mathrm{d}q_1\phi'(q_1)^2\exp[\beta\phi(q_1)]}{\int_0^\lambda \mathrm{d}q_1\exp[-\beta\phi(q_1)]} \ , \tag{3.29}$$

where we define $\epsilon = \kappa k_{\mathrm{B}}\Theta/\lambda^2$, a dimensionless compliance. As $\alpha > 0$, the coupling of the membrane to the colloid always increases its long-time diffusion constant D^*. Furthermore, at the compliant limit ($\kappa \to \infty$), D^* becomes:

$$\boxed{D^*(\kappa \to \infty) = k_{\mathrm{B}}\Theta\frac{\mu_{11}(\mu_{11}\mu_{22} - \mu_{12}^2)}{\lambda^{-2}Z_+Z_-(\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right)} \ , \tag{3.30}$$

where we define $Z_\pm$ as:

$$Z_\pm = \int_0^\lambda \exp[\pm\beta\phi(q)]\mathrm{d}q \ . \tag{3.31}$$

We notice, in that regime, the absence of a dependence on the compliance of the elastic mode.

3.4 Numerical modeling

This section presents the development of a numerical model to generate coupled Brownian trajectories according to Eqs. 3.3 and 3.4. We keep in mind that, owing to the stochasticity of the system, and the strong coupling between the elastic mode and the colloid, we need to generate a considerable amount of data to converge the statistics. Thus, we balance two conditions on our numerical simulations: we require a cheap integrator, which avoids

oversubscribing memory; and we must find a way to save data from hundreds of thousands of trajectories without cluttering memory. We begin by discretising Eqs. 3.3 and 3.4 using an Euler-Maruyama scheme [21] (See chapter 1, subsection 1.3.1).

3.4.1 Discretisation

We discussed, in chapter 1, the most simple way to integrate a stochastic process over time. This discretisation method is the Euler-Maruyama scheme. I generally prefer to attempt this method before trying more complex integrators, as it is the most convenient to implement, and is often sufficient for the processes I simulate. Naturally, other methods could work as well and be less time step-sensitive, but with these more complex methods comes an added cost on the calculations (namely, we debated using higher-order Runge-Kutta methods in this problem, and ultimately concluded it was not necessary upon analysing the first Euler-discretised trajectories we generated). Finally, I find a high pedagogical aspect to the Euler method. It can be intuited by anyone, with simple hand-waving arguments and a basic knowledge of numerics. The Euler-discretised versions of Eqs. 3.3 and 3.4 read:

$$q_1(t+\Delta t) - q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t - \frac{\mu_{12}}{\kappa} q_2(t)\Delta t + \xi_1\sqrt{2k_{\mathrm{B}}\Theta\Delta t}\ , \tag{3.32}$$

$$q_2(t+\Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa} q_2(t)\Delta t - \mu_{12}\phi'(q_1(t))\Delta t + \xi_2\sqrt{2k_{\mathrm{B}}\Theta\Delta t}\ , \tag{3.33}$$

where ξ_i are zero-mean Gaussian random variables, with the two point correlators given by:

$$\langle \xi_i \xi_j \rangle = \mu_{ij}\ . \tag{3.34}$$

Furthermore, in the rest of this numerical study, we choose $\phi(q_1)$ to be sinusoidal:

$$\phi(q_1) = \Delta U \sin\left(2\pi q_1/\lambda\right)\ . \tag{3.35}$$

3.4.2 Time step selection

In the present study, I identify two short time scales: the local relaxation time within a local minimum of the sinusoidal trap, and the relaxation time of the harmonic trap. Longer time scales associated with the diffusion of the colloid above the periodic trap exist, but we aim here at identifying the smallest time scale in the system.

Periodic trap time scale

Near a local minimum, we approximate the sinusoidal trap as harmonic. In that case:

$$\phi(q_1) \simeq \frac{1}{2} k_{\mathrm{eff}} q_1^2\ , \tag{3.36}$$

where k_{eff} is an effective stiffness, which is the local curvature of the potential:

$$k_{\text{eff}} = |\phi''(0)| \tag{3.37}$$

$$= \Delta U \left(\frac{2\pi}{\lambda}\right)^2 , \tag{3.38}$$

and we can finally identify the first time scale as:

$$\boxed{\tau_1 = (\mu_{11} k_{\text{eff}})^{-1} = \frac{\lambda^2}{4\pi^2 \Delta U \mu_{11}}} . \tag{3.39}$$

Elastic mode time scale

Similarly, we find the characteristic time scale of the harmonic trap associated to the elastic mode:

$$\boxed{\tau_2 = \frac{\kappa}{\mu_{22}}} . \tag{3.40}$$

Then, we choose a time step which satisfies:

$$\boxed{\Delta t = \frac{1}{100} \min \tau_1, \tau_2} . \tag{3.41}$$

Maximum simulation time

We also need to define an upper bound to the simulation time. This time must be sufficient to probe at long-time effects such as the colloid diffusing over the periodic trap it is subjected to. We define a time scale τ^* which depends on the long-time effective diffusion coefficient $D^*(\kappa)$ defined in Eq. 3.27 through:

$$\tau^*(\kappa) = \frac{\lambda^2}{D^*(\kappa)} . \tag{3.42}$$

I choose to run my simulations for a time T_{sim} which is much larger than τ^*: $T_{\text{sim}} = 100\,\tau^*(\kappa)$. This allows the probing of the long-time dynamics of the system, and the extraction of the effective diffusion coefficient from the mean-squared displacement of the colloid. This, in turn, sets the computational cost of a simulation, which I quantify using the total number of iterations in the time loop n_{steps}:

$$n_{\text{steps}} = \left\lfloor \frac{T_{\text{sim}}}{\Delta t} \right\rfloor = \left\lfloor \frac{100\tau^*}{\Delta t} \right\rfloor . \tag{3.43}$$

Note that τ^*, T_{sim} and τ_2 are $\kappa-$dependent. This means that simulations have different total durations and time step values. This must also be evaluated before launching a

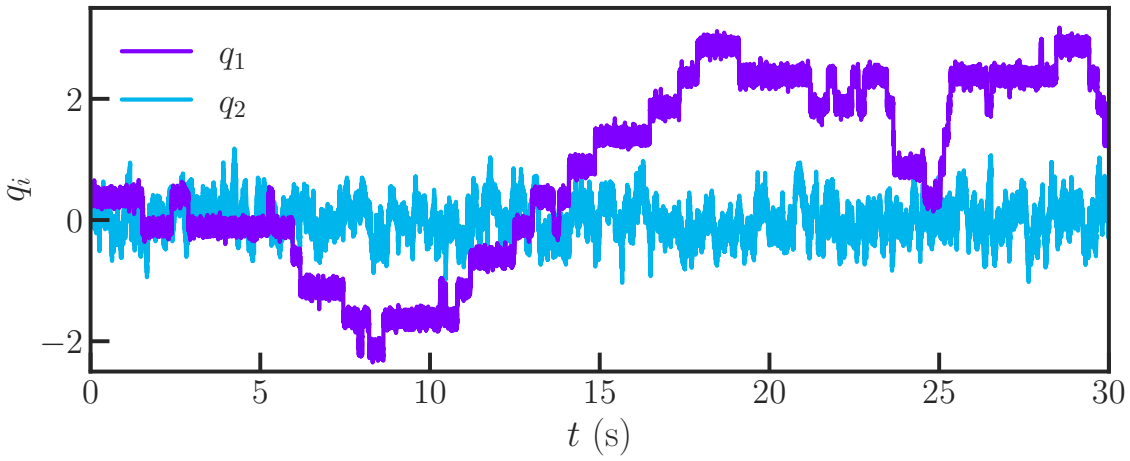

Figure 3.2: Trajectories of the colloid (purple) and the membrane (blue) for $\kappa = 0.01$. The other parameters are fixed to $\Delta U = 2.7 k_\mathrm{B}\Theta$, $k_\mathrm{B}\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1.0$, $\gamma_{22} = 1.0$ and $\gamma_{12} = 0.8$, $\Delta t = 10^{-4}$.

simulation, for example by killing processes of more than what the memory can hold at a time. Until a workaround is found, small values of κ may be impossible to probe due to the maximal simulation time T_sim being large in comparison to the simulation time step Δt.

3.5 Results

The interested reader can find a minimal version of a trajectory generation code in the GitHub associated to this manuscript. This simple code allows the user to tune parameters and test for themself their effect on the dynamics of the system.

3.5.1 Trajectories

I first generate trajectories of the two coupled colloids using Eqs. 3.32 and 3.33, for a given set of parameters. A sample of a single trajectory is shown in Fig. 3.2. The elastic mode fluctuates around its equilibrium position in $q_2 = 0$, whereas the colloid is trapped at short times within a local minimum, and escapes through random thermal fluctuations at longer times to explore space. This is a first verification that the simulations behave as expected.

3.5.2 Marginal equilibrium distributions

I verify the behaviour of the system at equilibrium by sampling the probability density functions of q_1 and q_2 from the generated trajectories, and comparing them to the

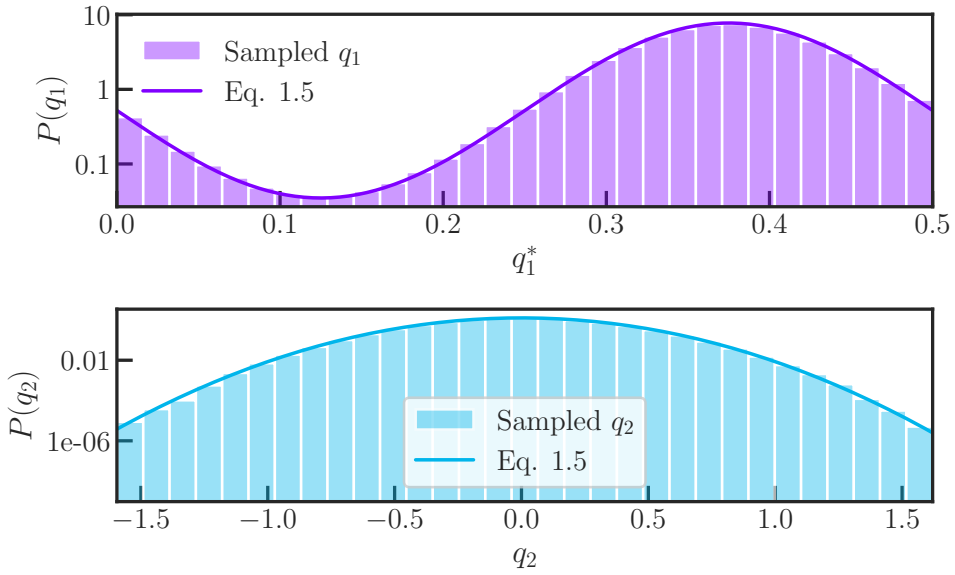

Figure 3.3: Numerical equilibrium probability density function of q_1 (top, for $q_1^* = q_1 \mod \lambda$) and q_2 (bottom) for $\kappa = 0.010$. The other parameters are fixed to $\Delta U = 2.7 k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1.0$, $\gamma_{22} = 1.0$, $\gamma_{12} = 0.8$, $\Delta t = 10^{-4}$s, for a single trajectory of $N = 1.0 \times 10^7$ points. The theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 3.5, solid lines) is also shown.

theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 3.5). This is shown in Fig. 3.3. We can see that the sampled probability density functions (histograms) and marginal theoretical prediction from the Gibbs-Boltzmann distribution (solid lines, Eq. 3.5) are in perfect agreement, even for low statistics (a single binned trajectory of 10^8 points suffices to reach a good match between theory and numerics).

3.5.3 Mean-Squared displacements

At this stage, we do not probe the dynamics of the system. To do so, we compute the MSD of q_1 and q_2 from the generated trajectories, and extract the effective diffusion coefficient of q_1 from the long-time behaviour of its MSD. We compare this long-time effective diffusion coefficient to the theoretical prediction from Eq. 3.27.

MSD from raw trajectories

In the present case below, the MSD is computed directly from the generated trajectories. We calculate the average of the squared displacements over all possible or time lags τ. This means that the memory stores whole trajectories q_1 and q_2. It then loops over all possible time lags τ, and for each of them, computes the average of the squared displacements $[q(t+\tau) - q(t)]^2$ over all possible time origins t. This technique is straightforward and sufficient for demonstration purposes.

MSD from the histogram of displacements

In our published work and in previous studies from our group [74], we showed that it is more robust to first compute the *histograms of displacements* at given time lags. We then extract the MSD from the width of the histogram of displacements as:

$$\mathrm{MSD}(\tau) = \sigma^2(\tau) \ , \tag{3.44}$$

where σ^2 is the variance of the histogram of displacements at time τ. The histogram of displacements has the advantage of being a stackable quantity across iterations of a same simulation. On a rolling basis, as multiple trajectories are generated, we add counts to the histogram of displacements. Furthermore, it is a much more general quantity than the MSD, as it allows for the computation of other moments of the distribution of displacements (skewness, kurtosis, etc.) if the study calls for it. The first step is the pre-definition of bin widths for each chosen time lag. This is automated from the theoretical value of the MSD at τ, according to Eqs. 1.2 and 3.44. We choose an odd number of bins (typically, $n_{\mathrm{bins}} = 31$) so that one bin is always centered on 0, and a bin width is given by:

$$w_{\mathrm{bin}} = \frac{10 \times 2D^*(\kappa)\,\tau}{n_{\mathrm{bins}}} \ , \tag{3.45}$$

with the 10 factor ensuring a large enough histogram width for all displacements to fall within the histogram. We can furthermore define a convergence criterion on the histograms of displacements as the relative distance between a "previous" histogram of cumulative displacements, and an "updated" histogram which contains all previous iterations plus the new additions from the latest simulated trajectories. When this relative distance is smaller than a given threshold for a given number of cumulative iterations, the simulation has converged, and it automatically stops. This hands-off technique permits much less supervision from the user over the simulation. I define the convergence criterion $\mathcal{C}$ as:

$$\mathcal{C} = \frac{\sum_i |P_i^{\mathrm{prev}} - P_i^{\mathrm{updated}}|}{\sum_i P_i^{\mathrm{prev}}} \ , \tag{3.46}$$

where P is the histogram of displacements, and i the index of the bin. I choose a threshold of $\mathcal{C} \leq 5 \times 10^{-3}$ for more than 3 iterations of 10 stacked trajectory statistics. The results of a typical MSD calculation for q_1 and q_2 are shown in Figs. 3.4 and 3.5. We can see that the MSD of q_1 (Fig. 3.4) is at first diffusive with a Δt exponent of 1 and a diffusion coefficient D_{m} which is directly proportional to the self-mobility of the colloid as:

$$D_{\mathrm{m}} = k_{\mathrm{B}}\Theta\mu_{11} \ . \tag{3.47}$$

It then falls into a slow crossover to reach a second diffusive regime of effective diffusivity D^* as defined in Eq. 3.27. The long-time diffusion coefficient is smaller than the short-time

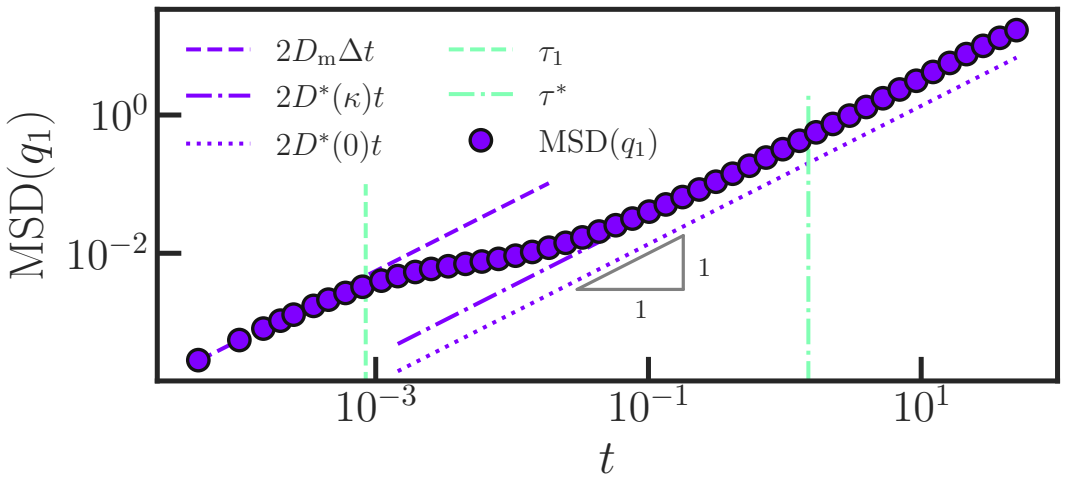

Figure 3.4: Mean-Squared Displacement of q_1 for $\kappa = 0.01$, $\Delta U = 2.7 k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\Delta t = 10^{-4}$, for a single trajectory of $n_{steps} = 1 \times 10^8$ points. The bulk diffusion coefficient D_m (Eq. 3.47) is shown for $\Delta t < \tau_1$ in purple dashes. The modified diffusion constant D^* (Eq. 3.30) is shown in a purple dash-dotted line for $\Delta t > \tau^*$. The dotted purple line indicates the expected long-time diffusion coefficient in the rigid limit (Eq. 3.27 for $\kappa = 0$).

diffusion coefficient, which is a signature of the trapping effect of the sinusoidal potential. The MSD of q_2 (Fig. 3.5) is also diffusive at short times with a diffusion coefficient D_2 which is directly proportional to the self-mobility of the elastic mode, as:

$$D_2 = k_B\Theta\mu_{22} \ , \tag{3.48}$$

and then crosses over to a confined regime at long times $\Delta t > \tau_2$ due to the harmonic trapping. We can predict the value of the long-time MSD plateau of q_2 as $\mathrm{MSD}(q_2) = 2k_B\Theta\kappa$, which we verify numerically.

3.5.4 Quantification of diffusion enhancement

Going beyond a simple graphical observation that numerical simulations agree with the predicted diffusivities, we now extract the values of the diffusivities associated to each diffusive regime from fitting the MSDs. In Fig. 3.4, we see that a diffusive regime corresponds to a linear regime with a time exponent of 1 as $\log(\mathrm{MSD}) = \log(2D) + \log(\Delta t)$, and the diffusion coefficient can be directly inferred from the offset of the affine fit. For a given set of parameters, we can then compare the effect of a varying compliance of a soft mode on the long-time diffusion coefficient of a colloid trapped nearby in a sinusoidal

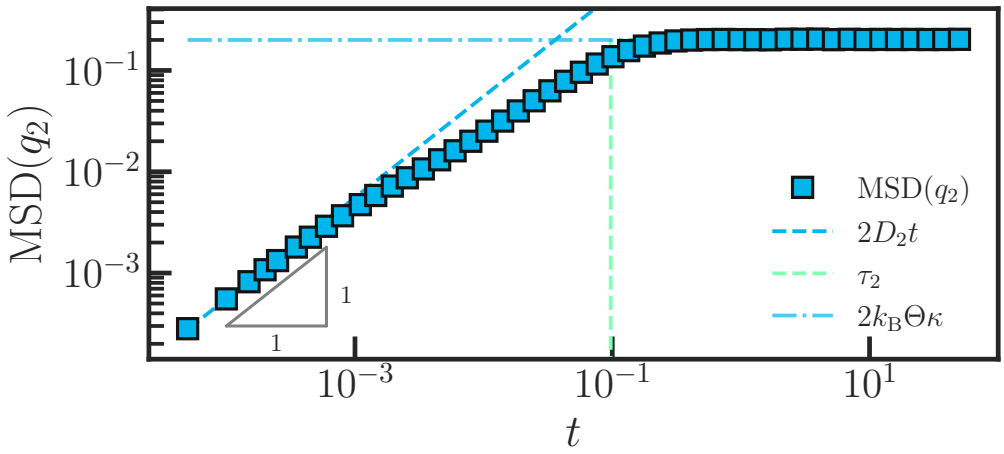

Figure 3.5: Mean-Squared Displacement of q_2 for $\kappa = 0.01$, $\Delta U = 2.7 k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1.0$, $\gamma_{22} = 1.0$, $\gamma_{12} = 0.8$, $\Delta t = 10^{-4}$, for a single trajectory of $N = 1 \times 10^8$ points. At small time lags $\Delta t < \tau_2$, the membrane presents a diffusion constant D_2 (Eq. 3.48, in a blue dashed line) which corresponds to a bulk, free diffusivity inside the minimum of the harmonic potential. The blue dash-dotted line $\Delta t > \tau_2$ corresponds to the long-time MSD plateau value $2k_B\Theta\kappa$.

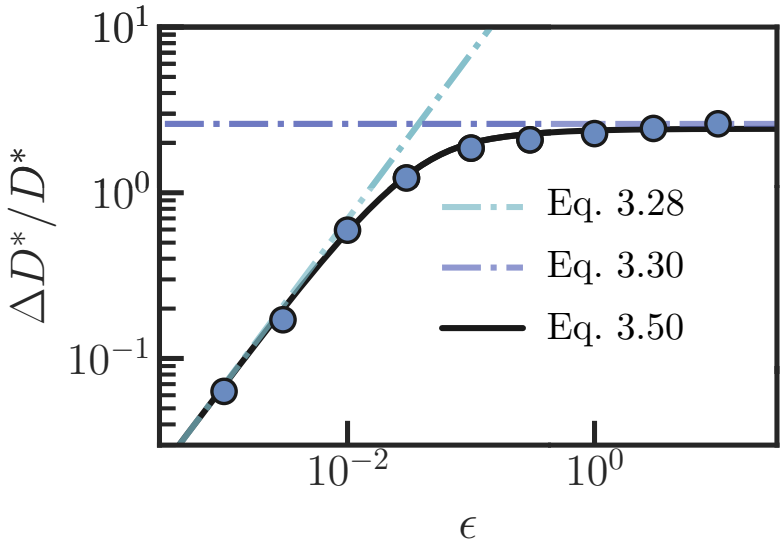

Figure 3.6: Enhancement factor $\Delta D^*(\epsilon)/D^*$ as a function of the coupling parameter ϵ for $\Delta U = 3.0 k_B\Theta$, $k_B\Theta = 1.0$, $\lambda = 1.0$, $\mu_{11} = 1$, $\mu_{22} = 1$, and $\mu_{12} = 0.85$. The Kubo analytical formula (Eq. 3.50) is in solid black. The rigid asymptote to the augmentation factor (Eq. 3.28) is shown in a light blue dash-dotted line, and the compliant plateau to the augmentation factor (Eq. 3.30) is in a dark blue dash-dotted line. Simulation results are shown in blue circles.

potential (Fig. 3.6). We define an augmentation factor $\Delta D^*(\epsilon)/D^*$ as:

$$\frac{\Delta D^*(\epsilon)}{D^*} = \frac{D^*(\epsilon) - D^*(0)}{D^*(0)} \ , \tag{3.49}$$

to compare the rigid and soft cases. The numerical results agree with the expected theoretical augmentation factors in the asymptotical regions $\epsilon \to 0$ and $\epsilon \to \infty$. Interestingly, one may think of ϵ as a ratio of time scales. At small values of ϵ, the elastic mode is stiff, and q_2 becomes the fast mode. At large ϵ, on the contrary, it comparatively becomes the slow mode. Around $\epsilon = O(1)$, however, the time scale separation is lost. The particle coordinate and the elastic degree of freedom now relax on comparable time scales, and thus, neither Lifson-Jackson expression is valid. To access this crossover regime, we solve the full two-dimensional Kubo problem.

Linear-response and Kubo formulation

We use a Kubo linear-response formulation to determine the effective diffusion coefficient beyond the stiff-limit and compliant-limit approximations. Simply put, the idea is to perturb the system with a weak constant force acting along the colloidal coordinate and to extract from the change in the stationary drift velocity the long-time diffusivity through the Einstein relation. The stationary probability density is then expanded to first order in the applied force, with $r(q, u)$ denoting the corresponding linear correction to the equilibrium distribution, and where q and u are the adimensionalised coordinates corresponding to q_1 and q_2, respectively. For each value of the dimensionless compliance ϵ, this correction is obtained by solving the associated stationary Fokker-Planck equation under periodic boundary conditions in q, vanishing flux in u, and the normalisation constraint that the integral of r is zero. Once $r(q, u)$ is known, the effective dimensionless diffusion coefficient follows from the Kubo expression:

$$D_{\mathrm{q}}^* = 1 - \int_0^1 \mathrm{d}q \int_{-\infty}^{\infty} \mathrm{d}u \left[v\psi'(q) + \frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}} u \right] r(q, u). \tag{3.50}$$

The FlexPDE code used to solve Eq. 3.50[1] is available in the GitHub related to this manuscript. It must be run for a panel of ϵ values, we can then be interpolated (Black curve on Fig. 3.6.)

3.5.5 Application with biologically plausible parametres

Let us consider a colloid of radius $a = 100\,\mathrm{nm}$ at a distance $d = 10\,\mathrm{nm}$ from a membrane of elastic modulus $E = 1\,\mathrm{kPa}$. The equivalent compliance of a membrane can be found

[1] This code works with inputting the matrix components of $\sqrt{\mu}$. The corresponding matrix is denoted **m**.

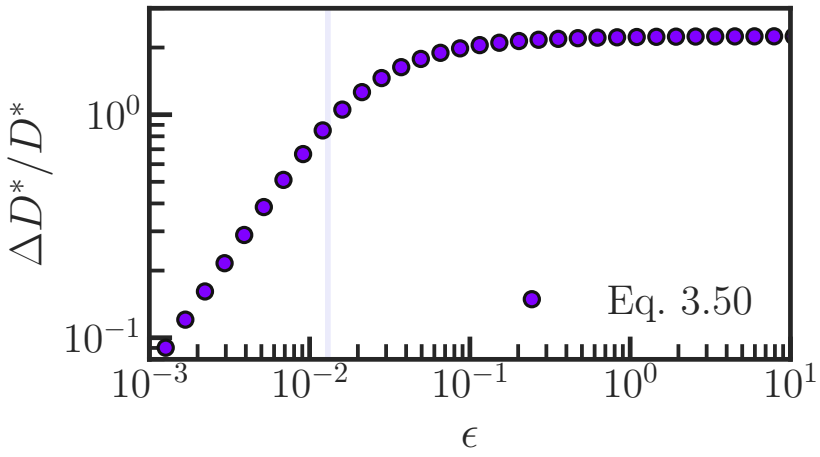

Figure 3.7: Estimation of the relative enhancement of the long-time diffusion coefficient of a colloid of radius $a = 100\,\mathrm{nm}$ in a periodic potential of amplitude $\Delta U = 4k_\mathrm{B}\Theta$ and spatial period $\lambda = 100\,\mathrm{nm}$ at a distance $d = 10\,\mathrm{nm}$ from a membrane of elastic modulus $E = 1\,\mathrm{kPa}$ or equivalent dimensionless compliance $\epsilon = 0.013$. The hydrodynamic constant μ_{12} is estimated using Oseen's model. Coloured disks correspond to the solutions of Eq. 3.50 for $\epsilon \in [0.0001, 100.0]$. The vertical line acts as a guide for the eye, indicating $\epsilon = 0.013$.

using Hertz's model to give $\kappa = (E\sqrt{ad})^{-1}$. The system is immersed in a fluid of viscosity $\eta = 10^{-3}\,\mathrm{Pa.s}$, at room temperature $\Theta = 300\,\mathrm{K}$. The colloid is subject to a periodic potential of amplitude $\Delta U = 4k_\mathrm{B}\Theta$ and spatial period $\lambda = 100\,\mathrm{nm}$, which could for example model a patchy adhesive surface. To extract mobility values, we assume $\mu_{11} = \mu_{22} = (6\pi\eta a)^{-1}$, and take Oseen mobility expressions (see the previous chapter, Eqs. 2.53) for μ_{12}, as $\mu_{12} = (4\pi\eta r)^{-1}$, where r is the centre-to-centre distance between the colloid and the elastic mode, *i.e.* $r = 2a + d$. With these values, ϵ falls exactly in the crossover region, where the system is not considered stiff or compliant. We estimate the expected enhancement of the long-time diffusion coefficient by solving Eq. 3.50 (Fig. 3.7) and measure at $\epsilon = 0.013$ an expected enhancement of about $\Delta D^*/D^* \approx 70\%$.

3.6 Take-home message

In this chapter, we have presented a toy model which encompasses qualitatively the physics of a colloid diffusing nearby a soft membrane. We have then developed a theoretical framework to predict the long-time diffusion coefficient of the colloid as a function of the compliance of the membrane, and we have implemented numerical simulations to verify these predictions. We have found that the coupling of the membrane to the colloid always increases the long-time diffusion constant of the latter given the framework of our model. However, this systematic increase should be heavily modified by the introduction of more physical ingredients in the model. For instance, the reader may have noticed

the mobility matrix is not position-dependent, which hydrodynamically speaking would describe colloids quite far apart compared to their respective radii, rendering the coupling extremely weak. Furthermore, the system is so far in only one dimension. Working in higher dimensions should modify quantitatively the expected enhancement on the diffusion coefficient of the colloid. We also work in the overdamped regime, but aknowledge the interest of keeping mass in the system to probe at added potential elastohydrodynamic-like mass effects. These matters are currently being investigated within our group.

Chapter 4

Dispersion of Brownian solutes in fluid flows

4.1 Introduction

Fluid flows are ubiquitous in many biological settings. The study of fluid dynamics provides insights on multiple ranges of applications. Oceanic currents and large-scale weather phenomena [75], the lift of an airplane foil [76], surfing [77] and drug delivery all rely at least partly on fluid dynamics. The latter example is particularly relevant to my thesis, as it is a perfect example of the interplay between a fluid flow and Brownian motion for the diffusive transport of drugs in the bloodstream. In this chapter, I focus on the study of Brownian motion in fluid flows, and more specifically in viscous flows, with dense suspensions and hydrodynamic interactions as a main problem.

4.1.1 The Reynolds number

We commonly use the Reynolds number Re to evaluate the relative importance of inertial effects in a fluid over viscous ones. It reads:

$$\mathrm{Re} = \frac{\rho U L}{\eta} \ , \tag{4.1}$$

where ρ is the fluid's density, U is a characteristic velocity, L is a characteristic length and η is the fluid's dynamic viscosity. Turbulence in a flow occurs at high Reynolds number, *i.e.* $\mathrm{Re} \gg 1$, whereas a fluid flow with small Reynolds number $\mathrm{Re} \ll 1$ is considered laminar, with a layered and constant profile. A classic example is that of cigarette smoke (Fig. 4.1). The smoke exits the cigarette's tip at low velocity. By taking air density at room temperature $\rho_{\mathrm{air}} = 1.2\,\mathrm{kg.m^{-3}}$, viscosity at about $\eta_{\mathrm{air}} = 1.8 \times 10^{-5}\,\mathrm{Pa.s}$, and a smoke characteristic velocity $U_{\mathrm{smoke}} = 1\,\mathrm{cm.s^{-1}}$ and characteristic plume width $L_{\mathrm{smoke}} = 1\,\mathrm{mm}$, the first regime has $\mathrm{Re} = 0.5$. Higher up, the smoke rises much faster and the plume is wider, with a characteristic speed of $U_{\mathrm{smoke}} = 10\,\mathrm{cm.s^{-1}}$ and a characteristic width of about $L_{\mathrm{smoke}} = 2\,\mathrm{cm}$, the ratio gives $\mathrm{Re} \approx 100$. Perturbations are suppressed in the first

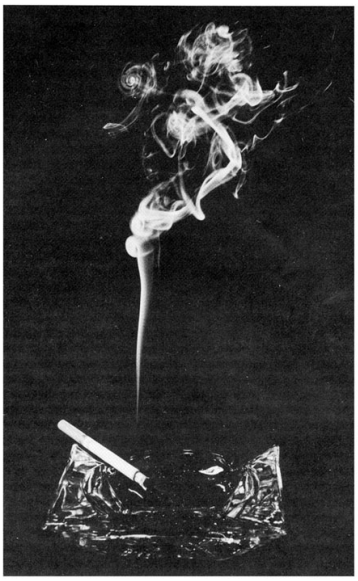

Figure 4.1: Smoke rising from a lit cigarette tip. At short distances from the tip, the smoke plume is narrow and the velocity low enough for the flow to be considered laminar. Higher up, the plume is wider and the velocity higher; the flow transitions to a turbulent one. Source: Physics Stack Exchange.

regime due to viscous effects. As the speed rises, perturbations can no longer be damped, and the typical turbulences arise.

Most microfluidic experiments and associated simulations are conducted in the low-Reynolds-number regime. Despite the apparent simplicity of Stokes flows, this regime exhibits a wide range of rich phenomena, including non-trivial transport, mixing, particle migration, and microswimmer dynamics [78, 79]. In the framework of this thesis, I am interested in understanding the coupling between hydrodynamically interacting colloids, fluid flows and confinement. Specifically, one question has already been well investigated in the last century: "What happens to a Brownian solute once it is advected by a low-Reynolds, viscous flow?" In this chapter, I first introduce the classical Taylor-Aris theory, developed to predict the dispersion of a Brownian solute under advection. Then, I present the results of a collaboration with Joshua McGraw, Masoodah Gunny, Alexandre Vilquin, Blaise Delmotte and Elodie Millan, investigating hydrodynamic interactions in a Brownian solute under flow advection.

4.1.2 Taylor-Aris dispersion

We consider a microchannel of width $2W$ (Fig. 4.2). The fluid in that channel is advected by a unidirectional flow $u(y)\boldsymbol{e}_x$, where x and y are the direction along and normal to the channel, respectively. The flow takes a characteristic Poiseuille parabolic shape. A solute of concentration $c(x, y, t)$ and bulk diffusivity D_0 is injected in that microchannel.

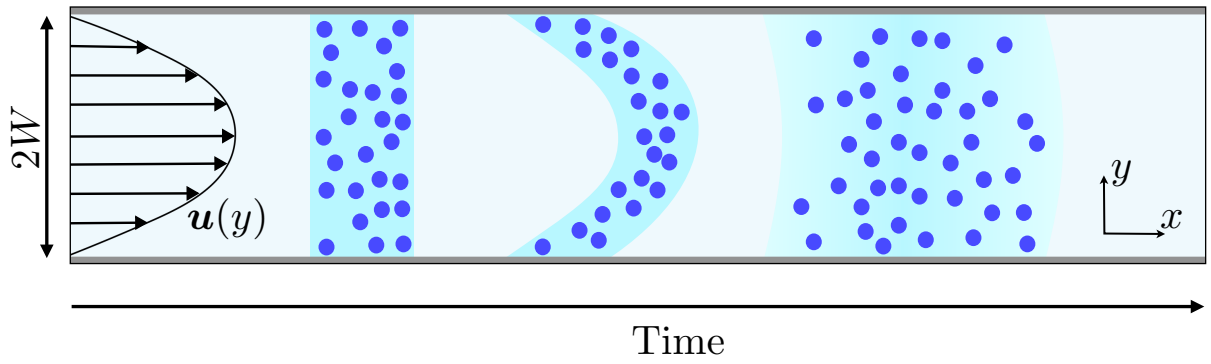

Figure 4.2: Schematic of the Taylor-Aris dispersion phenomenon. A plug of Brownian solute (darker blue) is injected in a microchannel of width $2W$. A viscous flow of velocity $u(y)$ advects the fluid and the plug of solute disperses in the direction along the channel length. Adapted from Gaëlle Audéoud's works.

It obeys the advection-diffusion equation:

$$\partial_t c + u(y)\partial_x c = D_0 \left(\partial_x^2 c + \partial_y^2 c\right) \ . \tag{4.2}$$

The concentration c is the sum of the cross-sectional averaged concentration $\bar{c}$ and a small transverse correction c' induced by the shear, as:

$$c(x, y, t) = \bar{c}(x, t) + c'(x, y, t) \ . \tag{4.3}$$

We define the transverse diffusion time τ_{D} as:

$$\tau_{\mathrm{D}} = \frac{W^2}{D_0} \ . \tag{4.4}$$

At times larger than τ_{D}, the concentration is equilibrated across the channel width. The effective one-dimensional advection-diffusion equation reads:

$$\partial_t \bar{c} + \bar{u}\, \partial_x \bar{c} = D_{x,\mathrm{TA}} \partial_x^2 \bar{c} \ , \tag{4.5}$$

where $D_{x,\mathrm{TA}}$ is an effective diffusion coefficient, called dispersion coefficient, $\bar{u}$ is the velocity average over y. The fluid flow velocity reads:

$$u(y) = \frac{3}{2}\bar{u}\left(1 - \frac{y^2}{W^2}\right) \qquad \text{with} \qquad -W < y < W, \tag{4.6}$$

and the dispersion coefficient is thus given by:

$$D_{x,\mathrm{TA}} = D_0 + \alpha \frac{\bar{u}^2 W^2}{D_0} \ , \tag{4.7}$$

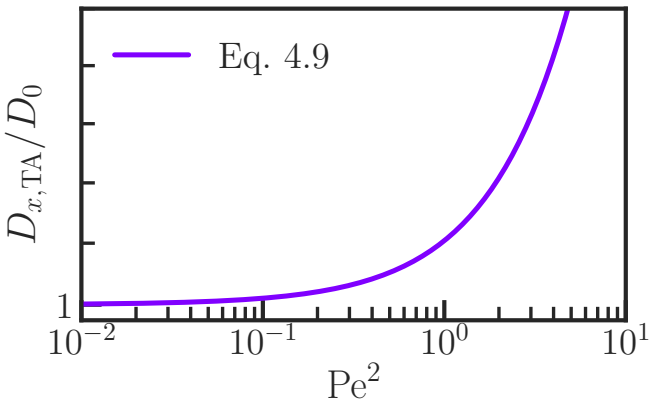

Figure 4.3: Predicted dispersion coefficient renormalised by the bulk diffusion coefficient D_{eff}/D_0 against Péclet number squared Pe^2 computed from Eq. 4.9 with $\alpha = 1/48$, corresponding to the case of a cylindrical tube.

where the $\alpha = 1/210$ in this case[1]. From there, we can identify the Péclet number Pe, which is given by:

$$\mathrm{Pe} = \frac{\bar{u}W}{D_0} \ . \tag{4.8}$$

The Péclet number compares advective to diffusive behaviour in the solute. Finally, we can write the effective diffusion coefficient (*dispersion*) along the channel length:

$$D_{\mathrm{eff}} = D_0 \left(1 + \alpha \mathrm{Pe}^2\right) \ , \tag{4.9}$$

which is the classical Taylor-Aris formula [80, 81, 82, 83]. This formula provides the flow-induced enhancement of the dispersion coefficient of a solute in a channel. For $\mathrm{Pe} = 0$, the dispersion should be minimal; beyond $\mathrm{Pe}^2 > 1/\alpha$, advection becomes dominant over thermal diffusion D_0 (Fig. 4.3). It is worth noting that, while I chose to speak in terms of concentration, the above computations remain valid for the probability of presence of a single colloid in a viscous flow. Taylor-Aris dispersion is still an open field of study. Recent studies have incorporated to the original Taylor-Aris theory a range of sophistications, from time-varying flow profiles [84, 85] to particle size determination [86, 87, 88, 89], and even to active swimmers [90], which will be the focus of the next chapter.

[1]The α coefficient is a geometrical factor, depending on the channel and flow shapes. While $\alpha = 2/105$ for a Poiseuille flow in a two-dimensional channel, a cylindrical channel with a Poiseille flow had $\alpha = 1/48$, whereas a plug flow yields $\alpha = 0$.

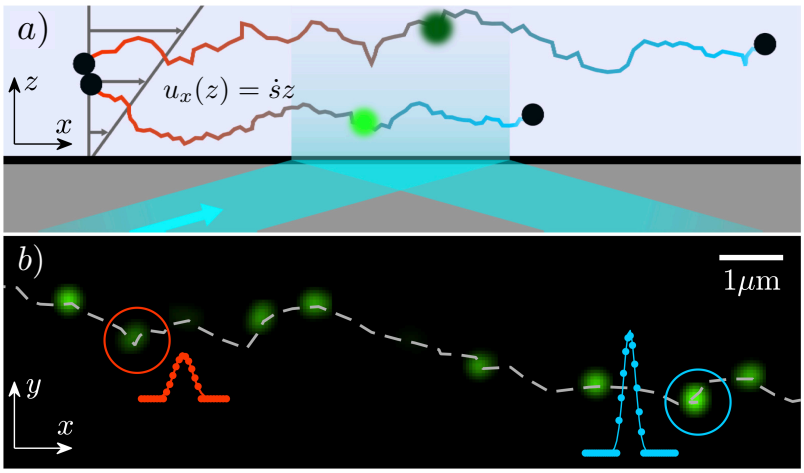

Figure 4.4: Schematic of the Taylor-dispersion total internal reflection fluorescence microscopy setup. *a*) Two colloids of radius $a = 55\,\text{nm}$ are advected by a shear flow of shear rate $\dot{s}$. As they penetrate the evanescent field, they fluoresce and their positions in all three dimensions are recovered. *b*) Experimental images for a single colloid. A succession of frames taken with lag time $\Delta t = 12.5\,\text{ms}$ is shown. Intensity profiles are represented in red and blue dots with Gaussian fits. The linked trajectory of the colloid is represented in a grey dashed line. Adapted from [91].

4.2 Hydrodynamic interactions in an advected solute

We present in this section a collaboration carried out with Masoodah Gunny, Alexandre Vilquin, Joshua McGraw and Blaise Delmotte, and building upon previous studies by Elodie Millan. Specifically, we wonder whether the volume fraction of a solute in a flow can influence the diffusion of this solute. This problem is highly nontrivial from an experimental standpoint. The hydrodynamic interactions between colloids, the fluid flow, and the confinement all contribute to the dynamics of the problem. Therefore, I conduct numerical simulations to replicate the experimental results while in control of every parameter and interaction. I present here briefly our experimental protocol and the results of our collaboration.

4.2.1 Experimental setup

This part of the chapter is largely based on Masoodah Gunny's PhD thesis manuscript [92]. Experiments are carried out with silica beads of radius $a = 55\,\text{nm}$ in salted water of concentration $[\text{NaCl}] = 5.4 - 54\,\text{mg.L}^{-1}$ (with corresponding Debye lengths $l_\text{D} = 35 - 10\,\text{nm}$). The solution is injected in a microfluidic channel of length $L = 8.8\,\text{cm}$, width $W =$

180 µm and thickness $H = 20\,\mu\text{m}$. A pressure gradient ΔP is then applied between the entrance and exit of the channel to start the flow. Very close to the wall, we approximate the Poiseuille profile of the flow as a linear shear $\dot{s}z$, with $\dot{s}$ a shear rate (Fig. 4.4). We take $\dot{s} = \Delta P H/(2\eta L)$. We are interested in the individual colloid behaviour, and thus need an imaging technique that allows for the visualisation of individual colloids rather than a global concentration profile. Imaging is done by Total Internal Reflection Microscopy (TIRFM) [93] (Fig. 4.5). TIRFM has the advantage of being a sub-wavelength imaging technique which still gives information on the three coordinates of a colloid's position within the observation range. The principle of TIRFM relies on evanescence to collect information on the colloid's altitude. An incident laser beam hits the wall of the microchannel with incident angle θ. This angle must obey:

$$\theta > \theta_\text{c} \ , \tag{4.10}$$

where θ_c is the critical angle above which no refraction is permitted:

$$\theta_\text{c} = \sin^{-1}\left(\frac{n_1}{n_2}\right) \ , \tag{4.11}$$

with n_1 and n_2 the optical indices of the water solution and the solid medium, respectively, and $n_1 > n_2$. Like this, refraction is prohibited, and almost all incident light is reflected. However, we know from Maxwell's equations that electromagnetic fields must remain continuous at the interface between the two media. Thus, the field must not end abruptly, but rather fade into a non-propagating electromagnetic field, which we call the evanescent wave. The penetration depth d of this wave is fairly limited, at about two colloid radii, and is given by:

$$d = \frac{\lambda_0}{4\pi\sqrt{n_1^2 \sin^2\theta - n_2^2}} \ , \tag{4.12}$$

where λ_0 is the vacuum wavelength of the laser. This field then excites fluorescent molecules at the surface of the colloids, and we retrieve the emitted light. We can infer the distance z between a colloid and the interface from the intensity a colloid emits, as:

$$I(z) = I_0 \exp\left(\frac{-z}{d}\right) \ , \tag{4.13}$$

where I_0 is the maximal intensity at the interface.

Snapshots of the solution under advection are taken with a camera. Trajectories are then linked from individual successive frames. From the trajectories, displacement distributions are computed, and the transverse diffusion coefficient is then defined as:

$$D_y = \frac{\sigma_y^2}{2\Delta t} \ , \tag{4.14}$$

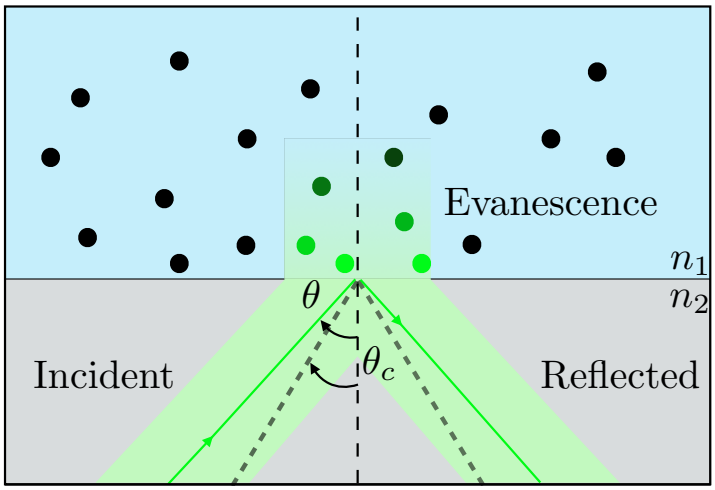

Figure 4.5: Principle of TIRFM. An incident beam hits the interface with an angle $\theta > \theta_c$. It is almost entirely reflected back from the interface. A small portion of the beam, however, does penetrate the sample in the form of an evanescent wave. The colloids in the sample can be visualised owing to their fluorescence when they are illuminated by the evanescent wave and their altitude is then recovered from the light intensity they emit.

where σ_y^2 is the variance of the distribution of displacements in the y-direction, and Δt is the time step between two consecutive frames, and is taken as $\tau = 1\,\text{ms}$. Then, solutions of various concentrations of silica colloids in salted water are tested. It is more intelligible to speak in terms of volume fraction ϕ, that is to say, the ratio of the volume occupied by colloids over the total volume occupied by the solution from there on. We define the volume fraction as:

$$\phi = \frac{4N\pi a^3}{3WLH} \,, \tag{4.15}$$

where N is the total number of colloids in solution. The transverse diffusion coefficients D_y computed from Eq. 4.14 are then extracted for various values of $\phi \in [5\times10^{-6}; 4\times10^{-5}]$ and ΔP and shown in Fig. 4.6. Experimental results seem to point towards a shear rate and volume fraction dependence on D_y (Fig. 4.6, bottom panels). Indeed, for high enough volume fractions ($\phi = 3.7 \times 10^{-5}$, or 0.000037 % of total volume occupied by colloids), D_y becomes larger than the bulk diffusion coefficient $D_0 = 3.9\,\mu\text{m}^2.\text{s}^{-1}$, and this effect is exacerbated with higher shear rates. There are a few potential causes that can easily be ruled out by carrying out numerical simulations. This has already been done by Élodie Millan for the case of electrostatic interactions. We now explore the role of hydrodynamic interactions between the colloids to understand their role in this experimental observation.

4.2.2 Numerical methods

As our collaborators hypothesised a hydrodynamic effect causing the enhancement in transverse diffusion D_y, I conduct all the forthcoming simulations using the RigidBlob-

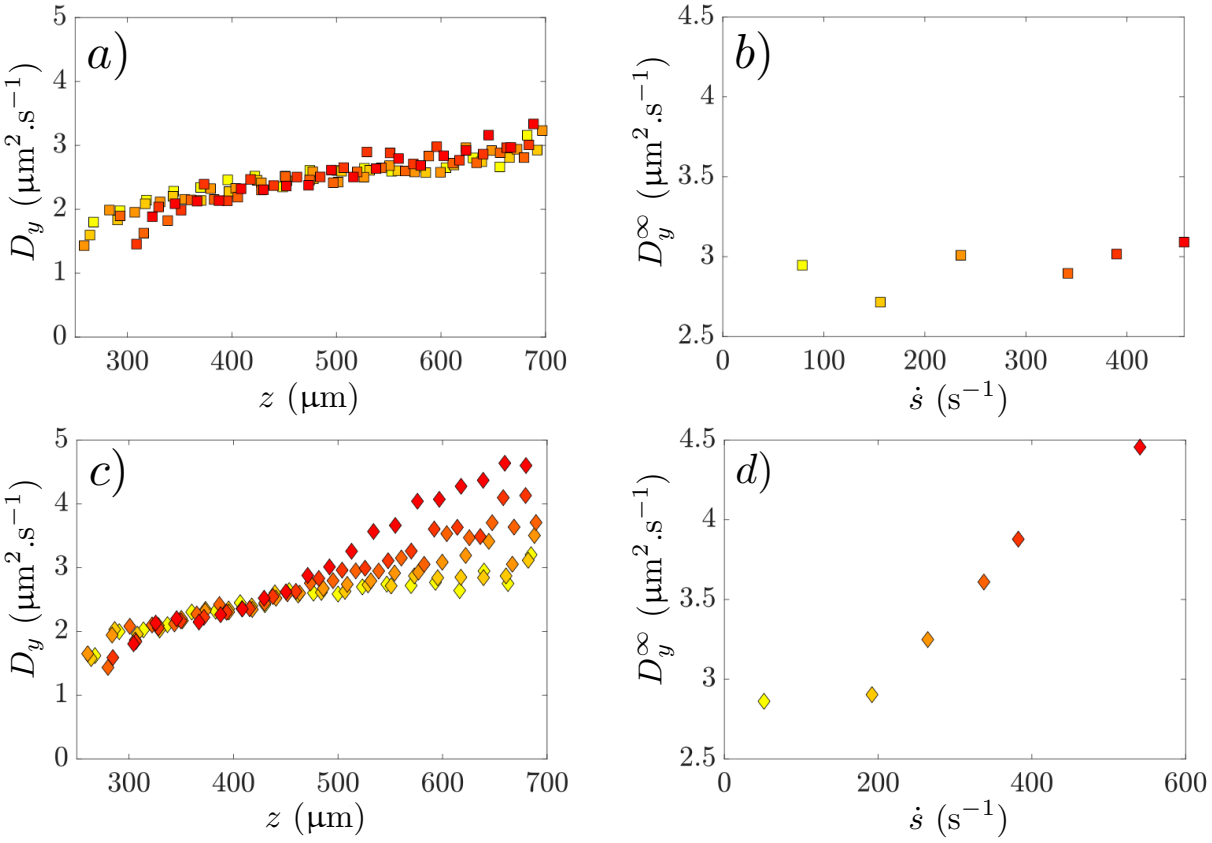

Figure 4.6: Transverse diffusion coefficients computed for $\phi = 5.4 \times 10^{-6}$ (a)) and $\phi = 3.7 \times 10^{-5}$ (c)) and for various pressure gradients ΔP (in colours). Left panels show the diffusion at binned heights z above the channel wall for $\phi = 5.4 \times 10^{-6}$ (b)) and $\phi = 3.7 \times 10^{-5}$ (d)). Credits: Masoodah Gunny and Alexandre Vilquin.

Walls library [46]. This library, described more in depth in chapter 2, is optimised for the simulation of colloids in hydrodynamic and electrostatic interactions, and allows boundary interactions and shear flows. The task involves reproducing the TIRFM setup numerically, generating trajectories in the same conditions as the experiments, and extract similar quantities from these trajectories. The point is to obtain data from a perfectly controlled environment, with perfect tracking of every colloid's position, and with all interactions known.

Simply put, a simulation has four main steps:

Parametrisation. The user provides the physical parameters (buoyant mass of a colloid Δm, colloid radius a, shear rate $\dot{s}$, fluid viscosity η, thermal energy $k_{\mathrm{B}}\Theta$, Debye length l_{D}, amplitude A of the electrostatic repulsion potential, evanescent limit d) and simulation parameters (time step Δt, total number of steps n_{steps}, periodic lengths L_x and L_y). This is done using the `inputfile.dat` and `parameters.py` scripts. The `parameters.py` script then reads `inputfile.dat` to convert, calculate, and pass parameters to the other scripts. As we discussed in the previous paragraphs, the two key quantities we must have full control over are the shear rate $\dot{s}$ and the volume fraction ϕ. The former is simply given as an input to the code in `inputfile.dat`, whereas the latter must be entered indirectly in the form of the total number N of *blobs*, which in this case are equivalent to colloids.

Initialisation. `sample_initial_positions_gb.py` reads from `parameters.py` the array of ϕ values to be probed. The volume fraction is actually ill-defined in our system if considered without care. Indeed, the simulation code is built to either work in triply periodic boundaries, or doubly periodic in (x, y) and half-infinite in z with a confining plane at $z = 0$. This means the maximum height above which colloids do not escape is set physically, not through user input directly. We can only decide the initial positions of the colloids, but without proper care once the system reaches equilibrium, colloids diffuse in the vertical direction to preferential positions (see Fig. 4.8). We recall from Chapter 3 that the distribution of the positions in the grand-canonical ensemble is predicted from the Gibbs-Boltzmann distribution:

$$P(z) = \frac{1}{Z} \exp\left(\frac{U(z)}{k_{\mathrm{B}}\Theta}\right) \ , \tag{4.16}$$

with $1/Z$ the partition function, and $U(z)$ the total potential energy, given here by:

$$\frac{U(z)}{k_{\mathrm{B}}\Theta} = A \exp\left(-\frac{z}{l_{\mathrm{D}}}\right) + \frac{z}{l_{\mathrm{B}}} \ , \tag{4.17}$$

with A the dimensionless magnitude of the electrostatic repulsion potential. The distri-

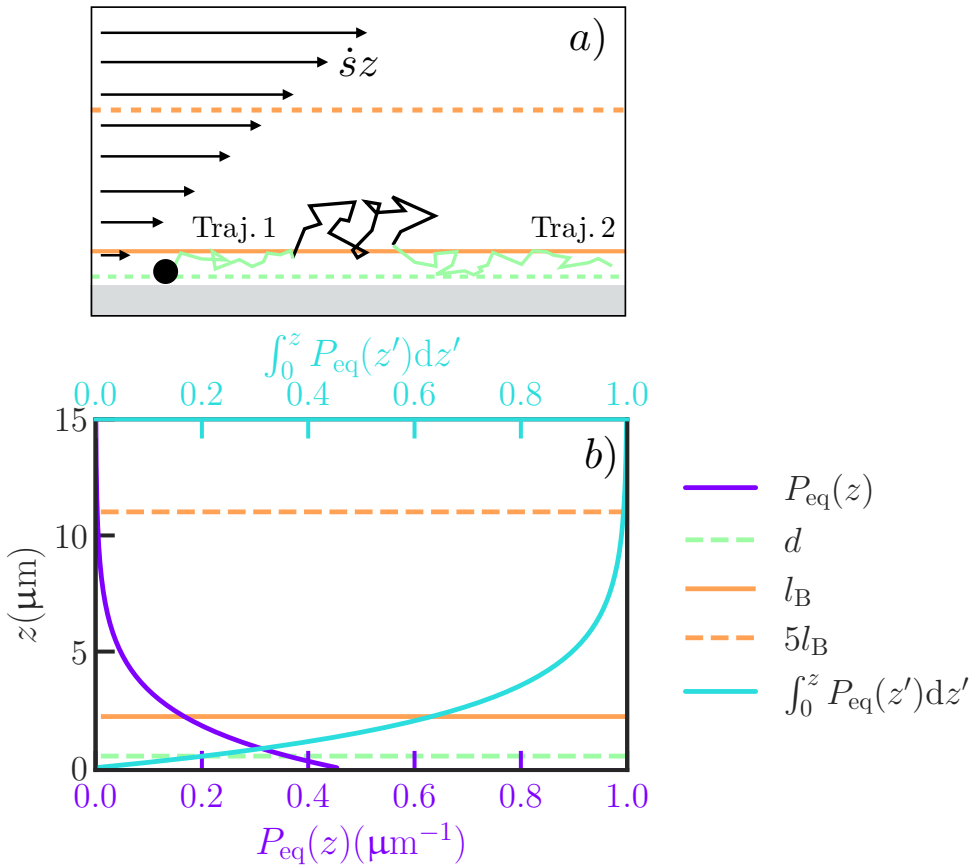

Figure 4.7: Numerical simulation setup. *a*) Colloids are initialised in position in a box of dimensions $L_x \times L_y \times 5l_{\mathrm{B}}$, with l_{B} the Boltzmann length. The box is periodic in the $x-$ and $y-$ directions, with a no-slip boundary at $z = 0$. A shear flow of shear rate $\dot{s}$ is applied in the x direction. Trajectories are recorded for a total time of $T = n_{\mathrm{steps}} \times \Delta t$, where n_{steps} is the total number of simulated steps, and Δt is the duration of one time step. Trajectories crossing the evanescent limit d are killed, and trajectories re-entering below d are tracked as new trajectories if they exist for more than three time steps. *b*) Equilibrium distribution of colloids in solution, as predicted by the Gibbs-Boltzmann distribution (purple line, Eq. 4.16) and corresponding cumulative sum (cyan line, Eq. 4.18). The evanescent limit d is represented in a green dashed line. The Boltzmann length l_{B} is represented with an orange line. We estimate that 99 % of the colloids lie under $5l_{\mathrm{B}}$ (orange dashed line).

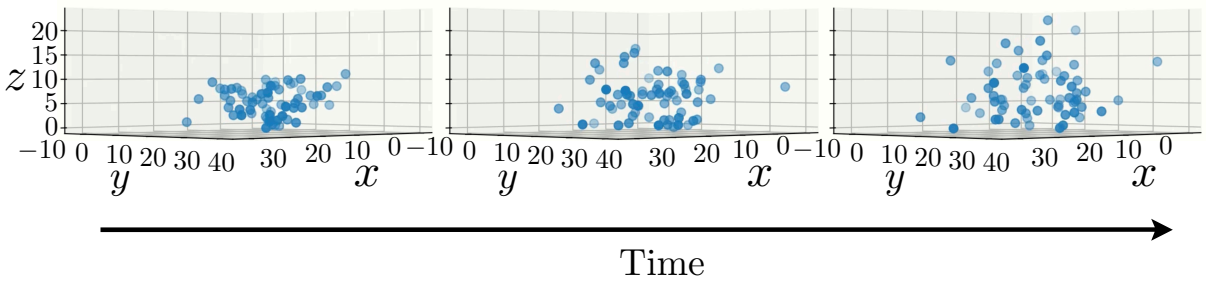

Figure 4.8: Numerical simulation results at initialisation and through time, for a bad initialisation of the system. Colloids were laced in a 10 µm tall box and left to thermalise with a Boltzmann length of $l_B = 1.1 \times 10^{-2}$ m.

bution of colloid positions along z can be integrated, as:

$$\int_0^z P_{\text{eq}}(z')\,\mathrm{d}z' \ , \tag{4.18}$$

to show that about 99 % of the colloids lie under $5l_B$, and about 60 % below l_B (Fig. 4.7b)). Under these conditions, two problems emerge: first, if we choose to lower the simulation cost by using a small box of height H_{box} with $H_{\text{box}} < 5l_B$ to place our colloids at the beginning of the simulations, the solution will inevitably homogenise to a lower volume fraction after some simulation time (Fig. 4.8). Secondly, one cannot simply choose a volume fraction for the whole box and homogeneously place the colloids in that box. Colloids diffuse in the vertical direction and reach the Gibbs-Boltzmann distribution, which is not homogeneous (Fig. 4.7 b)). Thus, the volume fraction is higher close to the wall and lower far from it (Fig. 4.7 b)). To compute the volume fraction ϕ_{5l_B} for the whole $5l_B$ slab in order to retrieve the target ϕ within the d slab, we use:

$$\phi_{5l_B} = \frac{\phi\lambda_e}{5l_B \frac{1}{Z}\int_0^{\lambda_e} P_{\text{eq}}(z)\mathrm{d}z} \ . \tag{4.19}$$

Then, we simulate the whole $5l_B$ slab and fill it with the adequate amount of colloids to reach the desired ϕ. Given that, for our system, $5l_B = 5.5 \times 10^{-2}$ m, and the TIRFM observation window is of 22 µm × 22 µm, we estimate the total amount of blobs in a simulation $N = 3.7 \times 10^7$ to reach a target volume fraction $\phi = 4 \times 10^{-4}$. This number is clearly high and expensive to simulate; we must find a workaround. A simple way of doing so is to artificially lower l_B. By taking a much higher value for the gravitational acceleration, at $g^* = 5000 \times g$, the new Boltzmann constant is $l_B^* = 2.20 \times 10^{-6}$ m, which in turn gives a simulation of $N = 152$ colloids. This is much more reasonable. `sample_initial_positions_gb.py` creates, in the current working directory, a new directory for each value of ϕ. In each new directory, it writes all the simulation scripts. Finally, and most importantly, it computes the initial colloid positions within the simulation box

from the Gibbs-Boltzmann distribution of Eq. 4.16 using a simple exclusion algorithm, and writes them to `sphere_array.clones`. This step is crucial to ensure that the system is prethermalised, and avoids a potentially costly prethermalisation step. This script can also be used to verify that the Gibbs-Boltzmann distribution is respected and that the system is properly thermalised.

Simulation. The script `multi_bodies.py` calls the entire simulation pipeline and launches the shear flow on the pre-thermalised colloidal suspension. It writes the trajectories at each time step to `run_blobs.sphere_array.config`. Other relevant simulation data, such as failed iterations and total computation time, are written to auxiliary output files.

Data treatment. From the trajectories stored in `run_blobs.sphere_array.config`, the trajectories are parsed, and the particle displacements are computed. We limit our studies to the same field of view as experiments. This means that we must kill in the treatment process the trajectories escaping d. Furthermore, if they re-enter the field of view for long enough time durations (we choose three consecutive time steps), their trajectories are treated as new trajectories by the treatment code. MSDs are computed from displacements on each trajectory and then binned by average height. Diffusion quantities are then extracted from the MSD if, and only if, a diffusive regime is identified, namely beyond the transverse diffusion time τ_{D} and when the log-log fits yield a time exponent equal to 1.

4.2.3 Results

Finally, we can compare the results of the simulations to the experimental results. We compute the normalised transverse diffusion coefficient D_y/D_0 as a function of the average trajectory height for various values of ϕ and $\dot{s}$ (Fig. 4.9). We notice that, in contrast to the experimental results, no clear dependence of D_y on ϕ or $\dot{s}$ is visible. The overall tendency of D_y/D_0 shows a slight increase the further from the contact with the wall, and quickly reaches a saturation around 1. This is in agreement with the expected behaviour of a Brownian solute in a viscous fluid, and with Brenner [27] and Faxen's [26]laws. The absence of any clear dependence on ϕ and $\dot{s}$ suggests that the experimental results may be due to an artefact, and that the observed enhancement of D_y may not be due to hydrodynamic interactions. This is in line with the fact that, at such low volume fractions, hydrodynamic interactions should be negligible. However, it is worth noting that experiments and numerical data were treated differently. Indeed, in experiments, trajectories were first linked from different frames, and displacements were computed from these trajectories. In simulations, we directly access the whole labeled trajectories, thus ruling out any potential tracking artifact. It is worth noting that the diffusion coefficient D_y itself has different definitions between our simulation results and the experiments. In experiments, D_y is computed from the variance of the distribution of displacements in the

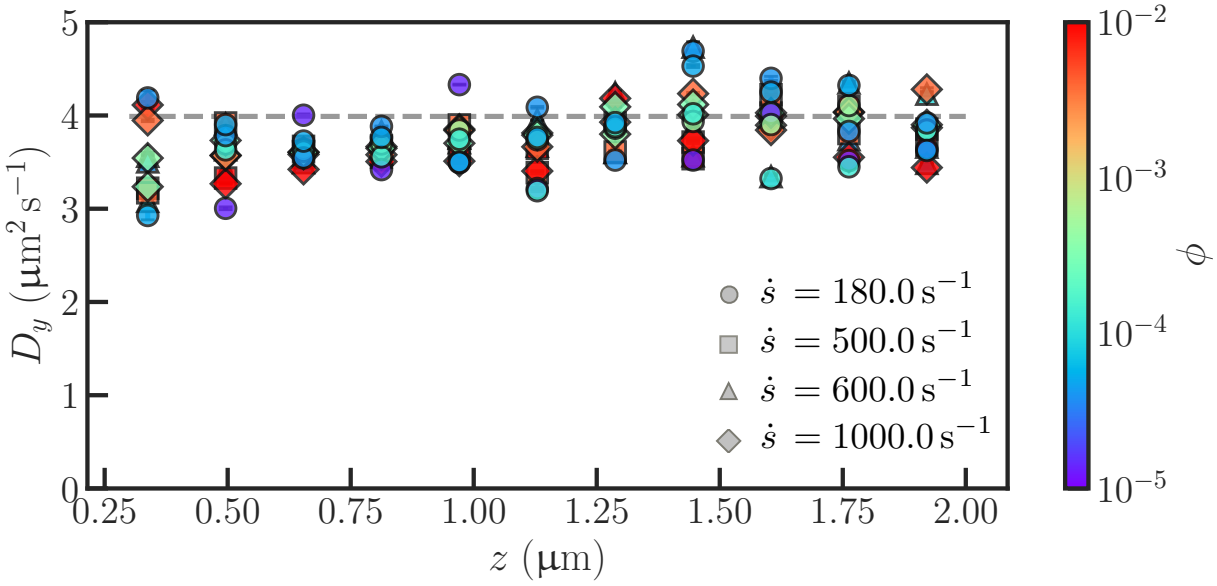

Figure 4.9: Transverse diffusion coefficient D_y as a function of average trajectory heights. Simulations were run once for each $(\phi, \dot{s})$ couple and for $N_{\text{steps}} = 50000$, with time step $\Delta t = 0.0001\,\text{s}$, fluid viscosity $\eta = 0.001\,\text{Pa.s}$, colloid radius $a = 55\,\text{nm}$, buoyant mass $\Delta m = 1.92 \times 10^{-16}\,\text{kg}$, gravity acceleration $g = 5000 \times 9.81\,\text{N.m}^{-2}$, thermal energy $k_B\Theta = 4.14 \times 10^{-21}\,\text{J}$, and $d = 1800\,\text{nm}$. The colormap represents volume fractions. Disks correspond to shear rate $\dot{s} = 100.0\,\text{s}^{-1}$, squares to $\dot{s} = 500.0\,\text{s}^{-1}$, triangles to $\dot{s} = 600.0\,\text{s}^{-1}$ and diamonds to $\dot{s} = 1000.0\,\text{s}^{-1}$. The dotted line indocates $D_0 = 3.9\mu\text{m.s}^{-1}$.

y-direction at a fixed time step τ, as shown in Eq. 4.14. If the system is not in a diffusive regime, the extracted coefficient may not be meaningful. From numerically generated data, we compute D_y from the slope of the MSD in the diffusive regime. Investigations are still being conducted to understand the origin of this discrepancy. We are currently analysing numerically generated TIRFM-like videos, with the same tracking and data analyses as the experiments used. If the effect then becomes visible, it would point towards a numerical artefact in the data extraction. However, if the effect is not visible, more investigations would be needed. We could for instance verify the actual volume fractions in experiments to fully rule out hydrodynamic effects.

4.3 Take-home message

In this chapter, we introduced the Taylor-Aris dispersion theory for Brownian solute in viscous fluid flows. We established that, in a Poiseuille flow, fluctuations permit a broadening of the concentration profile in time. This theory was the framework for our collaboration with Masoodah Gunny, Joshua McGraw, Blaise Delmotte and Alexandre Vilquin. We numerically replicated their experimental setup, consisting of a Poiseuille flow

advecting a Brownian solute in a rectangular microchannel. We measured the transverse diffusion coefficient, orthogonal to the flow direction. We observed no notable effect on this coefficient from the fluid velocity or the volume fraction of the solute. We concluded that, although our simulations seem not to indicate an effect of hydrodynamic interaction between colloids on the dispersion coefficient, a new treatment should be carried out both on the numerically generated and experimental data, using exactly the same protocol. This would thus definitely rule our either hydrodynamic effects, or treatment artifacts.

This chapter also served as in introduction to chapter 5, where we investigate the Taylor-Aris dispersion of an active solute.

Chapter 5

Dispersion of active microswimmers in shear flows

5.1 Introduction

The final part of this manuscript is dedicated to the study of active particles and their interplay with external fluid flows. This subject was actually the motivation for my thesis, and the first part of this chapter presents an introduction to the field of active matter. The end of this chapter treats the problem of active Taylor-Aris dispersion. The notion of Taylor-Aris dispersion is described in chapter 4. For the sake of clarity, I thus invite the reader to consult its developments before reading this part. We conducted this work in tandem with Marc Lagoin, Guirec de Tournemire, Ahmad Badr and Antoine Allard.

5.1.1 Activity

Authors define active matter as the first or both of the following:

1. individual units that extract free energy from internal or external sources and convert it to motion, forces, or mechanical stresses [94, 95, 96, 97]. They are thus inherently out-of-equilibrium systems;

2. individual self-propelled units which interact locally and exhibit emergent, collective behaviours, such as through hydrodynamic interactions (as it is the case for swimming microorganisms, squirmers, active droplets...) [96], phoretic interactions (catalytic Janus colloids, some chemotactic bacteria...) [98], mechanical force transmission (cellular tissue, cytoskeleton, active gels...) [94], and even social interactions (human crowds, bird flocks, fish schools, autonomous drones...) [99], to only cite a few. Some of these interactions can be accurately represented with minimal models, which approximate them as alignment interactions with some noise, such as the Vicsek model [100].

Numerous examples of active systems exist, and these systems can be found at any scale (Fig. 5.1). It is essential to not only classify them by size but also by Reynolds number.

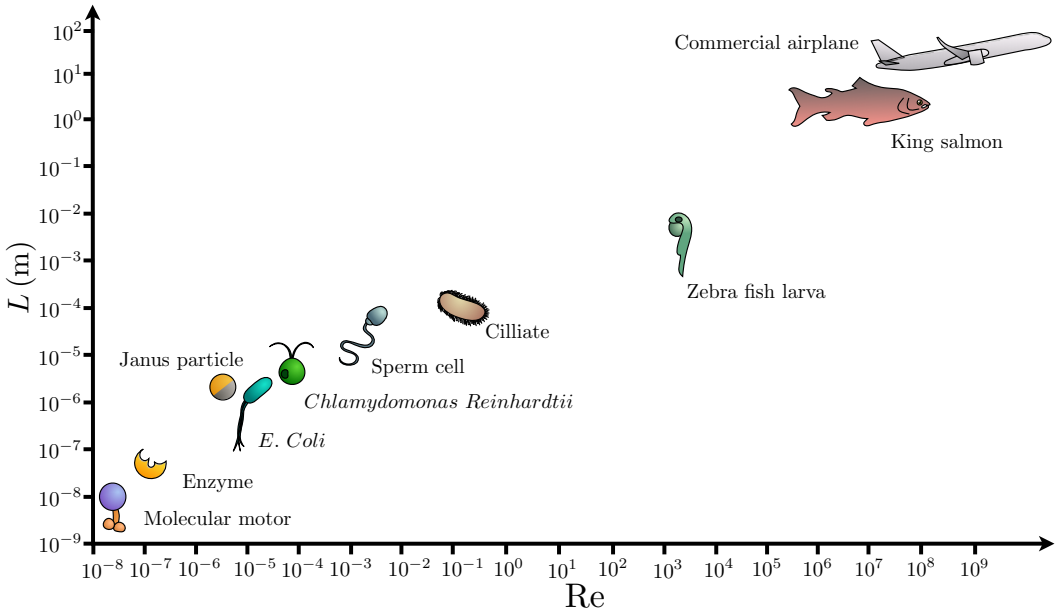

Figure 5.1: Comparison of typical length scales and Reynolds numbers for various active systems.

Interestingly, the definition of active matter is loose enough to even include human beings, robots and even large aircrafts. This thesis articulates around notions of small systems, and I thus restrict my arguments to these systems from there on.

5.1.2 *Chlamydomonas reinhardtii* as a model swimmer

A particularly convenient experimental model of a biological microswimmer is the unicellular green alga *Chlamydomonas reinhardtii* (Fig. 5.2) [101]. This tiny alga is the system we use in the study at the end of this chapter. It measures only a few tens of microns at most. However, through the beating of its anterior flagella, it self-propels in a fluid at surprisingly high speeds of ten body lengths per second. Yet, this is not sufficient for *Chlamydomonas* to be considered as high-Reynolds systems ($\mathrm{Re} \gg 1$), as it typically reaches a Reynolds number of about $\mathrm{Re} = 1 \times 10^{-4}$.

Furthermore, *Chlamydomonas reinhardtii* can orient towards or away from light sources by phototaxis, a response mediated by an "eye spot" rhodopsin photoreceptor, and controlled by both light intensity and directional sensing [102, 103, 104]. In the absence of a directional light signal, its swimming can display run-and-tumble-like dynamics, with random switches between straight swimming and abrupt reorientation events [101]. Under blue light and near a solid boundary, cells instead attach through their flagella and perform

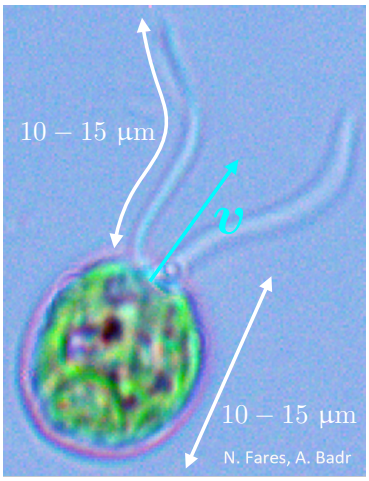

Figure 5.2: Photograph of a *Chlamydomonas reinhardtii* alga taken by Ahmad Badr and Nicolas Fares in our experimental rooms. The body and the flagella each measure about 10µm. The alga also features an eyespot which allows it to sense incoming light and adapt.

surface gliding, driven by surface transport instead of ordinary swimming [105, 106, 107, 108, 109]. Because of its simple geometry and well established and studied behaviour, *Chlamydomonas* is a standard model system for active matter in fluids.

5.2 Modelling approaches

Multiple models exist to study active matter. I describe here the ones I have come to know or use during my thesis, either for published works or personal interest. I also provide on the GitHub repository a simple Jupyter file to simulate Active Brownian Particles (ABP), Run-And-Tumble particles (RTP) and to compare them to the passive case. This file also contains a very small exploration of chiral ABP.

5.2.1 Active Brownian Particles

The thermal diffusion of a colloid affects both its translational degrees of freedom (x and y for a two-dimensional case) and rotational degree of freedoms (ϕ for a two-dimensional case). We recall from the Introduction of this manuscript the classical, passive bulk Brownian particle case. The translational thermal diffusion coefficient D_0 is defined as:

$$D_0 = \frac{k_\mathrm{B}\Theta}{6\pi\eta a} \ , \tag{5.1}$$

where $k_\mathrm{B}\Theta$ is the thermal energy of the bath, η the fluid viscosity and a the colloid radius. The thermal rotational diffusion coefficient D_R is inversely proportional to the colloid's

volume:

$$D_{\mathrm{R}} = \frac{k_{\mathrm{B}}\Theta}{8\pi\eta a^3} = \tau_{\mathrm{R}}^{-1} \ , \tag{5.2}$$

with τ_{R} the characteristic reorientation time scale.

We can describe the motion of the passive colloid with the following Langevin equations in the overdamped limit:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= \sqrt{2D_0}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= \sqrt{2D_0}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{R}}}\xi_\phi(t) \ , \end{aligned} \tag{5.3}$$

where $\xi_i(t)$ are Gaussian white noises. This process is heavily reliant on the thermal diffusion coefficients D_0 and D_{R}. This means that thermal Brownian motion is strongly hindered in the case of large colloids, where "large" means of the order of a few tens of microns. For example, if we take a colloid of a size that is comparable to a *Chlamydomonas*, that is to say $a = 10\,\mu\mathrm{m}$, we get $D_0 = 2.2 \times 10^{-3}\,\mu\mathrm{m}^2.\mathrm{s}^{-1}$. This value indicates that by pure thermal diffusion, a colloid of this size would need tens of hours to explore an area equivalent to its own size. As we discussed already, swimmers can circumvent this limitation by collecting energy from their surroundings to self propel with a given velocity v. A very simple model used to represent this scenario is the ABP model. We implement this model with the following Langevin equations:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D_0}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D_0}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{R}}}\xi_\phi(t) \ . \end{aligned} \tag{5.4}$$

The MSD in that case follows [110, 111, 112] (see Fig. 5.5):

$$\boxed{\mathrm{MSD}_x(t) = \mathrm{MSD}_y(t) = 2D_0 t + \frac{v^2}{D_{\mathrm{R}}^2}\left(D_{\mathrm{R}}t + \mathrm{e}^{-D_{\mathrm{R}}t} - 1\right)} \ . \tag{5.5}$$

We thus expect a thermally diffusing regime at very short times with diffusivity D_0, followed by a superdiffusive regime at intermediate times, and finally, after the characteristic time scale τ_{R}, a second, higher diffusive regime at very long times with a new effective diffusivity D_{eff}, as $D_{\mathrm{eff}} = v^2/(2D_{\mathrm{R}})$ (see Fig. 5.3 for a comparison of a passive Brownian colloid and an ABP).

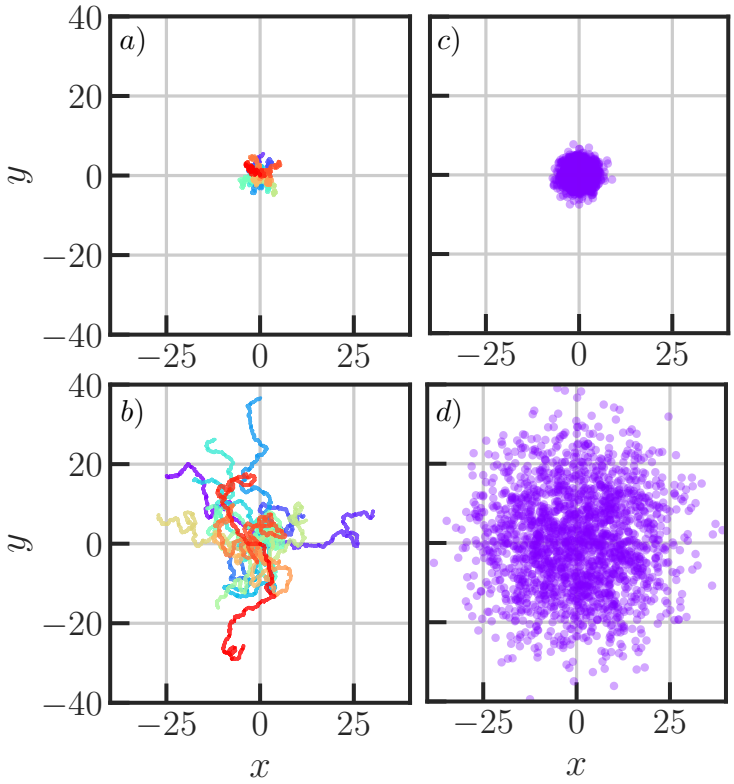

Figure 5.3: Comparison of passive and active Brownian colloid trajectories. Panels *a*) and *c*): Twenty stacked trajectories obtained from discretising Eqs. 5.3 and 5.4 respectively, with $\Delta t = 0.05$, $n_{\text{steps}} = 5000$, $D_0 = 0.05$, $D_{\text{R}} = 0.3$, $v = 0.1$, with all units in simulation units. Colours are only used for readability and correspond to separate trajectories. Panels *c*) and *d*): final position of 2000 simulations for the passive and active Brownian particle models, respectively.

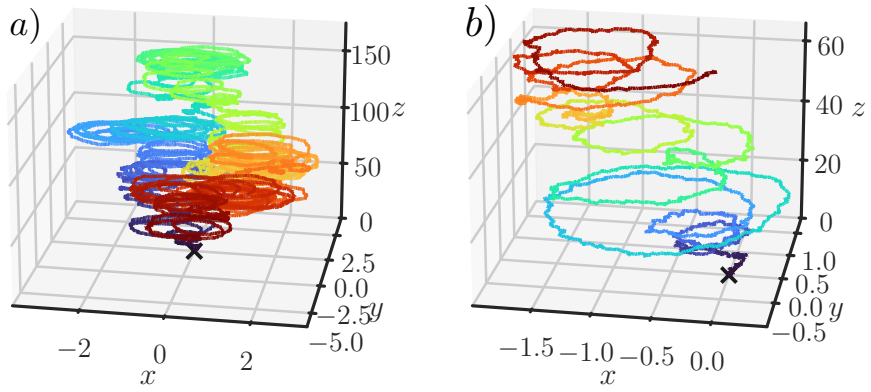

Figure 5.4: Simulated trajectories from the discretised Eqs. 5.7. *a*) Full trajectory for $n_{\text{steps}} = 80000$, $\Delta t = 0.01$, $D_0 = 0.005$, $D_{\text{R}} = 0.01$, $v = 1.0$, and $\omega = \pi/12$. All units are in simulation units. *b*) Same trajectory, for the first 800 steps. The colormap corresponds to elapsed time and is only indicative.

Chiral ABPs

Some microswimmers or clusters of microswimmers have irregular shapes or asymmetric propulsion mechanisms. It is possible to model the consequences of these asymmetries by adding a chiral component to the ABP model. While the ABP translation mechanisms remain unchanged, we add a constant rotation speed ω in the angular Langevin equation, as follows [113, 114, 115]:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D_0}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D_0}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \omega + \sqrt{2D_{\text{R}}}\xi_\phi(t)\ . \end{aligned} \tag{5.6}$$

It is therefore possible to play with the angular velocity ω to obtain different behaviours, from shrinking spirals to circles or expanding spirals for the two-dimensional case. A three-dimensional, helical motion can be generated starting from a three-dimensional

ABP model, such as:

$$\begin{aligned} \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} &= v\boldsymbol{n} + \sqrt{2D_0}\,\boldsymbol{\xi}_{\mathbf{T}}(t), \\ \frac{\mathrm{d}\boldsymbol{n}}{\mathrm{d}t} &= \boldsymbol{\Omega} \times \boldsymbol{n} + \sqrt{2D_{\mathrm{R}}}\,\boldsymbol{\xi}_{\mathbf{R}}(t) \times \boldsymbol{n}, \end{aligned} \tag{5.7}$$

with $\boldsymbol{r}$ the position vector $\boldsymbol{r} = (x, y, z)$, and $\boldsymbol{n}$ the orientation vector, which must be normalised, as $|\boldsymbol{n}| = 1$, and $\boldsymbol{\Omega}$ is the angular velocity vector defined as:

$$\boldsymbol{\Omega} = \omega\,\boldsymbol{e}_z = \begin{pmatrix} 0 \\ 0 \\ \omega \end{pmatrix} \ , \tag{5.8}$$

for a rotation about the $z-$ axis (Fig. 5.4).

5.2.2 Run and tumble

A run and tumble particle (RTP) is a process describing an active particle that alternates between two types of motion: long periods of approximately straight swimming, called runs, and sudden reorientation events, called tumbles [116, 117, 118]. During a run, the particle moves at a nearly constant self-propulsion speed v along its instantaneous orientation described by the angle $\phi(t)$. The position therefore evolves as:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D_0}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D_0}\xi_y(t) \ , \end{aligned} \tag{5.9}$$

which are the same equations as the ABP case. However, the orientation of the particle may suddenly change, usually according to a Poisson process with rate λ. This change is called a tumble, and happens for instance when an *E. coli* bacterium desynchronises its flagella. The probability that at least one tumble occurs during a given time Δt is:

$$P_{\mathrm{tumble}}(\Delta t) = 1 - \exp(-\lambda \Delta t) \ . \tag{5.10}$$

The typical time between two tumbles is then:

$$\tau_{\mathrm{run}} = \frac{1}{\lambda} \ . \tag{5.11}$$

At each tumble time t_i, the orientation is instantly changed as:

$$\phi(t_i^+) = \phi(t_i^-) + \Delta\phi_i \ , \tag{5.12}$$

where $\phi(t_i^+)$ is the orientation immediately after the tumble, $\phi(t_i^+)$ the orientation immediately before and $\Delta\phi_i$ is the reorientation. The particle has a persistent direction of motion, but this persistence is destroyed by two mechanisms: continuous rotational diffusion, controlled by D_R, and tumbling events, controlled by λ. In two dimensions, for fully random tumbles, the persistence timescale is given by:

$$\tau_\mathrm{p} = \frac{1}{D_\mathrm{R} + \lambda} \ . \tag{5.13}$$

At long times $t \gg \tau_\mathrm{p}$, the process thus becomes diffusive, much like the ABP case again, and with diffusivity:

$$D_\mathrm{eff} = D_0 + \frac{v^2}{2(D_\mathrm{R} + \lambda)} \ . \tag{5.14}$$

The MSD in that case is given by (see Fig. 5.5):

$$\boxed{\mathrm{MSD}_x(t) = \mathrm{MSD}_y(t) = 2D_0 t + \frac{v^2}{\lambda^2}\left(\lambda t + \mathrm{e}^{-\lambda t} - 1\right)} \ . \tag{5.15}$$

This equation is strictly equivalent to Eq. 5.5, except for a substitution of λ with D_R. The MSD is not able to distinguish between the two processes, and one must turn to other quantities (non-gaussianity, velocity autocorrelation, or even visually with the trajectory shape) to properly assess the difference between ABP and RTP processes. Now, we can identify two limiting cases for the RTP process: when $\lambda = 0$, we naturally retrieve the ABP process. When $D_\mathrm{R} = 0$, the motion is a pure run-and-tumble one where reorientation comes solely from the active process, with $D_\mathrm{eff} = v^2/(2\lambda)$.

5.2.3 Other conventional methods for swimmer modeling

The models I presented so far describe active swimmers through an ensemble of reduced sets of degrees of freedom. Therein, the swimmer is treated as a point-like particle through its leading far-field hydrodynamic signature. Though minimal, these models are efficient in terms of computation cost. However, they do not resolve the finite swimmer size, surface motion, full lubrication forces, or detailed flow structures that become important when swimmers come close to one another or to a boundary. A first step is to represent the swimmer as a finite-size body with surface activity. The classical example is the squirmer model introduced by James Lighthill in 1952 [119] but refined by Blake in 1971 [42], in which a spherical particle swims by imposing a tangential slip velocity on its surface. The propulsion mechanisms are then modeled through the flow generated at the swimmer's boundaries. It is then possible to recover different swimmer behaviours (Shaker, pusher, puller, etc.) by tweaking the system's parameters. More resolved approaches include solving the fluid, such as boundary-element, immersed-boundary, multiparticle

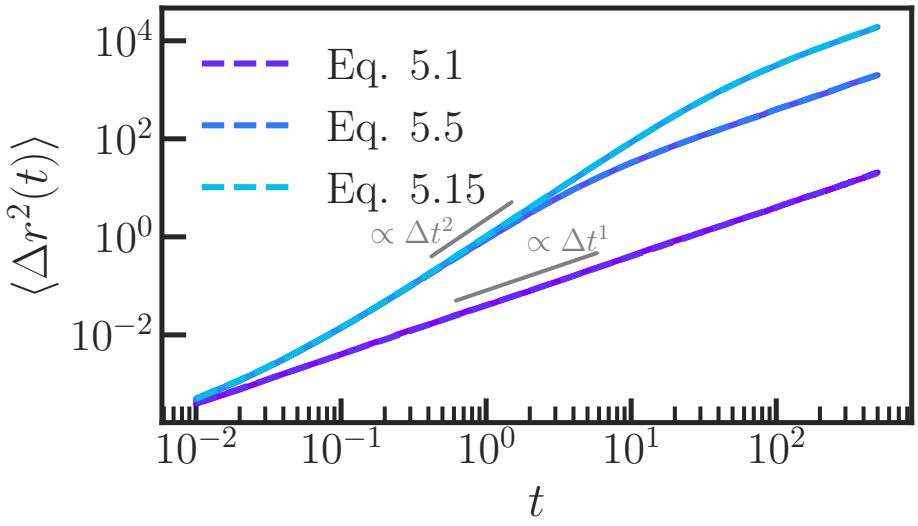

Figure 5.5: MSDs of an ABP (light blue line), a run-and-tumble particle (navy blue line) and a passive Brownian colloid (purple line), for trajectories of $n_{\text{steps}} = 5 \times 10^5$ steps, $\Delta t = 0.01$, $D_0 = 0.01$, $v = 1.0$, and $\lambda = 0.05$. Corresponding theoretical curves from Eqs. 5.1, 5.5 and 5.15 are shown in dashed lines in respective colours.

collision dynamics, or lattice-Boltzmann methods. These methods compute the fluid flow around the swimmer and can incorporate complex swimmer shapes, deformable boundaries, confinement, or no-slip walls. They are much more computationally demanding, but they provide access to regimes where the point-particle approximation breaks down. Point-particle models are well suited to identify generic non-equilibrium mechanisms, and finite-size and hydrodynamically resolved models are necessary when near-field interactions, confinement, or details in the propulsion mechanism are in control of the dynamics.

5.3 Active Brownian Particles in fluid flows

5.3.1 Context

In chapter 4, I introduced the notion of Taylor-Aris dispersion, originally developed for passive dilute solutions in Poiseuille flows. Extensions of this framework have been developed for biased and gyrotactic swimming microorganisms [120, 121, 122, 123], as well as for idealised active particles and active Brownian particles in confined flows [124, 125]. This study provides an experimental realisation with living *Chlamydomonas reinhardtii* in a confined, sinusoidal Poiseuille flow, which is rationalised numerically with Langevin simulations. It was published in the Proceedings of the National Academy of Sciences of the United States of America (PNAS) in 2025 [90].

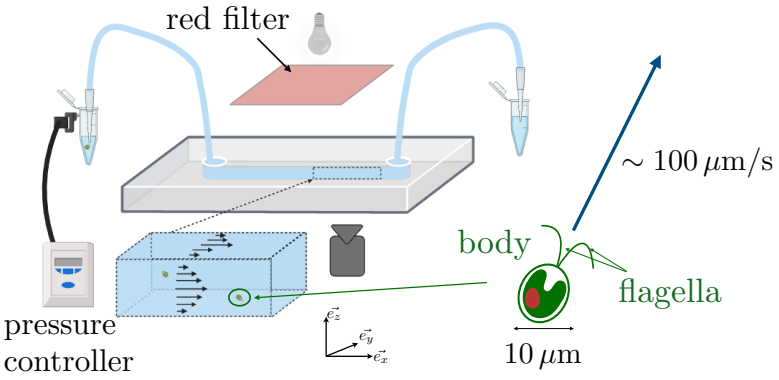

Figure 5.6: Schematic representation of the experimental setup for Active Taylor-Aris dispersion. A suspension of microswimmers is injected in a rectangular microchannel of width W, height H and length L. An oscillating flow is imposed in the microchannel using a pressure controller at one end, whereas pressure is maintained constant on the other end with a reservoir. Swimmers are illuminated with a red-filtered light, and imaged with a high-framerate camera through an objective.

5.3.2 Experimental setup

We create a microfluidic channel using soft lithography on a PolyDiMethylSiloxane (PDMS) chip. The channel has dimensions $(L, W, H) = (88\,\text{mm}, 180\,\mu\text{m}, 100\,\mu\text{m})$, with W the width, H the height and L the length of the channel (Fig. 5.6). The streamwise direction is defined along $\boldsymbol{e_x}$. The cell is then slowly filled with a dilute suspension of *Chlamydomonas reinhardtii* to ensure negligible interaction between individual algae. We suppress the phototacic response with a red filter [126]. A pressure controller is plugged at one end of the channel, on the entry point. On the exit point, constant pressure is maintained by a hydrostatic bath. We input a controlled pressure, which can be modulated by a sinusoid using the pressure controller. The pressure difference between the entry and exit points drives the flow of the suspension inside the microchannel. The fluid velocity is given by:

$$\boldsymbol{u}(y,z,t) = \frac{\Delta P}{R_\text{h}} f(y,z) \sin\left(\frac{2\pi t}{T} + \phi_0\right) \boldsymbol{e_x}\ , \tag{5.16}$$

where ΔP is the pressure difference between the amplitude of the pressure controller and the hydrostatic bath, R_h is the hydraulic resistance of the channel, T is the pressure oscillation period, ϕ_0 is a phase at origin, and $f(x,y)$ is the cross-sectional profile, which

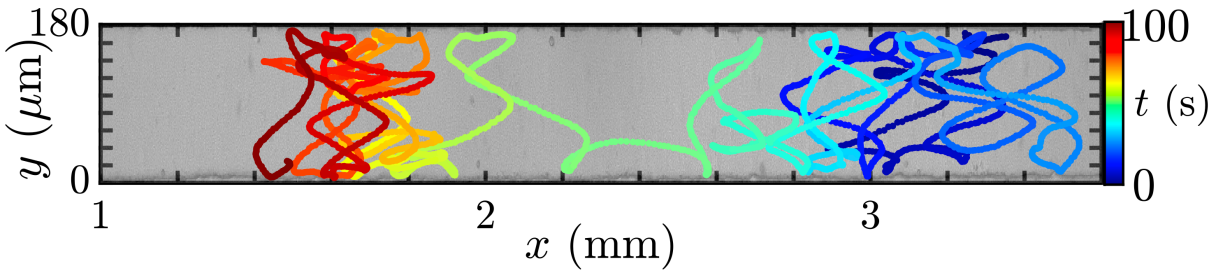

Figure 5.7: Typical experimental trajectory of a *Chlamydomonas reinhardtii* in an oscillating flow. The colormap corresponds to elapsed time.

in a quasi two-dimensional limit is expressed $f(y) = 4y/W\,(1 - y/W)$. Because the channel is sufficiently narrow and our observations flatten out the z coordinate, we use this expression of f for the study. We then image the channel from the top, over the $z-$plane. We record frames for 200 s at 40 frames per second. We then link individual trajectories using a linking algorithm, and only keep trajectories which last for more than 20 s in the field of view of the camera for treatment. The oscillatory flow has the advantage of forcing the algae into the field of view for longer durations (Fig. 5.7). The treatment of a single trajectory then involves demodulating the signal from the oscillatory signature of the flow velocity using a sliding windowed demodulation, assuming that the position of an alga in time along the $x-$axis can be expressed as:

$$x(t) = \hat{x}(t) + A[y(t), t] \cos\left(\frac{2\pi t}{T}\right) \ , \tag{5.17}$$

where $A[y(t), t]$ is an unknown function, and $\hat{x}(t)$ is the position of the alga in the flow's comoving frame.

5.3.3 Numerical model

To reproduce the experimental results from this study, I choose a two-dimensional ABP model with a rotational diffusion coefficient such that $D_{\mathrm{R}} = \tau_{\mathrm{R}}^{-1}$. Strictly speaking, this D_{R} is not a diffusion coefficient in the thermal sense; we define it as an active rotational diffusion coefficient. The equations are:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v\cos[\phi(t)] + \sqrt{2D_0}\xi_x + u(y,t) \ , \tag{5.18}$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = v\sin[\phi(t)] + \sqrt{2D_0}\xi_y \ , \tag{5.19}$$

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{1}{2}\omega(t) + \sqrt{2D_\phi}\xi_\phi \ , \tag{5.20}$$

with v the particle's self propelling speed, $D_\phi = \tau_{\mathrm{R}}^{-1}$ the rotation rate, ω the rotational velocity due to the vorticity of the flow as $\omega = \partial u(y,t)/\partial y$, and where fluid ve-

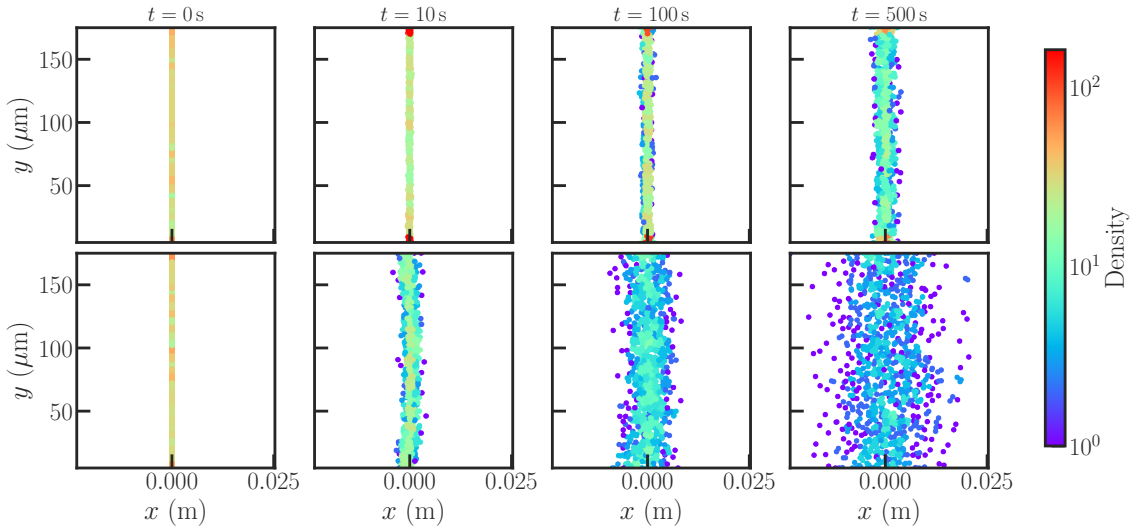

Figure 5.8: Snapshots of 100 numerically generated trajectories obtained from discretising Eqs. 5.20 with an Euler scheme with $n_{\mathrm{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\mathrm{s}$, $T = 2\,\mathrm{s}$, $k_{\mathrm{B}}\Theta = 4.14 \times 10^{-19}\,\mathrm{J}$, $v = 100\,\mu\mathrm{m.s}^{-1}\,\mathrm{m.s}^{-1}$, $D_{\mathrm{R}} = 3.35\,\mathrm{s}^{-1}$, $a = 5\,\mu\mathrm{m}$ for the active case without a flow (top), where $\Delta P = 0\,\mathrm{Pa}$, and with a flow (bottom), where $\Delta P = 73.0\,\mathrm{Pa}$. The simulations do not account for interactions. The colormap corresponds to the local density of ABPs in a binned (x, y) space.

locity is expressed, strictly equivalently to Eq. 5.16, as $\boldsymbol{u}(y,t) = 1/(2\eta_0)\,\Delta P/L\,y(H - y)\sin(2\pi t/T)\,\boldsymbol{e_x}$.

I then disretise Eqs. 5.20 using a forward Euler scheme. The results of solving the discretised Eqs. 5.20 is shown in Figs. 5.8 and 5.9. By comparing a case where particles are active but in no advection, and a case where they move both due to their own activity and to the fluid flow, one notices a very noticeable enhancement in the dispersion (Fig. 5.8). Indeed, in the absence of flow, particles tend to quickly reach one boundary where they accumulate, and then take a noticeable amount of time (of the order of τ_{R}) to reorient away from the wall through their active rotational process. In the case where a Poiseuille flow exists, particles tend not to approach the wall as much. I interpret this as a direct result of the stronger shear close to the wall: a swimmer will only need a tiny "push" in its orientation to escape contact with the wall. Along the $x-$axis, we see that the trajectory of a swimmer presents a strong signature of the oscillations of the fluid flow

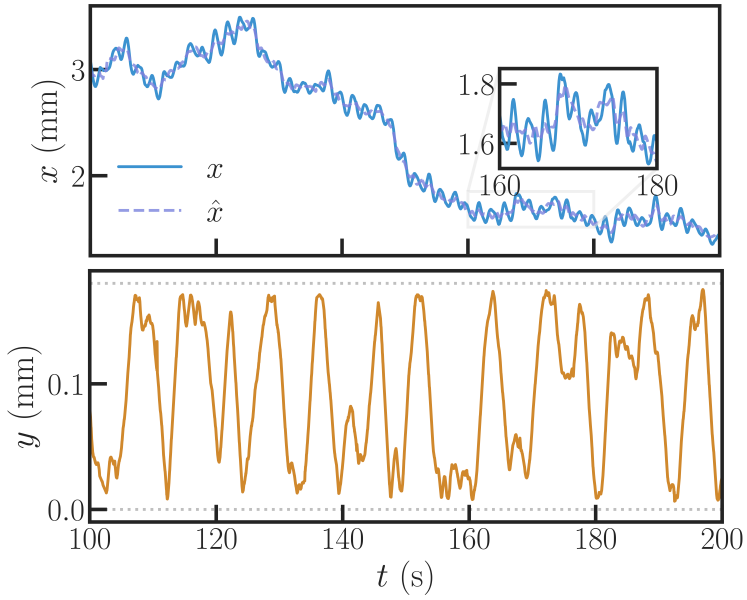

Figure 5.9: Numerical results obtained from discretising Eqs. 5.20 with an Euler scheme with $n_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}$ s, $T = 2$ s, $k_{\text{B}}\Theta = 4.14 \times 10^{-19}$ J, $v = 100\,\mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_{\text{R}} = 3.35\,\text{s}^{-1}$, $\Delta P = 73.0$ Pa, $a = 5\,\mu$m. Top: streamwise trajectory along the $x-$axis. The full blue curve shows the raw trajectory. The indigo dashed curve is the result of a demodulation of the signal obtained from a linear regression. The inset shows a zoom on times $t \in [160; 180]$ for better visualiation of the demodulation. Bottom: cross-streamwise trajectory along the $y-$axis (orange curve). The grey dotted lines represents the boundary of the channel, beyond which the ABP cannot wander.

(Fig. 5.9). To demodulate the signal, I use a different technique than the experimentalists on this study, with a lower computational cost: a simple linear regression, knowing that the parasitic drift has a known frequency. I feed to a linear regression model the smoothed $y-$ coordinate for each time step to avoid overfitting thermal noise. This is important, as we see in Eq. 5.16 that the amplitude of the sine wave depends on the $y-$position of the particle. Finally, the model takes the $x-$positions and returns a demodulated version, which we call $\hat{x}(t)$. Along the $y-$axis, I verify before conducting momentum analyses that the particle has had enough time to explore the whole channel cross-section fully, multiple times, a condition which ensures we are in the Taylor-Aris regime for passive Brownian colloids.

5.3.4 Results

We recall the Taylor-Aris dispersion formula from chapter 4, but keep in mind that the swimmer now has an effective diffusion coefficient at long times which is not D_0, the thermal diffusion coefficient, but D_{eff}, which we defined in the above sections as $D_{\text{eff}} = v^2/D_{\text{R}}$. The expected enhancement following the Taylor-Aris prediction thus reads:

$$D_{x,\text{TA}} = D_{\text{eff}}(1 + \alpha \text{Pe}^2) \ , \tag{5.21}$$

where α is a geometrical factor depending on the channel geometry (here, $\alpha = 1/210$) and Pe is the Péclet number which compares advection and diffusion, defined as:

$$\text{Pe} = \frac{\Delta P}{\sqrt{2}R_{\text{h}}} \frac{W}{D_{\text{eff}}} \ , \tag{5.22}$$

where the $\Delta P/\left(\sqrt{2}R_{\text{h}}\right)$ factor is the root-mean-square fluid velocity due to the oscillations of the pressure difference. To extract dispersion coefficients, we compute the dedridfted MSDs of multiple trajectories for a given set of parameters and average them. This is not the most computationally efficient method, but it is sufficient as we do not require a large amount of statistics for this study. We then repeat this process for multiple ΔP values, or equivalently, multiple Pe values. The results of these calculations is shown in Fig. 5.10. We already notice that every MSD follows the expected shape from Eq. 5.15, in a case where the thermal diffusion coefficient D_0 is too small to be visible at short times. The first visible regime is therefore the ballistic regime, where the particle has not had time yet to reorient. Beyond the typical reorientation time τ_{R}, the MSDs all take a diffusive behaviour. Furthermore, as the Péclet number grows, the MSDs retain their shape but are systematically higher, which is the hallmark of an enhanced dispersion coefficient due to fluid flow advection. We now turn to quantifying that enhancement of the dispersion. I present in Fig. 5.11 a typical plot in which dispersion coefficients are measured, from performing log-fits in the diffusive regime of the $\hat{x}-$wise MSD. In this treatment, I also retrieve the ballistic prefactors at short times, which correspond to the mean squared velocity of the particle in the $x-$ and $y-$ directions for later analyses. Finally, the treatment is repeated over a span of Péclet numbers. The enhancement factor on the diffusivity D_x/D_{eff}, where D_x is the dispersion coefficient extracted from numerical simulations, is plotted against Pe^2 in Fig. 5.12. We notice that the dispersion enhancement follows the trend set by Eq. 5.21, but may become smaller or larger by tuning the fluid oscillation period to lower or higher values respectively. The mean squared velocities (Fig. 5.13) on the other hand do not demonstrate a strong effect of the fluid oscillation period. This is expected as we chose our oscillation period much greater than τ_{R}, which means that in the ballistic regime, the particle never has time to feel the fluid oscillation before having reoriented. We also notice that $\langle V_y^2 \rangle$ remains unchanged

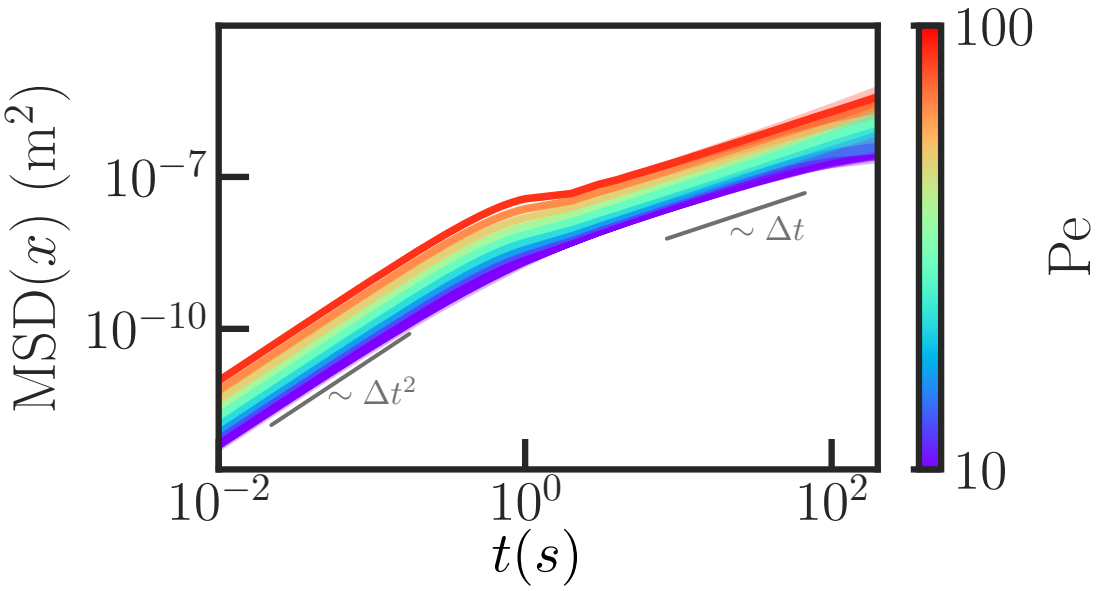

Figure 5.10: MSDs along the $x-$axis computed from demodulated numerical trajectories for various Péclet numbers, with $n_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}$ s, $T = 2$ s, $k_{\text{B}}\Theta = 4.14 \times 10^{-19}$ J, $v = 100\,\mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_{\text{R}} = 3.35\,\text{s}^{-1}$, $a = 5\,\mu$m. The colormap corresponds to the Péclet number of a given numerical experiment. The grey lines indicate a slope of 2 on the left and 1 on the right in log-scale, which correspond to the expected ballistic regime at short times and effective diffusive regime at long times of an ABP.

with Péclet number, whereas $\langle V_x^2 \rangle$ increases with Pe^2. This is consistent with speed composition, and we expect the ratio $\langle V_x^2 \rangle / \langle V_y^2 \rangle$ to follow:

$$\langle V_x^2 \rangle / \langle V_y^2 \rangle = 1 + \beta \frac{\text{Pe}^2}{{\text{Pe}_\text{A}}^2} \ , \tag{5.23}$$

where β is a geometric factor, and Pe_A is the zero-flow, active Péclet number [127, 128, 129, 130] such that $\text{Pe}_\text{A} = (W/D_{\text{eff}})\sqrt{\langle V_y \rangle^2} \simeq 3.2$.

5.4 Take-home message

In this chapter, I have investigated the dispersion of active microswimmers in confined viscous flows, using *Chlamydomonas reinhardtii* as a model swimmer. The aim was to test whether a classical theory for passive transport, namely the Taylor-Aris dispersion, can be extended to self-propelled particles whose motion is persistent and orientation variable. For this, microfluidic experiments were combined with single-cell tracking, statistical analysis, numerical Langevin simulations, and theoretical modeling.

In agreement with previous observations on swimming microorganisms, the cells were found to accumulate near the channel walls. In the direction along the flow, the MSD dis-

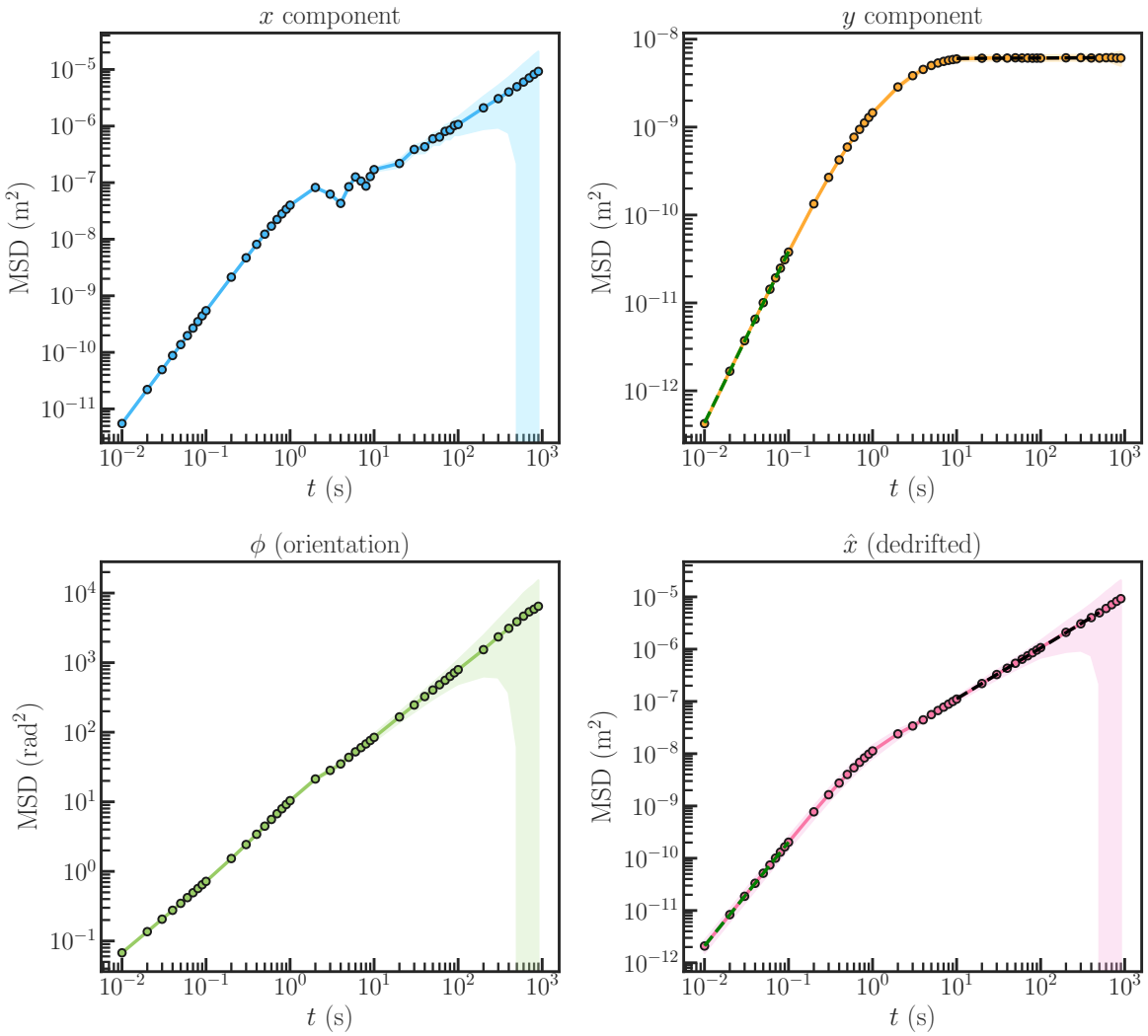

Figure 5.11: Typical MSDs computed from 100 stacked trajectories and averaged, with $n_{\text{steps}} = 10^6$, $\Delta t = 1 \times 10^{-3}\,\text{s}$, $T = 4\,\text{s}$, $k_\text{B}\Theta = 4.14 \times 10^{-19}\,\text{J}$, $v = 100\,\mu\text{m.s}^{-1}\,\text{m.s}^{-1}$, $D_\text{R} = 3.35\,\text{s}^{-1}$, $a = 5\,\mu\text{m}$, and $\Delta P = 9.9\,\text{Pa}$. The blue curve corresponds to the MSD along the $x-$axis, and presents a strong signature of the flow's oscillation. The orange curve corresponds to the $y-$axis. The light green curve corresponds to the angular MSD. Finally, the pink curve is the demodulated MSD on the $x-$axis. the green and black dash-dotted lines indicate the windows of the affine fits performed to retrieve the diffusion coefficient and mean squared velocity.

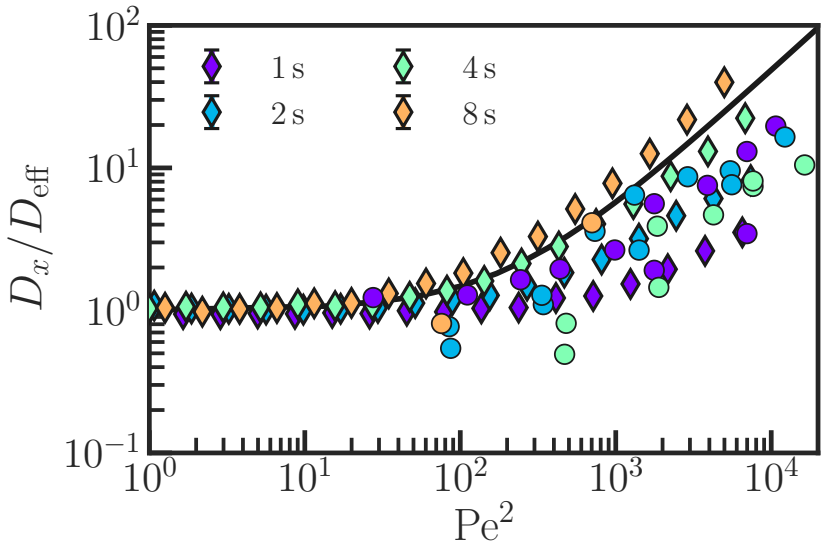

Figure 5.12: Comparison of experimental and numerical renormalised dispersion coefficients D_x/D_{eff} against Pe^2. Diamonds are the numerical simulation results, whereas circles correspond to the experimental data. The black curve shows Eq. 5.21. Colours correspond to various fluid oscillation periods.

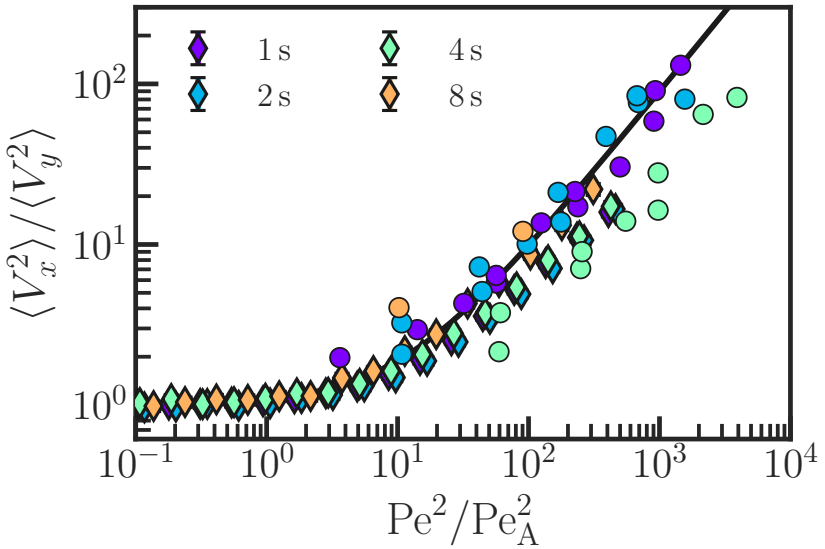

Figure 5.13: Comparison of experimental and numerical measurements of the ratio $\langle V_x^2\rangle/\langle V_y^2\rangle$ against Péclet number squared over active Péclet number squared. Diamonds are the numerical simulation results, whereas circles correspond to the experimental data. Colours correspond to various fluid oscillation periods. The black line is Eq. 5.23.

played a clear crossover from a short-time ballistic regime, dominated by persistent swimming and flow advection effects, to a long-time diffusive-like regime. The latter regime was clearly influenced by the amplitude and period of the imposed flow. A stronger fluid flow advection velocity led to a larger dispersion coefficient, whereas smaller periods of oscillation reduced the dispersion coefficient. These observations were quantified by a minimal theoretical description and by ABP Langevin simulations. Despite the complexity of the living system, the main behaviours could be captured by an effective active-particle description.

Several questions remain open. First, *Chlamydomonas reinhardtii* is a puller-type swimmer, and it would be interesting to test whether the same active Taylor-Aris mechanism persists, or is modified and how, for pusher-type microorganisms such as *E. coli*. Second, the present work focused on unbiased swimming in imposed flows. Incorporating external stimuli such as light, chemical gradients, or electromagnetic fields in general could reveal how taxis modifies active dispersion. Stronger confinement would require a more detailed treatment of near-wall hydrodynamic and lubrication effects, as well as possible transitions between swimming, adhesion, and gliding. Finally, the oscillatory flows used here were kept in a quasi-steady regime to obtain sufficient statistical resolution. Exploring higher oscillation frequencies, especially beyond the natural orientational decorrelation rate τ_{R} of the swimmers, could reveal new couplings which are relevant to biological settings.

Chapter 6

Concluding remarks

This thesis has presented the numerical study of Brownian motion in complex environments. Various cases have been considered, from the simple case of free Brownian motion, to rigid confinement and dense suspensions, soft and fluctuating confinement, fluid flow advection on a Brownian solute, and active Brownian motion in quiescent fluids and in fluid flows. These various cases were studied using a combination of analytical and numerical methods, with the latter being the main focus and tool. Three publications were produced, and two were presented in their respective chapters. Furthermore, collaborations regarding chapter 2 on hydrodynamic interactions in confined geometries are still ongoing, and a conclusion should be reached in the near future. A similar collaboration involving slip boundary conditions effects on the diffusivity of packed suspensions is also in preparation at the time of writing this manuscript.

Beyond the superimposition of hydrodynamic contributions

Diffusion is anisotropically and inhomogeneously modifies by confinement. This, in turn, modifies the statistical properties of Brownian displacements [74]. Furthermore, the presence of other colloids in the vicinity of a colloid of interest perturbs the fluid flow and modifies the diffusion of that colloid in a hydrodynamic coupling. In chapter 2, we have seen that the superimposition of hydrodynamic contributions from the confining walls is not sufficient to describe the dynamics of a Brownian particle in a confined geometry. The superimposition of hydrodynamic contributions, namely from Oseen [38] and Blake [42] approximations, is a common approximation used in the literature [30]. However, we have shown that this is not a satisfying theory for any system, and that it leads to significant errors in the prediction of the dynamics of a confined geometry where the colloid is both very close to confinement and to another colloid. The issue is that the superimposition of hydrodynamic contributions does not take into account the coupling between the two walls and the colloid, which is a three-body problem.

By using an image system, it becomes possible to reach higher orders of accuracy. This has been shown experimentally and theoretically already, and we reproduce these results numerically. However, this remains an approximation, and the question of the real dy-

namics in highly confined regimes remains open. We can reach higher fidelity by using asymptotic corrections to the image system. Still, the full three-body problem cannot be solved analytically, which is another argument for the use of numerical methods with good corrections and high computational efficiency. An experimental investigation is currently being carried out in an effort to strengthen the preliminary results in this chapter, and we are exploring more regimes and other numerical methods for comparison and efficiency purposes.

Softness can enhance diffusion

Confinement is not always rigid, and some studies, mostly experimental [69] but also theoretical and numerical [131], have demonstrated various new effects due to the deformability of either the confinement or the colloid itself. In chapter 3, we have seen that the softness of the confining walls can enhance the diffusion of a nearby trapped Brownian particle. Due to the fact that the soft walls can fluctuate, these fluctuations can lead to an effective diffusion coefficient that is larger than the one predicted by the confinement-free case. This has been shown both analytically and numerically. Our model is, however, largely minimal. Other studies based upon our findings should provide more insight on the effect of softness on diffusion, namely in more realistic, three-dimensional geometries with distance-dependent hydrodynamic interactions, or with higher dimensionality. A study involving inertia is also mandated, as our model is in the overdamped limit, and could reveal interesting added-mass effects.

Microswimmers disperse like Brownian particles

Extremely simple mathematical models can suffice to describe the dynamics of a suspension of microswimmers [96, 95, 97]. In Chapter 4, we describe the behaviour of a plain Brownian solute in advection, but these findings are quickly relevant in chapter 5, where we consider a similar dilute suspension of *Chlamydomonas reinhardtii* (Fig. 6.1). We find that the dispersion of such an active solution is predicted with the same laws as a passive solute [80, 82, 81, 83], with the only difference being the effective diffusivity, which results from a higher effective temperature in the case of the microswimmers. Thought not new conceptually, as active Brownian particle models are largely used to describe the behaviours of microswimmers, we still validate numerically what is observed in experiments and predicted by the passive the-

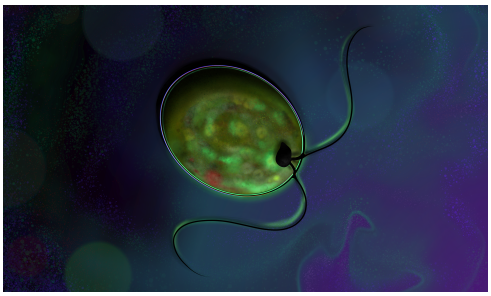

Figure 6.1: Artist's rendition of a *Chlamydomonas reinhardtii* alga swimming. Credits to Elodie Millan.

ory. This is extremely encouraging and invites us to explore more complex microswimmer behaviours. Indeed, one may wonder whether this validation stands for the case of a three-dimensional system rather than a quasi-two-dimensional one, where the existence of equilibrium points along the axis of symmetry of a cylindrical channel constitutes a major difference for puller microswimmers. One may also wonder what happens for organisms such as *E. coli*, a canonical example of a pusher microswimmer, where we know already that the far-field flow profile is quite different from that of a puller.

It is also common for swimmers to move in denser suspensions, with higher and more complex confinement effects from the wall and the presence of other swimmers. In that regard, it would be interesting to work with another model involving finite-size effects, and to fully resolve the fluid numerically and through Particle Image Velocimetry methods experimentally. Furthermore, this case would allow me to couple every bit of knowledge I have accumulated in this PhD, as it would involve the study of hydrodynamic interactions in confined geometries, the effects of softness on diffusion, and the behaviour of microswimmers in a dense suspension. With full control over each of these ingredients, I am certain we could reach a better global understanding of swimmer dynamics in complex environments, and tend to a more general theory of microswimmer dynamics in biological instances.

Chapter 7

Appendix: Publications

Characterising anomalous diffusion with a state-space deep-learning model

Hydrodynamic couplings between a colloid trapped in a periodic potential and a soft and fluctuating membrane enhance long-time diffusion

Active particles advected by viscous fluid flows undergo Taylor-Aris dispersion

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Journal of Physics: Photonics

PAPER

Bidirectional Mamba state-space model for anomalous diffusion

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Abstract

Characterizing anomalous diffusion (AnDi) is crucial in order to understand the evolution of complex stochastic systems, from molecular interactions to cellular dynamics. Here, we evaluate the performances regarding such a task of Bi-Mamba, a novel state-space deep-learning architecture articulated around a bidirectional scanning mechanism. Our implementation is tested on the AnDi-2 challenge datasets. Designed for regression tasks, the Bi-Mamba architecture infers efficiently the effective diffusion coefficient and anomalous exponent from single, short trajectories. As such, our results indicate the potential practical use of the Bi-Mamba architecture towards the characterization of AnDi.

1. Introduction

Anomalous diffusion (AnDi) [1] is ubiquitous in complex stochastic processes, and central in particular to the cell machinery, as classically shown by microrheology [2]. Its understanding provides insights on the transport of microparticles in heterogeneous media, such as molecules in biological cells [3, 4]. Methods such as microscopy combined with single-particle tracking allow for the detailed spatiotemporal analysis of single events [5, 6]. Over the last few decades, single-particle imaging methods have been steadily upgrading, increasing the amount of experimental data available on molecular interactions, cellular dynamics, microparticle behaviors [7, 8], and nanoparticles [9] in various complex media. Other approaches, such as dynamic light scattering [10, 11] and differential dynamic microscopy [12] have provided valuable insights on dense particle suspensions.

However, in most biological cases, trajectories are typically scarce, short and noisy. As such, it is often hard, if not impossible, to infer meaningful information using traditional tracking methods. Deep-learning methods for advanced microscopy have thus emerged as a promising change of paradigm [13]. As such, the work we present here results from the AnDi challenge [14], and in particular from our participation to its second edition [15]. We took a step aside from the standard methods used by the other participants in order to improve the collective knowledge on advanced artificial-intelligence methods for complex particle tracking. Our results indicate that the novel Bi-Mamba model infers efficiently the effective diffusion coefficient and anomalous exponent from single, short and noisy trajectories.

In a typical two-dimensional anomalous-diffusion process, the mean-squared displacement (MSD) is determined over time t by:

$$\mathrm{MSD}\,(t) = 4Kt^{\alpha}\,, \tag{1}$$

where K is the effective diffusion coefficient and α is the AnDi exponent. For Brownian particles, $\alpha = 1$. Besides, a particle is in a sub-diffusive regime when $0 < \alpha < 1$, and in a super-diffusive regime when $1 < \alpha < 2$. In figure 1, typical samples of these different diffusion types are shown with their respective trajectories and MSDs. For example, in complex cell environments a particle can exhibit AnDi due to:

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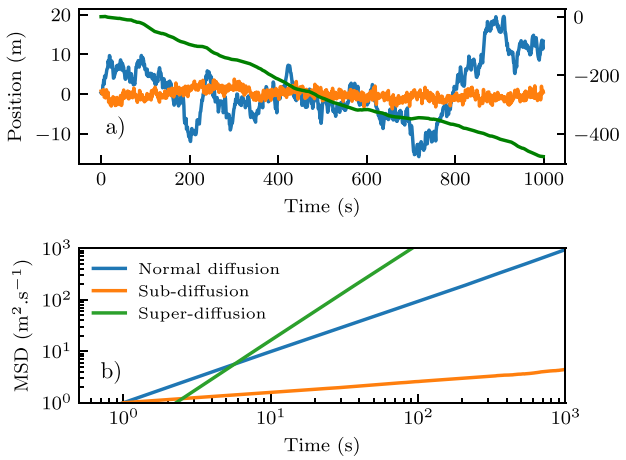

Figure 1. (a) Typical trajectories numerically simulated for a Brownian particle ($\alpha = 1$), a sub-diffusive particle with $\alpha = 0.2$, and a super-diffusive particle with $\alpha = 1.98$. (b) Corresponding mean-squared displacements of the trajectories (see equation (1)).

- *Obstacles and crowding:* in the presence of obstacles, diffusion can be hindered, leading to a sub-diffusive behavior. Moderate concentrations of obstacles can cause AnDi over short distances, and the effect becomes more pronounced as the obstacle concentration approaches the percolation threshold [16];
- *Binding and trapping:* temporary binding of particles to fixed sites [17] leads to a sub-diffusive behavior;
- *Lipid rafts and membrane heterogeneity:* interactions of particles with lipid rafts in cell membranes have been shown to lead to complex diffusive behaviors [18].

The first AnDi Challenge took place in 2020 and aimed at assessing the performances of various methods in quantifying AnDi [14]. The main focus was dedicated to the inference of K and α, for various simulations of bio-mimetic cases. The second AnDi challenge took place in 2024 and aimed at evaluating the performances of various methods for detecting and quantifying changes in single-particle motion [15]. As such, the focus was not only on the inference of the effective diffusion coefficient and exponent, but also on trajectory segmentation, with the trajectory statistics being described by a maximum of two K and two α values. One typical goal is for example to measure at what rate a particle binds and unbinds to/from a given cellular site. The AnDi datasets involve five phenomenological models:

- *Single-state:* particles diffusing according to a single diffusion state, as observed for some lipids in the plasma membrane [19–21];
- *Multi-state:* particles diffusing according to two diffusion states, and undergoing transient changes of K and α, as observed for proteins due to allosteric changes or ligand binding [22–25];
- *Dimerization:* particles diffusing according to two diffusion states, and undergoing transient changes of K and α, induced by encountering other diffusing particles, as observed in protein dimerization and protein-protein interactions [26, 27];
- *Transient confinement:* particles undergoing transient diffusion changes when entering or leaving given areas, as observed in the confinement induced by clathrin-coated pits on cell membranes [28];
- *Quenched trap:* two-state model of diffusion representing proteins being transiently immobilized at specific locations, induced by binding to immobile structures, as observed in cytoskeleton-induced molecular pinning [29, 30].

2. Bidirectional state-space model

Deep-learning models for inference from time series are generally based on recurrent neural networks (RNNs) or convolutional neural networks (CNNs). RNNs process the data sequence by updating a hidden state after each element. However, as the gradient also needs to go through each element one by one, long-term information may be lost. CNNs use convolutional kernels to give more weight to local information, as they do not need to process the sequence one by one. CNNs benefit from parallel computing and faster training speed but also limit the retrieval of long-term global information.

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Algorithm 1. Bi-Mamba.

Input: $x \in \mathbb{R}^{(B,L,D)}$
Output: $y \in \mathbb{R}^{(B,L,D)}$
$x \leftarrow \text{Norm}(x)$;
for $i \in \{forward, backward\}$ **do**
  if $i = forward$ **then**
    $x_i \leftarrow x$;
  else
    $x_i \leftarrow \text{Flip}(x)$;
  end
  $x_i \leftarrow \text{MambaBlock}(x_i)$;
  if $i = forward$ **then**
    $x_i \leftarrow \text{Flip}(x_i)$;
  end
  $x_i \leftarrow \text{Norm}(x_i + x)$;
end
$y_M = x_{\text{forward}} + x_{\text{backward}}$;
$y \leftarrow \text{FeedForward}(y_M)$;
$y \leftarrow \text{Norm}(y + y_M)$;
return y;

The Mamba architecture [31] is a state-space model designed to overcome the limitations of traditional deep-learning models like RNNs and CNNs by handling long-term dependencies with a high training efficiency. The key innovations of the Mamba architecture include a selection mechanism and hardware optimization for GPU usage. The Mamba architecture is essentially a reformulation of RNNs and CNNs as selective state-space models. The novelties brought by this architecture consist in:

- *A selection mechanism:* the selection mechanism in the Mamba model enables it to dynamically focus on relevant information and ignore irrelevant data. This mechanism is akin to the attention mechanism used in Transformer models. It allows the model to adjust its focus based on the input data, enhancing its ability to capture important features and dependencies;
- *Hardware awareness:* the Mamba architecture is optimized for GPU usage, which allows it to scale linearly with sequence length. This optimization ensures that the model can handle long sequences efficiently and benefit from parallel computing, resulting in faster training.

The Mamba architecture was originally developed for use in large language models (LLMs). However, recent studies in the literature have demonstrated its capability to perform segmentation on 3D medical images, showcasing its versatility and robustness in handling complex data structures. Therefore, the present study aims at determining whether a Mamba-based model can be effectively applied to complex time-series datasets, such as anomalous Brownian trajectories. The approach we chose was to evaluate the performances of Bi-Mamba, i.e. the novel state-space deep-learning architecture (Mamba) articulated with a bidirectional scan mechanism. Indeed, to the best of our knowledge, the Mamba architecture has not been used so far in the case of physical studies, and AnDi data in particular.

2.1. The Bi-Mamba block

Originally, the Mamba architecture has been designed to process 1D data sequence, in one direction. Considering the complex patterns that can occur in anomalous Brownian trajectories, we have implemented a bidirectional version of the Mamba block. As described in the algorithm 1, the block is made as an encoder and takes as an input $x \in \mathbb{R}^{(B,L,D)}$ with B the number of trajectories to be analyzed in parallel, L the length of the trajectories and D the number of features extracted from the trajectories. The output of the Bi-Mamba block is $y \in \mathbb{R}^{(B,L,D)}$. The Bi-Mamba block is composed of two Mamba blocks, one for the forward pass and the other for the backward pass. The forward Mamba block processes the input data in the forward direction, while the backward Mamba block processes the input data in the reverse direction. We denote the time-reversal operation as Flip, and the variables of each side as x_{forward} and x_{backward}, respectively. The outputs of the two Mamba blocks are then added and passed to a feedforward layer for further processing. To reduce the risk of vanishing gradients, residual connection between the input and the output of each step is applied [32].

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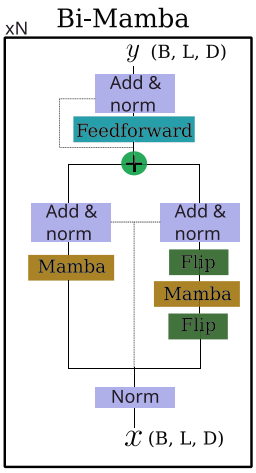

Figure 2. Bi-Mamba block implementation. Norm corresponds to the normalization layer, Flip corresponds to the time-reversal operation, and dashed lines correspond to residual connections.

Algorithm 2. Our Model—MambaBrown.

Input: $x \in \mathbb{R}^{(B,L,D)}$
Output: (y_S, y_K, y_α)
$y_S \leftarrow$ BiMambaBlock(x);
$y_S \leftarrow$ Dense(y_S); // $(B, L, 2)$
$y_K \leftarrow$ BiMambaBlock(x, y_S);
$y_K \leftarrow$ Dense(y_K); // $(B, 2)$
$y_K \leftarrow$ ReLu(y_K);
$y_\alpha \leftarrow$ BiMambaBlock(x, y_S);
$y_\alpha \leftarrow$ Dense(y_α); // $(B, 2)$
$y_\alpha \leftarrow$ Sigmoid(y_α);
return (y_S, y_K, y_α);

2.2. Our model—MambaBrown

To achieve the different tasks of the AnDi-2 challenge, we have implemented a model using the Bi-Mamba block, as shown in figure 2. The input data is first processed by a Bi-Mamba block for the segmentation of the diffusion state of the trajectory. As the trajectories are simulated using two different diffusion states, the segmentation block output is then processed by a dense layer to reduce the output dimensionality to $(B, L, 2)$, with each dimension corresponding to the probability of the trajectory to be in one of the two states. Then, the original features and the segmentation result are passed into two separate blocks, one for the K regression and the other for the α regression. Therefore, each task is handled by a dedicated block, allowing each block to specialize in a single function. Each of the outputs of the regression blocks is then processed by a dense layer to reduce the dimensionality of the output to $(B, 2)$, with each dimension corresponding to the predicted value of K and α respectively. The output of the α-regression block is passed to a sigmoid activation function and multiplied by 2 to ensure that the output is between 0 and 2. The output for the K-regression block is passed to a ReLu activation function to ensure that the output is positive.

3. Training and results

The data ensemble consists of sets of two-dimensional trajectories with coordinates $r(t) = [r_x(t), r_y(t)]$, with a maximum length of 200 time steps. The features computed using given trajectories are: the displacements for the first lag time $\Delta t = 1$, $\Delta r_i\,(\Delta t = 1) = r_i(t + \Delta t) - r_i(t)$, with $i \in \{x, y\}$, the one-dimensional MSDs $\langle \Delta {r_i}^2 \rangle (\Delta t)$, the displacement angle between two consecutive displacements [33], and the total displacement from the origin $d = \sqrt{x(t)^2 + y(t)^2}$. Trajectories smaller than 200 data points are zero-padded. The loss is then calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. The α loss is evaluated using mean absolute error (MAE), the K loss is evaluated using mean-squared

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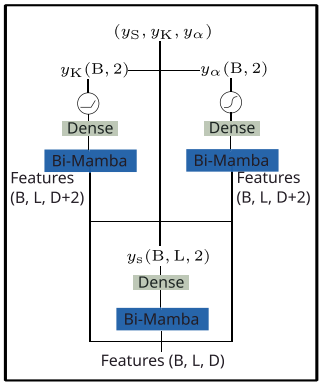

Figure 3. MambaBrown model implementation. Dense corresponds to the dense layer reducing the dimensionality of the Bi-Mamba block output. Circles correspond to the activation functions ReLu and Sigmoid, on the left and right respectively.

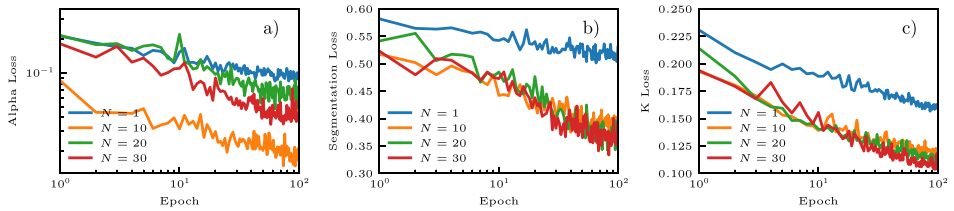

Figure 4. Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for different depths. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for α inference, (b) the Mean-Squared Logarithmic Error (MSLE) loss for K inference, and (c) the Weighted Cross-Entropy (WCE) loss for diffusion-state segmentation.

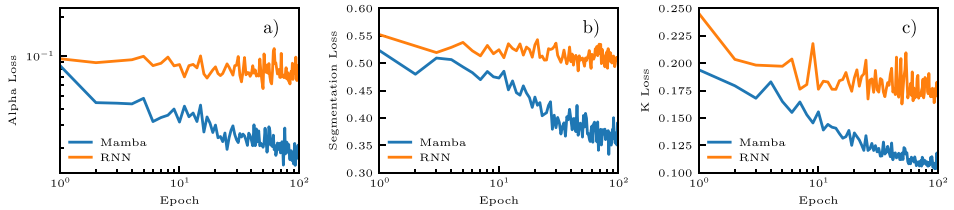

Figure 5. Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for our Mamba-based and RNN-based models. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for α inference, (b) the mean-squared logarithmic error (MSLE) loss for K inference, and (c) the weighted cross-entropy (WCE) loss for diffusion-state segmentation.

logarithmic error (MSLE), and the segmentation loss is evaluated using the weighted cross-entropy error (WCE). The training is carried out on a subset of 200 000 trajectories generated using the AnDi-2 challenge simulation tools. The loss is calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. Segmentation is evaluated using the WCE loss, while the K regression is evaluated using the MSLE loss, and the α regression is evaluated using the MAE loss. In figure 4, we evaluate the loss as a function of the depth N of the different blocks, from which we determine our best model to have a depth of $N = 10$ for α regression and $N = 30$ for segmentation and K regression.

For fair benchmarking purposes, we compare our model (see figure 3) to a similar architecture but using bidirectional RNN instead of our Bi-Mamba block (see figure 2). As shown in figure 5, the Bi-Mamba model shows promising performances, scoring better overall in each category than a RNN-based model. Moreover, it shows its capabilities to train for a larger number of epochs without overfitting and with a smaller loss variance.

4. Conclusion

In this work, we implemented the structure and evaluated the performances of the Mamba architecture—a novel application of state-space models—towards the characterization of AnDi. Using a bidirectional scan mechanism, we demonstrated a notable efficiency in diffusion-state segmentation, as well as for effective-diffusion-coefficient and anomalous-exponent regression tasks. Improvement could be achieved through the use of specialized models, fine-tuned using simulations generated near the predictions of the generally-trained model presented in this work. Our contribution to the AnDi-2 challenge was designed in order to specifically evaluate the performances of Bi-Mamba models for AnDi. Our first model developed during the competition phase ranked 7th in α inference (MAE of 0.27), 9th in K inference (MSLE of 0.05), 3rd in diffusion-type measurement [i.e. trapped with ($\alpha < 0.2$), directed with $\alpha > 1.8$, or normal with $\alpha = 1$ and with a F1 error of 0.91], and 10th on the change-point detection (root-mean-squared error of 2.7). Moreover, we expect improvements on the segmentation task by using Mamba-based Unet methods [34]. All together, this work contributes to the knowledge on, and continued development of, deep-learning methods for AnDi and physics in general.

Data availability statement

The data that support the findings of this study are openly available at the following URL: https://github.com/EMetBrown-Lab/Mamba-EMetBrown-ANDI2.

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Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode

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In many physical systems, degrees of freedom are coupled *via* hydrodynamic forces, even in the absence of Hamiltonian interactions. A particularly important and widespread example concerns the transport of microscopic particles in fluids near deformable boundaries. In such a situation, the influence of elastohydrodynamic couplings on Brownian motion remains to be understood. Unfortunately, the temporal and spatial scales associated with the thermal fluctuations of usual surfaces are often so small that their deformations are difficult to monitor experimentally, together with the much slower and larger particle motion at stake. Here, we propose a minimal model describing the hydrodynamic coupling of a colloidal particle to a fluctuating elastic mode, in presence of an external periodic potential. We demonstrate that the late-time diffusion coefficient of the particle increases with the compliance of the elastic mode. Our results reveal and quantify two features: first, spontaneous microscopic transport in complex environments can be affected by soft boundaries – a situation with numerous practical implications in nanoscale and biological physics; and second, the effects of fast and tiny surface deformations are imprinted in long-term and large-distance colloidal mobility, and are therefore measurable in practice.

Brownian motion refers to the random movement of a colloid at equilibrium, induced by its collisions with the molecules of the surrounding fluid. It has played a foundational role in modern statistical physics since its discovery[1] and theoretical description[2,3] over a century ago. The celebrated experiments of Perrin on colloids in a gravitational field[4] validated further the theoretical description and confirmed the atomic nature of matter. Extending the classical framework of Brownian motion in bulk Newtonian fluids to more realistic systems, such as particles diffusing in complex environments, opens up new challenges for nanoscale and biological physics. One common route is to study Brownian motion in viscoelastic fluids or intracellular media, as typically performed through microrheology[5], which revealed the emergence of anomalous diffusion[6]. Another source of complexity lies in the introduction of topological constraints and boundaries. For instance, a simple-yet-canonical situation where interfaces matter is the diffusion near rigid walls, that has been extensively studied both theoretically and experimentally[7–15]. Besides the practical interest of such a configuration towards the sensing of surface forces[16], one of the key aspects lies perhaps in the apparition of anisotropic and space-dependent mobilities[17] due to the frictional boundary condition at the walls. As a consequence, multiplicative noises emerge, leading to non-Gaussian diffusion and the probability enhancement of rare events[18–22].

However, in most practical situations where molecules or microorganisms diffuse near surfaces, the latter are typically soft. Therefore, the dynamical properties of Brownian particles near deformable boundaries must be understood. Along this line of thought, a few studies, mostly theoretical, have addressed the case of tracers diffusing near soft surfaces. Examples include liquid interfaces[23,24], and fluctuating ones[25]. The case of elastic interfaces was also addressed in the contexts of contactless microrheology[26] and biological membranes[27]. Nevertheless, these works focused on the low-coupling regime, where the particles are distant or point-like, and thus unable to generate rectified flows that lead to surface deformation. Interestingly, though, within a purely deterministic context, near-contact elastohydrodynamic couplings have been studied in the past decades. Examples include contactless colloidal-probe rheology of soft substrates in normal mode[28,29], as well as the emergence of lift forces at low Reynolds numbers[30,31], that were first revealed theoretically[32–36], and then confirmed experimentally[37–41]. The leading-order corrections induced by elasticity to the full mobility tensor near a rigid wall[7] were obtained for various geometries and rheologies[42–44], and revealed a zoology of counter-intuitive inertial-like solutions.

In view of the above state of the art, the arising challenge is now to understand the influence of soft boundaries on the spontaneous motion of

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nearby finite-sized Brownian particles. Molecular-dynamics simulations were performed in such a context[45]. Besides, a recent experimental study indicated that soft colloids diffusing near rigid walls feel transient, non-conservative forces that could be related to an intricate combination of thermal fluctuations, soft boundaries and viscous flows[46]. Finally, at a practical level, a natural question to ask is whether the effects of fast and tiny surface fluctuations can be observed over the long times and large distances associated with colloidal diffusion.

Here, we introduce a minimal, but generic, theoretical model in order to investigate Brownian elastohydrodynamic effects. Specifically, we combine the hydrodynamic coupling of a colloidal particle with an elastic mode[47,48] to an external periodic trapping[49–52]. We solve the problem both analytically and numerically, and demonstrate that the late-time diffusion coefficient of the particle increases notably with the compliance of the elastic mode.

Results

Model

As depicted in Fig. 1, we consider an overdamped system made of two coupled degrees of freedom: i) a principal degree of freedom $q_1(t)$, corresponding to the spatial position of a test colloid over time t, which is free to explore space while being subject to a periodic trapping potential $\phi(q_1)$ with spatial period λ (in practice, such trapping potential could represent a surface with patchy, adhesive patterns); and ii) an auxiliary degree of freedom $q_2(t)$, corresponding to an elastic mode, which is harmonically confined with a potential energy $q_2^2/(2\kappa)$, where κ is the compliance. The compliance, i.e., inverse stiffness, represents the softness parameter of the elastic mode.

Thus, the total energy of the system is given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa}. \tag{1}$$

At thermal equilibrium, the probability density function (PDF) of q_1 and q_2 is given by the Gibbs-Boltzmann distribution

$$P_{\rm eq}(q_1, q_2) = \frac{\exp\left[-\beta H(q_1, q_2)\right]}{Z}, \tag{2}$$

where Z is the normalizing partition function, and where $\beta = 1/(k_{\rm B}T)$ with T the temperature and $k_{\rm B}$ Boltzmann's constant. The equilibrium statistics of q_1 and q_2 are independent due to the additivity of the Hamiltonian, which implies, in particular, that q_2 does not affect the equilibrium distribution of q_1.

The system dynamics is described by the coupled overdamped Langevin equations

$$\gamma_{11}\frac{{\rm d}q_1}{{\rm d}t} + \gamma_{12}\frac{{\rm d}q_2}{{\rm d}t} = -\phi'(q_1) + \eta_1(t), \tag{3}$$

$$\gamma_{12}\frac{{\rm d}q_1}{{\rm d}t} + \gamma_{22}\frac{{\rm d}q_2}{{\rm d}t} = -\frac{q_2}{\kappa} + \eta_2(t), \tag{4}$$

where the constants γ_{ij}, with $i, j \in \{1, 2\}$, are the components of the friction tensor γ, which is related to the mobility tensor μ through $\mu = \gamma^{-1}$; and where the functions $\eta_i(t)$ are zero-mean Gaussian white noises, with a correlation function $\langle \eta_i(t)\eta_j(t')\rangle = 2k_{\rm B}T\delta(t-t')\gamma_{ij}$, where $\langle\cdot\rangle$ is the ensemble average. This choice of noise correlation is self-consistent with the equilibrium distribution of q_1 and q_2 being given by Eq. (2).

Theoretical analysis

Let us now define

$$\epsilon = \frac{\kappa k_{\rm B}T}{\lambda^2} \tag{5}$$

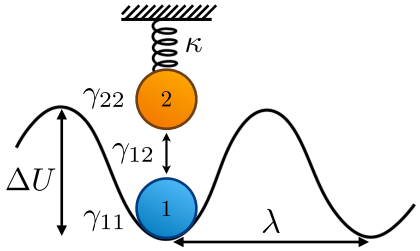

Fig. 1 | Schematic of the system. A test Brownian particle (1) is trapped in a periodic energy potential of amplitude ΔU and spatial period λ. It is coupled via a hydrodynamic-like interaction of constant damping coefficient γ_{12} to a thermally-fluctuating elastic mode, represented by another Brownian particle (2) trapped in a harmonic trap of compliance (i.e., inverse softness)κ. Besides, both particles feel individual hydrodynamic drag forces represented by their constant self-drag coefficients γ_{11} and γ_{22}, and are described within the overdamped Langevin regime.

the dimensionless compliance. In the weakly-compliant regime ($\epsilon \ll 1$), one can compute the late-time diffusion coefficient $D^*(\epsilon)$ of the colloid using a perturbation analysis at $\mathcal{O}(\epsilon)$, together with an exact representation of the effective diffusion coefficient obtained via the Stokes-Einstein relation, and a multiscale analysis[53,54]. Doing so (see details in SI Note 3), we obtain

$$D^*(\epsilon) \simeq D^*(0)(1+\alpha), \tag{6}$$

with

$$\alpha = \beta\epsilon\frac{\lambda^2\gamma_{12}^2}{\gamma_{11}^2}\frac{\int_0^\lambda {\rm d}q_1\, \phi'(q_1)^2\, \exp\left[\beta\phi(q_1)\right]}{\int_0^\lambda {\rm d}q_1\, \exp\left[\beta\phi(q_1)\right]}, \tag{7}$$

and

$$D^*(0) = \frac{k_{\rm B}T\lambda^2}{\gamma_{11}Z_+Z_-}, \tag{8}$$

where

$$Z_\pm = \int_0^\lambda {\rm d}q_1\, \exp\left[\pm\beta\phi(q_1)\right]. \tag{9}$$

Since $\alpha \geq 0$, the hydrodynamic coupling of the colloid to the elastic mode always increases the late-time diffusion coefficient of the colloid. Moreover, the effect is larger at larger compliance κ. We also see that a non-flat periodic potential $\phi(q_1)$ is required for the effect to exist, within the assumptions of the model. Such an influence of external potentials on hydrodynamic friction is reminiscent of a molecular-dynamics study in the literature[55].

In the particular case of a sinusoidal potential, where $\phi(q_1) = \Delta U \sin(2\pi q_1/\lambda)$, with ΔU the amplitude of the potential, we have

$$\alpha = 4\pi^2\epsilon\frac{\gamma_{12}^2}{\gamma_{11}^2}\frac{\nu I_1(\nu)}{I_0(\nu)}. \tag{10}$$

with $\nu = \beta\Delta U$, where I_n denotes the order-n modified Bessel function of the first kind.

As such, the relative increase of the late-time diffusion coefficient is larger for smaller λ. This is attributed to the fact that for a smaller λ the coupling to the elastic mode increases the jump rate of the colloid between neighboring minima, instead of being averaged out as would be the case at large λ. We also see that the relative increase of the late-time diffusion coefficient depends linearly on the function $\nu I_1(\nu)/I_0(\nu)$, which is a monotonically increasing function of ν. This means that the relative magnitude of the effect increases with the depth of the trap – the obvious counterpart

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being that larger times are then needed to reach the late-time diffusive regime. Finally, the relative magnitude of the effect increases with the relative magnitude γ_{12}/γ_{11} of the hydrodynamic coupling to the elastic mode, as expected.

The multiscale analysis can also be employed in the limit where $\epsilon \to \infty$, which corresponds to a very compliant elastic mode. In this case, we also find (see details in SI Note 4) that the late-time diffusion coefficient is increased relatively to the rigid-case value, and saturates to a maximum value at large ϵ, as

$$D^* = \frac{k_B T \mu_{11}(\mu_{11}\mu_{22} - \mu_{12}^2)}{\lambda^{-2} Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt{\epsilon}}\right). \quad (11)$$

Once again, we first note that when the periodic potential $\phi(q_1)$ is flat, the late-time diffusion coefficient is not modified by the hydrodynamic coupling to the elastic mode. However, interestingly, such a plateau value can be notably higher than the rigid-case value, indicating the potential strength of the effect in practical systems relevant to soft matter and biophysics. For instance, let us consider a spherical particle of radius $a = 100$ nm within a fluid of viscosity $\eta = 10^{-3}$ Pa.s and at temperature $T = 300$ K. The particle is assumed to be at a distance $d = 10$ nm from a soft tissue with a Young modulus $E = 1$ kPa, or equivalent stiffness $\kappa^{-1} \sim E\sqrt{ad}$, and is placed in a trapping potential of amplitude $4\,k_B T$ with spatial period $\lambda = 100$ nm. Taking $\mu_{11} = \mu_{22} = (6\pi\eta a)^{-1}$ for the self-mobilities, and Oseen's $\sim\mu_{12} = (4\pi\eta r)^{-1}$ for the coupling mobility, with $r = 2a + d$, one expects a 100% relative increase of the long-term diffusivity with respect to the classical Lifson-Jackson value.

Heuristic argument

In order to understand the softness-induced late-time diffusivity enhancement from a physical point of view, we now present a much simpler heuristic argument giving the same result as the rigorous analysis in the small-ϵ regime (see details in SI Note 2). The intrinsic time scale $\tau_2 = \kappa/\mu_{22}$ of the stochastic process $q_2(t)$ is assumed to be small with respect to all the other time scales. In this limit, the effective zero-temperature overdamped Langevin equation for $q_1(t)$ is

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_e(q_1)\phi'(q_1), \quad (12)$$

where we have introduced the dressed mobility

$$\mu_e(q_1) = \frac{1}{\gamma_{11}\left[1 + \frac{\kappa\gamma_{12}^2\phi''(q_1)}{\gamma_{11}^2}\right]}. \quad (13)$$

The equilibrium marginal PDF in position q_1 is $P_{\mathrm{meq}}(q_1) = (Z_-)^{-1}\exp[-\beta\phi(q_1)]$. Therefore, assuming that the time-dependent marginal PDF $P_1(q_1, t)$ satisfies an effective Fokker-Planck equation, the latter must have the form

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\left\{\mu_e(q_1)\left[k_B T\frac{\partial P_1(q_1,t)}{\partial q_1} + \phi'(q_1)P_1(q_1,t)\right]\right\}, \quad (14)$$

where the T-independent term on the right-hand side is determined from the zero-temperature dynamics, while the T-dependent term on the right-hand side is constructed to ensure convergence of P_1 to P_{meq} at infinite time. Note that we have implicitly assumed here that the dressed mobility $\mu_e(q_1)$ does not exhibit any temperature dependence.

The dressed diffusion coefficient is given by the Stokes-Einstein relation, $D_e(q_1) = k_B T\mu_e(q_1)$, which differs from the bare (or "microscopic") diffusion coefficient $D_m = k_B T\mu_{11}$ of the colloid. Interestingly, the spatial dependence of the dressed diffusion coefficient implies that the effective Langevin equation must have a multiplicative noise, leading to a supplementary source of non-Gaussian displacements[16,18–22,56] in addition to the trivial one induced by the trapping potential.

We now invoke the Lifson-Jackson formula[57], providing the late-time diffusion coefficient in a periodic potential, which reads

$$D^*(\epsilon) = \frac{\lambda^2}{\int_0^\lambda \mathrm{d}q_1 \frac{\exp[\beta\phi(q_1)]}{D_e(q_1)}\int_0^\lambda \mathrm{d}q_1 \exp[-\beta\phi(q_1)]}. \quad (15)$$

From this expression, we can recover Eqs. (6) and (7). We emphasize that the heuristic argument we presented here is not rigorous. Nevertheless, it can also be applied to evaluate the effective diffusion coefficient of an underdamped Brownian particle near the overdamped limit (see details in SI Note 6), for which we once again recover the results obtained via rigorous multiscale analysis[53,54]. The same method also yields the first-order correction to the effective diffusion coefficient for a particle carrying a dipole moment which interacts with a random or periodic electric field when the dipole dynamics is assumed to be much faster than the particle diffusion[58].

Comparison to numerical simulations

In order to verify the above predictions, Eqs. (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$ (see details in Methods). Typical raw trajectories are shown in Fig. 2a). We observe that the elastic mode $q_2(t)$ fluctuates around its equilibrium value ($q_2 = 0$), within the harmonic potential it is subject to, whereas the position $q_1(t)$ of the test colloid fluctuates at short times in a minimum of the sinusoidal potential $\phi(q_1)$ before randomly jumping from one minimum to the other by passing the potential barrier. From such trajectories, we can first compute the late-time PDF in positions. As an example, Fig. 2b) shows the equilibrium marginal PDF of the periodized variable q^*, defined by $q_1 = q^* \mathrm{mod}(\lambda)$, which satisfies

$$P_{\mathrm{meq}}(q^*) = \frac{\exp[-\beta\phi(q^*)]}{\int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)]}. \quad (16)$$

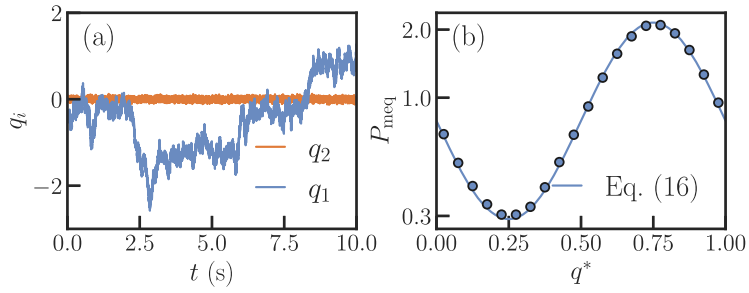

Fig. 2 | Trajectories and equilibrium distribution. **a** Typical trajectories $q_i(t)$ of the test particle ($i = 1$) and the elastic mode ($i = 2$), simulated by numerically solving Eqs. (3) and (4), for mobility matrix elements $\mu_{11} = 1.0$, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, with $\Delta U = 1.0$, $\lambda = 1.0$, $\kappa = 1.0 \times 10^{-3}$, time increment $\Delta t = 1.0 \times 10^{-4}$, and $k_B T = 1.0$. Simulations are initialized from equilibrium. **b** Late-time marginal PDF P_{meq} for the periodized position q^*, given by $q_1 = q^* \mathrm{mod}(\lambda)$, of the test particle, obtained by binning 10^2 trajectories of 10^7 points. The symbols are the results of the numerical simulations. The solid blue line corresponds to the Boltzmann distribution given in Eq. (16).

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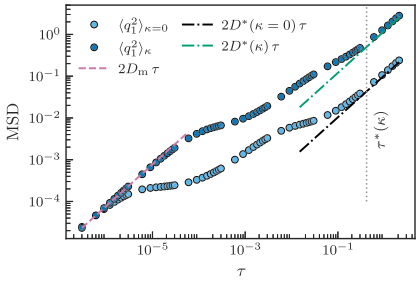

Fig. 3 | Effect of softness on the mean-squared displacement. Mean-squared displacements $\langle q_1^2\rangle_\kappa$ and $\langle q_1^2\rangle_0$ of the test particle as functions of the time increment τ, for $\kappa = 0.1$ (dark blue disks) and $\kappa = 0.0$ (light blue disks) respectively, for damping matrix elements $\gamma_{11} = 1.309$, $g_{12} = 1.296$, $\gamma_{22} = 1.309$, with $\Delta U = 2.7$, $\lambda = 0.5$, time increment $\Delta t = 3.0 \times 10^{-7}$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted black line indicates $2D^*(0)\tau$, where $D^*(0)$ is the classical Lifson-Jackson late-time diffusion coefficient[57]. The dash-dotted green line indicates $2D^*(\kappa)\tau$, where $D^*(\kappa)$ is the late-time diffusion coefficient given by Eq. (6). The dashed pink line indicates $2D_{\rm m}\tau$, where $D_{\rm m} = k_B T \mu_{11}$ is the bare diffusion coefficient. The characteristic time $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$ is indicated with the light-gray vertical dotted line.

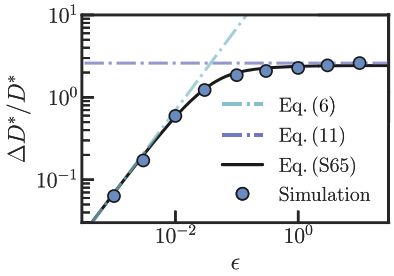

Fig. 4 | Effect of softness on diffusivity. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ, for $\mu_{11} = 1.0$, $\mu_{12} = 0.85$, $\mu_{22} = 1.0$, $\Delta U = 3.0$, $\lambda = 1.0$, and $k_B T = 1.0$. The symbols are the results of the numerical simulations. The dash-dotted sky blue line corresponds to the small-ϵ prediction of Eq. (6). The dash-dotted navy blue line corresponds to the large-ϵ prediction of Eq. (11). The solid black line corresponds to the Kubo formula of Eq. (S65) (see details in SI Note 5).

In order to probe the dynamical effects of the hydrodynamic coupling of the trapped test particle to the elastic mode, we now turn to the analysis of the mean-squared displacement (MSD), defined as

$$\langle q_i^2\rangle(\tau) = \langle [q_i(t+\tau) - q_i(t)]^2\rangle, \tag{17}$$

where $\langle . \rangle$ denotes the ensemble average, and τ is a time increment. As shown in Fig. 3, the MSD of the test colloidal particle exhibits two distinct diffusive regimes separated by a crossover region.

We compute from the MSD the short-time diffusive regime, characterized by the bare diffusion coefficient $D_{\rm m}$, and the late-time diffusive regime, which is characterized by the late-time diffusion coefficient D^*, and is reached when the particle has diffused over many periods of the trapping potential, i.e., when its MSD is large compared to λ^2. This sets a time scale $\tau^*(\kappa) = \lambda^2/D^*(\kappa)$, which must be exceeded to see the late-time diffusive regime. We extract the relative excess in late-time diffusion coefficient, i.e., $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$, as a function of ϵ, from log-log fits of the MSD, excluding: data below $5\tau^*$, which belongs to a different dynamical regime, as well as from the last decade, which lacks statistics for proper data processing. The results are shown in Fig. 4 for a given set of parameters. We observe that the numerical-simulation data agree well with the small-ϵ and large-ϵ asymptotic predictions of Eqs. (6) and (11), respectively. Furthermore, the Kubo formula can be used to evaluate D^* for any value of ϵ (see details in SI Note 5, and Eq. (S65)). We see that the numerical-simulation data agree at all ϵ values with the Kubo formula, and that the latter interpolates well the two asymptotic predictions.

Conclusions

We theoretically and numerically studied the Brownian motion of colloid-like particles near elastic interfaces using minimal ingredients. We showed that a Brownian particle trapped in a periodic potential sees its late-time diffusivity dressed and significantly enhanced by hydrodynamic coupling to a hidden elastic mode. Despite being minimalist, our description reveals generic, original and large effects, relevant for the transport of colloidal particles near complex boundaries. In particular, our results suggest the intriguing possibility of understanding, and even tuning, the diffusivity of colloids near deformable and patchy interfaces—with key implications for biophysics and nanofluidics.

Methods

Equations (3) and (4) are numerically solved for a sinusoidal trapping potential $\phi(q_1)$, using an Euler scheme, through a compiled *Cython*[59] code. The discretized Langevin equations read

$$q_1(t+\Delta t) - q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t - \frac{\mu_{12}}{\kappa} q_2(t)\Delta t + \xi_1\sqrt{2k_B T\Delta t}, \tag{18}$$

$$q_2(t+\Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa} q_2(t)\Delta t - \mu_{12}\phi'(q_1(t))\Delta t + \xi_2\sqrt{2k_B T\Delta t}, \tag{19}$$

where ξ_i represent zero-mean Gaussian random variables, characterized by the correlator

$$\langle \xi_i \xi_j\rangle = \mu_{ij}. \tag{20}$$

We carefully choose the time step Δt for numerical stability and accuracy, satisfying $\Delta t \le 10^{-2}\tau_2$. The mobility tensor μ_{ij} must be positive-definite. To enforce this, we write noises ξ_i as $\xi_i = m_{ij}\eta_j$, with η_j zero-mean, independent Gaussian random variables satisfying $\langle \eta_i \eta_j\rangle = \delta_{ij}$, with $\delta_{ij} = 1$ for $i = j$ and 0 otherwise. This yields a decomposition of the mobility tensor $\mu_{ij} = m_{ik}m_{kj}$. The matrix elements m_{ij} ensure that noise correlations are respected. For computational efficiency, the noise array is pre-generated using the `numpy.random.normal` function. The simulation recursively updates the system state at each discrete time step. Initial positions of both modes are sampled from Eq. (2) to ensure thermalization. Trajectories are computed iteratively, and their corresponding displacement PDFs are binned. Convergence is evaluated with the relative distance between iterations of the stacked displacement PDFs, with a criterion of 1% relative distance between the stacked histogram and new iterations, for more than three consecutive iterations.

Data availability

All data are available upon reasonable request to the corresponding authors. A Supplementary Information (SI) document provides details on: the analytical derivation, and numerical estimations.

Code availability

The simulation scripts are open-source and available on the GitHub repository https://github.com/EMetBrown-Lab/Lacherez2024_Codes. The repository provides a *Jupyter* notebook for single-trajectory simulations. For large-scale parameter sweeps, the `setup.py` script enables optimized

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pre-compilation using *Cython*, and the `automate_simulation.py` script supports systematic exploration of the parameter space. These tools are designed for efficient single-node execution and ensure reproducibility of all results presented in this study. We invite the reader to consult the README.md file. Additional treatment codes are available upon reasonable request to the authors.

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Author contributions

J.L. conducted numerical simulations, data analysis and drafted the manuscript. M.L. helped with the optimization of the numerical simulations and commented on the manuscript. Y.A. conceived the project, numerically computed the Kubo formula and commented on the manuscript. D.S.D. conceived the project, developed the analytical model and edited the manuscript. T.S. conceived the project, led the project, and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode – *Supplementary Information* –

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(Dated: October 14, 2025)

I. SUPPLEMENTARY NOTE 1: DESCRIPTION OF THE MODEL

We consider an overdamped system made up of two degrees of freedom, the principal degree of freedom q_1 which is free to explore a large region of space, for example subjected to a periodic potential $\phi(q_1)$, and the second degree of freedom q_2, highly constrained and taken to be harmonically confined with potential energy $\frac{1}{2\kappa}q_2^2$, where we assume that κ the spring compliance is small. The total energy of the system is thus given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{1}{2\kappa}q_2^2 \, . \tag{S1}$$

In thermal equilibrium, the steady state of the probability distribution of q_1 and q_2 is given by the Gibbs-Boltzmann distribution

$$P_{\rm eq}(q_1, q_2) = \frac{\exp\left[-\beta H(q_1, q_2)\right]}{Z} \, , \tag{S2}$$

where Z is the normalizing partition function and $\beta = 1/(k_{\rm B}T)$ is the inverse thermal energy, where T is the temperature and $k_{\rm B}$ is Boltzmann's constant. In equilibrium, the statistics of q_1 and q_2 are independent due to the additivity of the Hamiltonian H, *i.e.* $P_{eq}(q_1, q_2) = P_{1eq}(q_1)P_{2eq}(q_2)$. However, dynamically, they can be coupled; for instance this model can be seen as a toy model for a colloidal particle near an elastic compliant surface. As such, any single time observation in equilibrium of the measurable quantity q_1 will be independent of the effect of q_2 and hence will not show any effect due to the hydrodynamic coupling with q_2. This means that one would have to measure at least two time correlation functions; but, due to the rapid dynamics of q_2, one would have to resolve the dynamics on very short time scales to detect the hydrodynamic coupling.

The simplest form for the over-damped Langevin dynamics of this system is

$$\frac{{\rm d}q_1}{{\rm d}t} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa}q_2 + \eta_1(t) \tag{S3}$$

$$\frac{{\rm d}q_2}{{\rm d}t} = -\frac{\mu_{22}}{\kappa}q_2 - \mu_{12}\phi'(q_1) + \eta_2(t) \, , \tag{S4}$$

where the mobility coefficients μ_{ij} are taken for simplicity to be constant, and the noise correlation function is chosen to ensure that this Langevin system has a steady state distribution given by Eq. (S2), explicitly

$$\langle \eta_i(t)\eta_j(t)\rangle = 2k_{\rm B}T\delta(t-t')\mu_{ij} \, . \tag{S5}$$

From a mathematical standpoint, the constraints on the mobility matrix are that it is symmetric and has positive eigenvalues. However, from a physical point of view, in the absence of coupling, we must have

$$\mu_{11} > 0, \text{ and } \mu_{22} > 0 \, . \tag{S6}$$

The constraints on the eigenvalues of the mobility matrix then translate to

$$\det(\mu) = \mu_{11}\mu_{22} - \mu_{12}^2 > 0 \, . \tag{S7}$$

In an experimental context, it may be possible to directly observe the motion of the variable q_1, but not necessarily on very short time scales, while the direct observation of the second variable q_2 may be impossible for it cannot be

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resolved spatially nor temporally. If the variable q_1 is subject to a confining potential and a measurement is made of its equilibrium statistics, the variable q_1 has no effect whatsoever, and changing the value of κ for instance will not affect the measurements. Observing the dynamics of the variable q_1 may yield information about the variable q_2 but if the time scale over which q_2 equilibrates is too short, again this information is censored by being averaged out.

Here, we propose a solution to demonstrate the effect of the variable q_2 by examining its effect on the late time diffusion constant of q_1 when it is subjected to a periodic potential. Given that we are examining diffusion at large times and over large distances, neither the temporal nor the spatial resolutions of the experiment are an issue. The effect of the variable q_2 is determined by its influence on how q_1 moves in the potential ϕ and how this influence integrates over time to change the diffusion constant of q_1. We will see that there is an intrinsic time scale associated with the dynamics of the elastic variable q_2, and we assume that this time scale is small and thus difficult to resolve. Measurements of q_1 made over time scales larger than τ_2 will not see the effect of q_2 due to averaging. If the variable q_1 diffuses in a periodic potential and the typical time to cross the potential barriers (*i.e.* jump from minimum to minimum) is of the order of τ_2, then the first passage time from minimum to minimum will be affected by the dynamics of q_2 and thus show up in the late time diffusion constant of q_1, yielding a mechanism for revealing the hydrodynamic coupling. Rather than trying to resolve short time scales directly, we are thus imposing a short distance, and this can be done with an optical trap.

II. SUPPLEMENTARY NOTE 2: HEURISTIC ARGUMENT FOR THE CORRECTION TO THE LATE TIME DIFFUSION CONSTANT

To determine the effective Langevin dynamics for the variable q_1 in the limit where κ is small, and so the variable q_2 is small, we consider the dynamics of the system without any noise. Using the subsequently derived effective dynamics for q_1 at zero temperature, we then add a white noise to the system in order to ensure that the resulting Langevin equation has the correct marginal equilibrium probability distribution

$$P_{\rm meq}(q_1) = \frac{\exp\left[-\beta\phi(q_1)\right]}{Z_-} \,, \tag{S8}$$

where $\beta = \frac{1}{k_{\rm B}T}$ is the inverse thermal energy, and where we use the notation (made dimensionless as compared to the main text)

$$Z_\pm = \int_0^\lambda {\rm d}q \exp[\pm\beta\phi(q)] \,, \tag{S9}$$

where λ is the period of the potential. This is clearly not a rigorous procedure, and we will justify the result rigorously below using multiscale analysis. This method can also be applied to compute the correction to diffusion of a Brownian particle in a periodic potential due to inertia (the deviation from the strongly over-damped limit) [1, 2] and actually recovers first order correction (for completeness, we give a derivation later in this SM). It can also be applied to compute the correction to the diffusion constant of a Brownian particle carrying an electric dipole in a periodic electric field; and again, the simple analysis agrees with a more rigorous one using stochastic analysis [3].

We proceed by formally solving Eq. (S4) at $T = 0$ (that is to say, without noise) to give

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}}\left(1 + \frac{\kappa}{\mu_{22}}\frac{\rm d}{{\rm d}t}\right)^{-1}\phi'(q_1) \,, \tag{S10}$$

where $\tau_2 = \frac{\kappa}{\mu_{22}}$ is a short time scale due to the fact that κ is small. Carrying out an operator expansion for small τ_2, we find

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}}\left[\phi'(q_1) - \frac{\kappa}{\mu_{22}}\phi''(q_1)\frac{{\rm d}q_1}{{\rm d}t} + O(\tau_2^2)\right] . \tag{S11}$$

Now, substituting this in Eq. (S3) gives

$$\frac{{\rm d}q_1}{{\rm d}t} = -\left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}}\right)\phi'\left(q_1(t)\right) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2}\phi''(q_1(t))\frac{{\rm d}q_1}{{\rm d}t} \,, \tag{S12}$$

and from this

$$\frac{{\rm d}q_1}{{\rm d}t} = -\mu_{\rm e}(q_1)\phi'(q_1) \tag{S13}$$

where $\mu_e(q_1)$ is the effective mobility

$$\mu_e(q_1) = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}\left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2}\right)} \tag{S14}$$

In terms of the friction matrix $\gamma = \mu^{-1}$, the effective mobility can be written in a simpler form

$$\mu_e(q_1) = \frac{1}{\gamma_{11}\left(1 + \frac{\kappa\phi''(q_1)\gamma_{12}^2}{\gamma_{11}^2}\right)} . \tag{S15}$$

Rather than discussing the noise necessary to render this into a Langevin equation, we note that to have the correct equilibrium distribution, the corresponding Fokker-Planck equation must read

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\mu_e(q_1)\left[k_B T\frac{\partial P_1(q_1,t)}{\partial q_1} + \phi'(q_1)P_1(q_1,t)\right] , \tag{S16}$$

and we see that the effective diffusion constant is given by $D_e(q_1) = k_B T\mu(q_1)$. Note that in doing this, we have added multiplicative noise and a temperature dependent drift if we use the Ito convention of stochastic calculus; however we avoid this discussion by working directly with the Fokker-Planck equation. Notice that when $T = 0$, Eq. (S16) is the transport equation for Eq. (S13).

Now, if the potential is periodic with period λ, we can use the well known Lifson-Jackson formula for the late time effective diffusion constant for Langevin processes in a potential and with varying diffusivity, both of period λ:

$$D^* = \frac{\lambda^2}{\int_0^\lambda \mathrm{d}q \frac{\exp[\beta\phi(q)]}{D_e(q)} \int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)]} . \tag{S17}$$

In this case, we can easily show by integration by parts that

$$D^*(\kappa) = \frac{k_B T\lambda^2}{\gamma_{11}\int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)] \int_0^\lambda \mathrm{d}q \exp[\beta\phi(q)]\left(1 - \frac{\kappa\gamma_{12}^2\phi'^2(q)}{k_B T\gamma_{11}^2}\right)} . \tag{S18}$$

We thus see that in the regime where κ is small, the effect of a finite value of κ is to increase the long time diffusivity of the particle. We can also write $\phi(q) = \Delta U\psi\left(\frac{q}{\lambda}\right)$, where ΔU is an adjustable energy scale (for instance, it could be related to the laser intensity and in which case λ would be a wavelength). This allows us to write

$$D^*(\kappa) = \frac{k_B T}{\gamma_{11}\int_0^1 \mathrm{d}p \exp[-\beta\Delta U\psi(p)] \int_0^1 \mathrm{d}p \exp\left[\beta\Delta U\psi(p)\right]\left(1 - \frac{\kappa\Delta U^2\psi'^2(p)\gamma_{12}^2}{k_B T\lambda^2\gamma_{11}^2}\right)} . \tag{S19}$$

Therefore, even if κ is very small, increasing the value of ΔU or decreasing the value of L can be used to make the change in the long time diffusion constant measurable.

In the absence of any compliance, $\kappa = 0$, and no potential we note that

$$D^*(0) = \frac{k_B T}{\gamma_{11}} , \tag{S20}$$

this corresponds to the modification of the diffusivity of q_1 when the *surface* q_1 is rigid but still exerts a hydrodynamic interaction.

From a physical standpoint, the fluctuations of q_2 can only influence the dynamics of q_1 on the time scale τ_2 before they are averaged out. However, if the typical time that q_1 takes to climb over the potential barrier to the next period is of order τ_2, then q_2 can effect the long time diffusion of q_1. In this way, fast and small degrees of freedom coupled to a slower degree of freedom can be detected even at low temporal resolution, provided one has sufficiently good statistics. By changing variables, we obtain

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]\frac{\kappa\Delta U^2\psi'^2(p)\gamma_{12}^2}{k_B T\lambda^2\gamma_{11}^2}}{\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]}\right)^{(-1)} . \tag{S21}$$

Now, setting

$$v = \beta \Delta U \tag{S22}$$

gives

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\kappa k_{\rm B} T v^2 \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{\int_0^1 {\rm d}p \exp[v\psi(p)]\psi'^2(p)}{\int_0^1 {\rm d}p \exp[v\psi(p)]}\right)^{(-1)} . \tag{S23}$$

If we take

$$\psi(u) = \sin(2\pi u) , \tag{S24}$$

we find

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{4\pi^2 \kappa k_{\rm B} T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)}\right)^{(-1)} , \tag{S25}$$

where $I_n(z)$ denotes the modified Bessel function of the first kind.

This can be written as

$$\frac{D^*(\kappa)}{D^*(0)} \simeq (1 + \alpha) , \tag{S26}$$

where

$$\alpha = \frac{4\pi^2 \kappa k_{\rm B} T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)} . \tag{S27}$$

In the next section, we will see that it is natural to write

$$\alpha = 4\pi^2 \epsilon \frac{\gamma_{12}^2}{\gamma_{11}^2} \frac{v I_1(v)}{I_0(v)} , \tag{S28}$$

where

$$\epsilon = \frac{\kappa k_{\rm B} T}{\lambda^2} , \tag{S29}$$

is the effective dimensionless compliance parameter that must be small for the above analysis to be valid.

The enhancement of the relative diffusion constant is clearly maximized by choosing small values of λ so that the variable q_2 can influence the jumping rate between neighboring minima rather than being averaged out if λ were large. We also see that the relative change depends on the function of the energy scale of the periodic potential

$$g(v) = \frac{v I_1(v)}{I_0(v)} , \tag{S30}$$

which is a monotonically increasing function of v. This means that the relative change is increased with the energy scale ΔU of the trap.

The Fokker-Planck equation in Eq. (S16) corresponds to the Ito stochastic differential equation

$$\frac{{\rm d}q_1}{{\rm d}t} = -\mu_{\rm e}(q_1)\phi'(q_1) + T\mu_{\rm e}'(q_1) + \sqrt{2T\mu_{\rm e}(q_1)}\eta(t) , \tag{S31}$$

where $\eta(t)$ is a zero-mean white noise with correlation function $\langle \eta(t)\eta(t')\rangle = \delta(t - t')$.

In the next section, we will see that this process has exactly the same diffusion constant as the process q_1 in the original hydrodynamically-coupled equations (S3 S4). Clearly, Eq. (S31) represents a temporal coarse graining of the original process q_1 on time scales greater than τ_2. Finally, at very short time scales $t < \tau_2$, the microscopic time diffusion constant can similarly be deduced from Eq. (S3) and is given by

$$D_{\rm m} = k_{\rm B} T \mu_{11} = \frac{k_{\rm B} T \gamma_{22}}{\gamma_{11}\gamma_{22} - \gamma_{12}^2} . \tag{S32}$$

The expression $D_{\rm m}$ is simply the diffusion constant for the tracer in the absence of any hydrodynamic interactions with q_2 and no periodic potential; or, said otherwise, the bulk diffusivity of q_1.

III. SUPPLEMENTARY NOTE 3: CORRECTION TO THE DIFFUSION BASED ON EXACT FORMULA FOR DIFFUSIVITY AND HOMOGENISATION THEORY

For small κ, q_2 is dominated by its fluctuations which scale *via* the equipartition of energy as

$$\delta q_2 \sim \sqrt{\kappa k_{\rm B} T} \ . \tag{S33}$$

Therefore, from now on, we will use the dimensionless position variables

$$q_1 = \lambda q \text{ and } q_2 = \sqrt{\kappa k_{\rm B} T} u \ , \tag{S34}$$

and again we also write $\beta\phi(q_1) = v\psi\left(\frac{q_1}{\lambda}\right)$. The coupled Langevin equations can be rewritten in terms of the dimensionless time

$$s = \frac{\mu_{11} k_{\rm B} T}{\lambda^2} t \tag{S35}$$

to yield

$$\frac{{\rm d}q}{{\rm d}s} = -v\psi'(q) - \frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}} u + \eta_1'(s) \tag{S36}$$

$$\frac{{\rm d}u}{{\rm d}s} = -\frac{\mu_{22}}{\mu_{11}}\frac{u}{\epsilon} - v\frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}}\psi'(q) + \frac{1}{\sqrt{\epsilon}}\eta_2'(s) \ , \tag{S37}$$

where $\eta_1'(s)$ and $\eta_2'(s)$ are Gaussian white noises with correlation functions

$$\langle \eta_i'(s)\eta_j'(s')\rangle = 2\frac{\mu_{ij}}{\mu_{11}}\delta(s-s') \ . \tag{S38}$$

and ϵ is defined in Eq. (S29).

The late time diffusion coefficient for q_1 is then given by via

$$2D^* t \simeq \langle q_1^2(t)\rangle = \lambda^2\langle q^2(s)\rangle \simeq 2\lambda^2 D_{q_1}^* s = 2D_q \mu_{11} k_{\rm B} T t \ . \tag{S39}$$

This means that the late time diffusion constant for q_1, D^*, is related to the late time diffusion constant of the scaled process q by

$$D^* = \mu_{11} k_{\rm B} T D_q^*\left(\epsilon, v, \frac{\mu_{22}}{\mu_{11}}, \frac{\mu_{12}}{\mu_{11}}\right) \ , \tag{S40}$$

where we have given the explicit functional dependencies for D_q^*. We now carry out a multiscale analysis along the lines of [1, 2] where the correction of diffusion of a Brownian particle in a periodic potential due to inertial effects was given. Here, we will use the Stokes-Einstein relation to compute the diffusion constant and add an additional force h acting on the variable q in addition to the periodic potential. The Fokker-Planck equation for the variables q and u is then given by

$$\frac{\partial P}{\partial s} = HP \ , \tag{S41}$$

where H is

$$HP = \left(H_0 + \frac{1}{\sqrt{\epsilon}}H_1 + \frac{1}{\epsilon}H_2\right)P \ , \tag{S42}$$

and the corresponding terms are given by

$$H_0 P = \frac{\partial}{\partial q}\left(\frac{\partial P}{\partial q} + [v\psi'(q) - h]P\right) \tag{S43}$$

$$H_1 P = \frac{\mu_{12}}{\mu_{11}}\left[u\frac{\partial P}{\partial q} + 2\frac{\partial^2 P}{\partial q \partial u} + (v\psi_1'(q) - h)\frac{\partial P}{\partial u}\right] \tag{S44}$$

$$H_2 P = \frac{\mu_{22}}{\mu_{11}}\frac{\partial}{\partial u}\left(\frac{\partial P}{\partial u} + uP\right) \ . \tag{S45}$$

The expansion we will make is based on small ϵ but also depends on the parameters μ_{22}/μ_{11} and μ_{12}/μ_{11}, the separation of scales needed to implement the expansion is more precisely states as

$$1 \ll \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\epsilon}\frac{\mu_{22}}{\mu_{11}} \,, \tag{S46}$$

and notably the second inequality implies that

$$\frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\sqrt{\epsilon}}\frac{\mu_{22}}{\mu_{11}} \,. \tag{S47}$$

It will be useful later to employ the adjoint $H^\dagger$ of H given by

$$H^\dagger = H_0^\dagger + \frac{1}{\sqrt{\epsilon}}H_1^\dagger + \frac{1}{\epsilon}H_2^\dagger \,, \tag{S48}$$

where

$$H_0^\dagger f = \frac{\partial^2 f}{\partial q^2} - (v\psi'(q) - h)\frac{\partial f}{\partial q} \,, \tag{S49}$$

$$H_1^\dagger f = \frac{\mu_{12}}{\mu_{11}}\left(-u\frac{\partial f}{\partial q} + 2\frac{\partial^2 f}{\partial q \partial u} - [v\psi'(q) - h]\frac{\partial f}{\partial u}\right) \,, \tag{S50}$$

$$H_2^\dagger f = \frac{\mu_{22}}{\mu_{11}}\left(\frac{\partial^2 f}{\partial u^2} - u\frac{\partial f}{\partial u}\right) \,. \tag{S51}$$

We proceed by computing the average velocity (denoted here by V) of q due to the applied field h for a tracer particle which is started with the local equilibrium distribution within a single period of the potential. This is given by

$$V = \langle \frac{\mathrm{d}q}{\mathrm{d}s} \rangle = \int_{-\infty}^{\infty} \mathrm{d}q\, \mathrm{d}u\, \mathrm{d}u_0 \int_\Lambda \mathrm{d}q_0 \; q \frac{\mathrm{d}P(q,u|q_0,u_0;s)}{\mathrm{d}t} P_{\mathrm{peq}}(q_0,u_0) \,, \tag{S52}$$

where Λ the integration range over q_0 is over a single period of length 1 and $P_{\mathrm{peq}}(q,u)$ is the equilibrium distribution for the variable u and the periodized variable $q^* = q \bmod(1)$. Using the Fokker-Planck equation (S41) then gives

$$V = \int_{-\infty}^{\infty} \mathrm{d}q\, \mathrm{d}u\, \mathrm{d}u_0 \int_\Lambda \mathrm{d}q_0 q \left[HP(q,u|q_0,u_0;s)\right] P_{\mathrm{eq}}(q_0,u_0) = \int_{-\infty}^{\infty} \mathrm{d}q\, \mathrm{d}u\, \mathrm{d}u_0 \int_\Lambda \mathrm{d}q_0 [H^\dagger q] P(q,u|q_0,u_0;s) P_{\mathrm{peq}}(q_0,u_0) \,, \tag{S53}$$

where

$$H^\dagger q = -v\psi'(q) + h - \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}u \,. \tag{S54}$$

We notice that $H^\dagger q$ is periodic in q, so we can write

$$\int_{-\infty}^{\infty} \mathrm{d}u\, \mathrm{d}u_0\, \mathrm{d}q \int_\Lambda \mathrm{d}q_0 P(q,u|q_0,u_0;s) P_{\mathrm{peq}}(q_0,u_0) = \int_{-\infty}^{\infty} \mathrm{d}u\, \mathrm{d}u_0 \int_\Lambda \mathrm{d}q\, \mathrm{d}q_0 [H^\dagger q] \sum_{n=-\infty}^{\infty} P(q+n,u|q_0,u_0;s) P_{\mathrm{peq}}(q_0,u_0) \,. \tag{S55}$$

However, $\sum_{n=-\infty}^{\infty} P(q+n,u|q_0,u_0;s) = P_p(q,u|q_0,u_0;s)$ is simply the transition density for the periodic variable q^* on Λ. Clearly $P_p(q,u|q_0,u_0;s)$ satisfies the same Fokker-Planck equation as P but restricted to $q \in \Lambda$ and with periodic boundary conditions.

As we have assumed that q^* starts in equilibrium we find have

$$\int_{-\infty}^{\infty} \mathrm{d}u_0 \int_\Lambda \mathrm{d}q_0 P_p(q,u|q_0,u_0;s) P_{\mathrm{peq}}(q_0,u_0) = P_{\mathrm{peq}}(q_0,u_0) \,, \tag{S56}$$

and we thus obtain the Stratonovich formula

$$V = \int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q [H^\dagger q] P_{\mathrm{peq}}(q,u) \,, \tag{S57}$$

where

$$HP_{\rm peq}(q,u)=0\ , \tag{S58}$$

and has the periodic boundary condition $P_{\rm peq}(q+1,u)=P_{\rm peq}(q,u)$. The Stokes-Einstein formula then gives the diffusion constant for the variable q as

$$D_q^*=\frac{\partial V}{\partial h}=\int_{-\infty}^{\infty}{\rm d}u\int_\Lambda {\rm d}q\ \left[\frac{\partial H^\dagger}{\partial h}q\right]P_{\rm peq0}(q,u)+[H^\dagger q]\frac{\partial}{\partial h}P_{peq}(q,u)\ . \tag{S59}$$

In Eq. (S59), everything is evaluated at $h=0$, and the integral over q is on a single period Λ of the periodic potential ψ. We now denote

$$\frac{\partial}{\partial h}P_{\rm peq}(q,u)=r(q,u)\ , \tag{S60}$$

which obeys

$$Hr(q,u)-\frac{\partial}{\partial q}P_{\rm peq0}(q,u)-\frac{\mu_{12}}{\mu_{11}}\frac{1}{\sqrt{\epsilon}}\frac{\partial}{\partial u}P_{\rm peq0}(q,u)=0\ , \tag{S61}$$

and must in addition obey the integral constraint, due to the conservation of probability,

$$\int {\rm d}q\,{\rm d}u\frac{\partial}{\partial h}P_{\rm peq}(q,u)=\int {\rm d}q\,{\rm d}u\ r(q,u)=0\ . \tag{S62}$$

Now we note that Eq. (S54) gives

$$\frac{\partial H^\dagger}{\partial h}q=1\ , \tag{S63}$$

and consequently

$$D_q^*=\int_{-\infty}^{\infty}{\rm d}u\int_\Lambda {\rm d}q\ P_{\rm peq0}(q,u)+[H^\dagger q]r(q,u)\ . \tag{S64}$$

This simplifies to

$$D_q^*=1-\int {\rm d}q\,{\rm d}u\left(v\psi'(q)+\frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}u\right)r(q,u)\ . \tag{S65}$$

The above expression can also be derived using established results on the theory of transport processes [4], or using a general Kubo-type formalism [5, 6]. In principle, the partial differential equation Eq. (S61) can be numerically solved and the integral in Eq. (S65) can be evaluated using the solution to give a numerically exact result for D^* for all values of ϵ, and we will show how this can be done in section (V). Here we compute the corrections due to finite κ perturbatively for small ϵ while assuming that the variables μ_{ij} are of the same order to validate the expansion we make. We work in the adjoint representation writing

$$r(q,u)=P_{\rm peq0}(q,u)w(q,u)\ , \tag{S66}$$

which gives

$$H^\dagger w(q,u)+v\psi'(q)+\frac{u\mu_{12}}{\mu_{11}\sqrt{\epsilon}}=0\ , \tag{S67}$$

along with the integral constraint

$$\int_{-\infty}^{\infty}{\rm d}u\int_\Lambda {\rm d}qP_{\rm peq0}(q,u)w(q,u)=0\ . \tag{S68}$$

Setting $h=0$ in Eq. (S54), we see that Eq. (S67) has a solution $w(q,u)=q+A$, where A is a constant. The integral constraint means that we must choose

$$A=-\langle q\rangle_{\rm eq}=\int_{-\infty}^{\infty}{\rm d}u\int_\Lambda {\rm d}q\ qP_{\rm peq0}(q,u)\ . \tag{S69}$$

However, we must ensure periodicity in q. We thus write

$$w(q,u) = q - \langle q \rangle_{\rm eq} + s(q,u) \ , \tag{S70}$$

where $s(q,u)$ has the boundary condition

$$1 + s(1,u) = s(0,u) \ , \tag{S71}$$

and obeys the equation

$$H^\dagger s(q,u) = 0 \ . \tag{S72}$$

Plugging this into the formula for D_q^*, we find

$$D_q^* = 1 - \int_{-\infty}^{\infty} {\rm d}u \int_\Lambda {\rm d}q P_{\rm peq0}(q,u) \left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u \right) (q - \langle q \rangle_{\rm eq} + s(q,u)) \ . \tag{S73}$$

The shift due to the term $\langle q \rangle_{\rm eq}$ gives zero contribution to the integral. The part of the integrand containing q can be integrated by parts (the second term giving zero) to give

$$D_q^* = \frac{1}{Z_-} \exp\left[-v\psi(1)\right] - \int {\rm d}q \, {\rm d}u P_{\rm peq0}(q,u) \left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u \right) s(q,u) \ , \tag{S74}$$

and here we have explicitly taken $\Lambda = [0,1]$. In what follows, we denote

$$Z_\pm = \int_0^1 {\rm d}q \ \exp(\pm v\psi(q)). \tag{S75}$$

The first term in the integral on the right-hand side of Eq. (S74) can be integrated by parts to give the computationally useful formula

$$D_q^* = - \int_{-\infty}^{\infty} {\rm d}u \int_\Lambda {\rm d}q \, P_{\rm peq0}(q,u) \left(\frac{\partial s(q,u)}{\partial q} + \frac{1}{\sqrt{\epsilon}} \frac{\mu_{12}}{\mu_{11}} u s(q,u) \right) \ . \tag{S76}$$

Now, we look for a solution of the form

$$s(q,u) = s_0(q,u) + \sqrt{\epsilon} s_1(q,u) + \epsilon s_2(q,u) + \sqrt{\epsilon}^3 s_3(q,u) + O(\epsilon^2) \ . \tag{S77}$$

This yields, expanding to order ϵ,

$$\begin{aligned}
H_2^\dagger s_0 &= 0 && \text{(S78)} \\
H_1^\dagger s_0 + H_2^\dagger s_1 &= 0 && \text{(S79)} \\
H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 &= 0 && \text{(S80)} \\
H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3 &= 0 && \text{(S81)} \\
H_0^\dagger s_2 + H_1^\dagger s_3 + H_2^\dagger s_4 &= 0 \ . && \text{(S82)}
\end{aligned}$$

The resulting diffusion constant is then given by

$$\begin{aligned}
D_q^* \ = \ & -\frac{1}{\sqrt{\epsilon}} \int {\rm d}q \, {\rm d}u \, P_{\rm peq0}(q,u) \frac{\mu_{12}}{\mu_{11}} u s_0(q,u) - \int {\rm d}q \, {\rm d}u \, P_{\rm peq0}(q,u) \left[\frac{\partial s_0(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_1(q,u) \right] \\
& - \sqrt{\epsilon} \int {\rm d}q \, {\rm d}u \, P_{\rm peq0}(q,u) \left[\frac{\partial s_1(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_2(q,u) \right] \\
& - \epsilon \int {\rm d}q \, {\rm d}u \, P_{\rm peq0}(q,u) \left[\frac{\partial s_2(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_3(q,u) \right] \ ,
\end{aligned} \tag{S83}$$

and for notational simplicity, we have not explicitly written the integration ranges which are $u \in (-\infty, \infty)$ and $q \in [0,1]$. The equation (S79), $H_0^\dagger s_0(q,u) = 0$, shows that $s_0(q,u) = s_0(q)$ is independent of u. Note that in terms of

the parity with respect to u, the operators $H_0^\dagger$ and $H_2^\dagger$ are even and $H_1^\dagger$ is odd. From the structure of the equations, this means that $s_n(q,u)$ is even in u for n even and odd in u for n odd. As a consequence we find that

$$D_q^* = -\int \mathrm{d}q\,\mathrm{d}u\,P_{\mathrm{peq0}}(q,u)\left[\frac{\partial s_0(q,u)}{\partial q}+\frac{\mu_{12}}{\mu_{11}}us_1(q,u)\right]-\epsilon\int \mathrm{d}q\,\mathrm{d}u\,P_{\mathrm{peq0}}(q,u)\left[\frac{\partial s_2(q,u)}{\partial q}+\frac{\mu_{12}}{\mu_{11}}us_3(q,u)\right]\,, \tag{S84}$$

and it is easy to see that the higher order corrections are also integer powers of ϵ. We now consider Eq. (S79): $H_1^\dagger s_0 + H_2^\dagger s_1 = 0$. From this we see that solution for s_1 exists if the Fredholm alternative condition holds (the solvability condition)

$$\int \mathrm{d}u P_{\mathrm{eq}}(u) H_1^\dagger s_0(q) = 0\,, \tag{S85}$$

where $P_{\mathrm{eq}}(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)$, the marginal equilibrium probability distribution of u, is in the kernel of the adjoint of H_2, that is to say $H_2^\dagger$. We find that

$$H_1^\dagger s_0(q) = -\frac{\mu_{12}}{\mu_{11}}u\frac{\partial s_0(q)}{\partial q}\,, \tag{S86}$$

and this gives

$$s_1(q,u) = -\frac{\mu_{12}}{\mu_{22}}u\frac{\partial s_0(q)}{\partial q}\,. \tag{S87}$$

Now, consider the equation Eq. (S80), $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$. Here, we find that

$$H_1^\dagger s_1 = \frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\left([u^2-2]\frac{\partial^2 s_0(q)}{\partial q^2}+v\psi'(q)\frac{\partial s_0(q)}{\partial q}\right)\,. \tag{S88}$$

Integrating Eq. (S80) with respect to $P_{\mathrm{eq}}(u)$ then gives the solvability condition for s_2 as

$$H_0^\dagger s_0 + \frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\left(-\frac{\partial^2 s_0(q)}{\partial q^2}+v\psi'(q)\frac{\partial s_0(q)}{\partial q}\right) = 0\,, \tag{S89}$$

which is simply

$$\left(1-\frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\right)H_0^\dagger s_0 = 0\,. \tag{S90}$$

This has the solution

$$s_0(q) = A\int_0^q \exp[v\psi(p)]\,\mathrm{d}p\,, \tag{S91}$$

where A is a constant. Note that we can choose $s_0(0) = 0$ without loss of generality as D_q^* is independent of any constant term in $s(q)$. Using the boundary condition $s_0(1) = -1$ then gives

$$s_0(q) = -\frac{\int_0^q \exp[v\psi(p)]\,\mathrm{d}p}{Z_+}\,, \tag{S92}$$

and so

$$s_0'(q) = -\frac{\exp[v\psi(q)]}{Z_+}\,. \tag{S93}$$

This then yields

$$s_1(q,u) = \frac{\mu_{12}}{\mu_{22}}u\frac{\exp[v\psi(q)]}{Z_+}\,. \tag{S94}$$

Importantly, we see that $s_1(q,u)$ is periodic in q and so the solution for s at order $\sqrt{\epsilon}$ obeys the full boundary condition *via* the term $s_0(q)$ solely, all the other corrections must consequently be periodic. This gives the $O(0)$ contribution to D_q^*

$$D_q^*(0) = \frac{1}{Z_+ Z_-}\left[1 - \frac{\mu_{12}^2}{\mu_{22}\mu_{11}}\right] . \tag{S95}$$

Returning to Eq. (S80) $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$ we find that

$$\frac{\mu_{12}^2}{\mu_{11}\mu_{22}}[u^2-1]\frac{\partial^2 s_0(q)}{\partial q^2} + H_2^\dagger s_2 = 0 . \tag{S96}$$

This satisfies the solvability condition, and has solution

$$s_2(q,u) = u^2 \frac{\mu_{12}^2}{2\mu_{22}^2}\frac{\partial^2 s_0(q)}{\partial q^2} + s_{20}(q) , \tag{S97}$$

where $s_{20}(q)$ is undetermined (but must be periodic). Now solving Eq. (S81), $H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3$, we find the periodic solution

$$s_3(q,u) = -u\frac{\mu_{12}}{\mu_{22}}s_{20}'(q) + \frac{\mu_{12}u\exp[v\psi(q)]\left(\mu_{12}^2u^2v^2\psi'(q)^2 + v\psi''(q)[6(\mu_{11}\mu_{22}-\mu_{12}^2) - u^2\mu_{12}^2]\right)}{6Z_+\mu_{22}^3} . \tag{S98}$$

To determine $s_{20}(q)$, we use the Fredholm alternative for solvability for Eq. (S82), $H_0^\dagger s_2 + H_1^\dagger s_3 + H_0^\dagger s_4 = 0$, which yields after some lengthy algebra

$$s_{20}(q) = \int_0^q \exp[v\psi(p)]\,\mathrm{d}p\left[A' + \frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+}\right] , \tag{S99}$$

where we have again taken $s_{20}(0) = 0$, and so the (periodic) boundary condition at $q = 1$ is $s_{20}(1) = 0$ (recalling that the inhomogeneous boundary condition is satisfied by $s_0(q)$), we thus have

$$A'Z_+ + \int_0^1 \exp[v\psi(p)]\,\mathrm{d}p\,\frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+} = 0 , \tag{S100}$$

and so

$$A' = -\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2}\int_0^1 \exp[v\psi(p)]v^2\psi'(p)^2\mathrm{d}p . \tag{S101}$$

This then yields

$$s_{20}'(q) = \left[-\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2}\int_0^1 \exp[v\psi(p)]v^2\psi'(p)^2\mathrm{d}p + \frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+}\right]\exp[v\psi(q)] . \tag{S102}$$

From the above, we find that the correction to D_q^* to $O(\epsilon)$ is

$$\Delta D_{q2}^* = \epsilon v^2 \frac{(\mu_{11}\mu_{22}-\mu_{12}^2)}{\mu_{22}\mu_{11}}\frac{\mu_{12}^2}{\mu_{22}^2}\frac{\int_0^1 \exp[v\psi(p)]\psi'(p)^2\mathrm{d}p}{Z_- Z_+^2} . \tag{S103}$$

Using this to compute D^* then gives the formula derived from the heuristic argument presented in section (II).

IV. SUPPLEMENTARY NOTE 4: THE COMPLIANT LIMIT $\epsilon \gg 1$

In the limit $\epsilon \gg 1$, one can carry out a similar analysis to the limit $\epsilon \ll 1$. The Kubo-type formulae derived in the previous section still hold and the equation for $s(q,u)$ is now written as

$$\left[\epsilon H_0 + \sqrt{\epsilon}H_1 + H_2\right] s(q,u) , \tag{S104}$$

and the formula for D_q^* is unchanged.

Here, we find that one need to write a perturbative expansion of the form

$$s(q,u) = \sqrt{\epsilon} s_0(q,u) + s_1(q,u) + \frac{1}{\sqrt{\epsilon}} s_2(q,u) + O(\frac{1}{\epsilon}) \ . \tag{S105}$$

We find that

$$H_0^\dagger s_0 = 0 \tag{S106}$$

$$H_0^\dagger s_1 + H_1^\dagger s_0 = 0 \tag{S107}$$

$$H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0 = 0 \ . \tag{S108}$$

The equation $H_0 s_0 = 0$ then gives $s_0(q,u) = s_0(u)$, and so, to leading order, we find

$$D_q^* = -\int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q \, P_{\mathrm{peq}0}(q,u) \left(\frac{\partial s_1(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} s_0'(u) \right) + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S109}$$

We impose the non-periodic boundary condition on the $O(1)$ term $s_1(q,u)$

$$s_1(0,u) = 0 \ ; \ 1 + s_1(1,u) = 0 \ . \tag{S110}$$

The term s_1 obeys the equation $H_1 s_0 + H_0 s_1 = 0$, where $H_1 s_0 = -\frac{\mu_{12}}{\mu_{11}} v\psi'(q) s'(u)$. We notice that

$$\int_\Lambda \mathrm{d}q \, P_{\mathrm{eq}}(q) H_1 s_0 = 0 \ , \tag{S111}$$

where

$$P_{\mathrm{eq}}(q) = \frac{\exp[-v\psi(q)]}{Z_-} \ , \tag{S112}$$

and so this equation obeys the solvability criterion. The solution is given by

$$s_1(q,u) = -q \frac{\mu_{12} s_0'(u)}{\mu_{11}} + \frac{1}{Z_+} \left[\frac{\mu_{12} s_0'(u)}{\mu_{11}} - 1 \right] \int_0^q \mathrm{d}p \ \exp[v\psi(p)] \ . \tag{S113}$$

Using this result in Eq. (S109) then gives to leading order

$$D_q^* = \frac{1}{Z_+ Z_-} \left[1 - \frac{1}{\sqrt{2\pi}} \frac{\mu_{12}}{\mu_{11}} \int_{-\infty}^{\infty} \mathrm{d}u \exp(-\frac{u^2}{2}) s_0'(u) \right] + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S114}$$

Finally, the equation for $s_0(u)$ can be found by using the solvability condition for s_2

$$0 = \int_\Lambda \mathrm{d}q \ P_{\mathrm{eq}}(q) \left[H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0 \right] = H_2^\dagger s_0(u) + \int_\Lambda \mathrm{d}q \ P_{\mathrm{eq}}(q) H_1^\dagger s_1(q,u) \ . \tag{S115}$$

After some algebra, the solution which is integrable with respect to the equilibrium measure $P_{\mathrm{eq}}(u)$ for u is

$$s_0(u) = \frac{u \mu_{12} \mu_{11}}{(\mu_{11}\mu_{22} - \mu_{12}^2) Z_+ Z_- + \mu_{12}^2} \ . \tag{S116}$$

Using this in Eq. (S114) then gives

$$D_q^* = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S117}$$

Note that when there is no periodic potential ($v = 0$) one has $Z_+ = Z_- = 1$, and we find

$$D_q^*(\epsilon = \infty) = k_{\mathrm{B}} T \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}} = \frac{k_B T}{\gamma_{11}} \tag{S118}$$

which is the same result for the diffusion constant in the non-compliant case $\epsilon = 0$ and with $v = 0$. In terms of the unscaled variables this then gives

$$D^* = k_{\mathrm{B}} T \mu_{11} \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\lambda^{-2} Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S119}$$

V. SUPPLEMENTARY NOTE 5: NUMERICAL COMPUTATION OF D^* FROM THE KUBO FORMULA

The diffusion constant D_q^* related to D^* by Eq. (S40) can be numerically computed by solving Eq. (S61) with periodic boundary conditions and the integral constraint Eq. (S62). This can be carried out by using the FlexPDE Multiphysics Software For Partial Differential Equations. The numerical domain is finite and taken over $[-1/2, 1/2]$ for the variable q with periodic boundary conditions. The variable u is unbounded, but we take a finite domain with $u \in [-L, L]$ (typically, it suffices to take $L = 5$), and impose no-flux boundary conditions at the surfaces $u = \pm L$. This means that the effective potential for u becomes infinite at the edges, but in practice the variable $r(q, u)$ has decayed to zero at these values of u. Examples are shown in Figs. 4, S1, and S2, for various parameters.

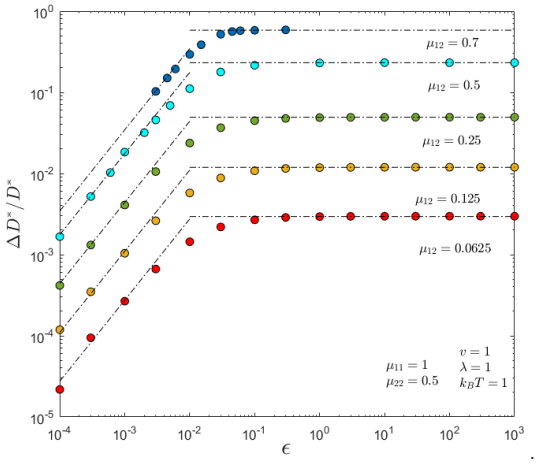

FIG. S1. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ, for μ_{11}= 1.0, $\mu_{22} = 0.5$, $v = 1.0$, $\lambda = 1.0$, $k_B T = 1.0$, and various μ_{12} as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small-ϵ prediction of Eq. (5), and the large-ϵ prediction of Eq. (11).

VI. SUPPLEMENTARY NOTE 6: HEURISTIC ARGUMENT FOR MODIFICATION OF DIFFUSION WITH INERTIA

Here, we show how the heuristic method proposed beforehand can be used to compute the effective diffusion constant for a particle due to first order inertial effects, recovering the result of [1, 2]. This calculation also demonstrates how a periodic potential can be applied to detect the signature of inertia on late-time diffusion.

We consider the zero-temperature equation

$$m\frac{\mathrm{d}v}{\mathrm{d}t} + \gamma v = -\phi'(q) \ , \tag{S120}$$

where $v = \frac{\mathrm{d}q}{\mathrm{d}t}$ is the particle's velocity.

In operator notation, this formally gives

$$v(t) = -\frac{1}{\gamma}\left(1 + \tau\frac{\mathrm{d}}{\mathrm{d}t}\right)^{-1}\phi'(q) \ , \tag{S121}$$

where $\tau = \frac{m}{\gamma}$ is the short timescale over which the velocity relaxes and which would be difficult to resolve experimentally.

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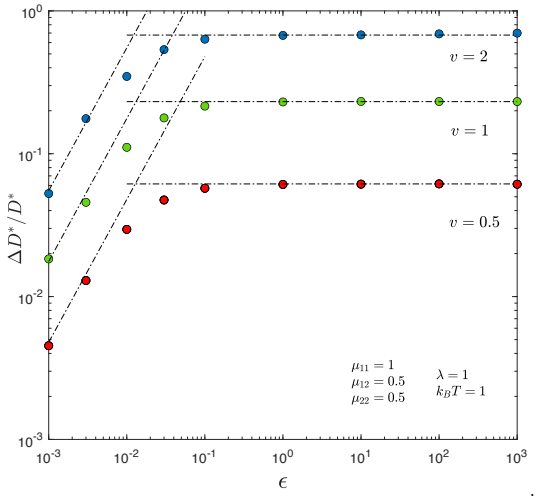

FIG. S2. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient D^* of the test particle as a function of the dimensionless compliance ϵ, for μ_{11}= 1.0, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, $\lambda = 1.0$, $k_{\rm B}T = 1.0$, and various v as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small-ϵ prediction of Eq. (5), and the large-ϵ prediction of Eq. (11).

Now, expanding to first order in τ, we find

$$v(t) = -\frac{1}{\gamma}\phi'(q) + \frac{\tau}{\gamma}\phi''(q)v(t) \ , \tag{S122}$$

and so

$$\frac{{\rm d}q}{{\rm d}t} = v(t) = -\mu_{\rm e}(q)\phi'(q) \ , \tag{S123}$$

where

$$\mu_{\rm e}(q) = \frac{1}{\gamma\left[1 - \frac{\tau}{\gamma}\phi''(q)\right]} \ . \tag{S124}$$

The diffusion constant is thus

$$D_{\rm e}(q) = \frac{k_{\rm B}T}{\gamma\left[1 - \frac{\tau}{\gamma}\phi''(q)\right]} \ . \tag{S125}$$

Now, applying the Lifson-Jackson formula, we obtain

$$D^* = \frac{k_{\rm B}T\lambda^2}{\gamma\int_0^\lambda {\rm d}q\exp[\beta\phi(q)]\left[1 - \frac{m}{\gamma^2}\phi''(q)\right]\int_0^\lambda {\rm d}q\exp[-\beta\phi(q)]} \ . \tag{S126}$$

This can be rewritten as

$$D^* = \frac{k_{\rm B}T\lambda^2}{\gamma\int_0^\lambda {\rm d}q\exp[\beta\phi(q)]\left[1 + \frac{m}{k_{\rm B}T\gamma^2}\phi'(q)^2\right]\int_0^\lambda {\rm d}q\exp[-\beta\phi(q)]} \ , \tag{S127}$$

or to the same order

$$D^* = \frac{k_{\rm B}T\lambda^2}{\gamma\int_0^\lambda {\rm d}q\exp[\beta\phi(q)]\int_0^\lambda {\rm d}q\exp[-\beta\phi(q)]}\left(1 - \frac{m}{k_{\rm B}T\gamma^2}\frac{\int_0^\lambda {\rm d}q\exp[\beta\phi(q)]\phi'(q)^2}{\int_0^\lambda {\rm d}q\exp[\beta\phi(q)]}\right) \ , \tag{S128}$$

which recovers the result given in [1, 2] via multiscale analysis. Notice that in this particular case, the effect of inertia is to reduce the late time diffusion constant, and again, it is necessary to impose an external periodic potential to produce a measurable effect on the late time diffusivity.

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PNAS RESEARCH ARTICLE | BIOPHYSICS AND COMPUTATIONAL BIOLOGY
PHYSICS

Enhanced dispersion of active microswimmers in confined flows

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In the presence of a laminar shear flow, the diffusion of passive colloidal particles is enhanced in the direction parallel to the flow. This classical phenomenon is known as Taylor–Aris dispersion. Besides, microorganisms, such as active microswimmers, exhibit an effective diffusive behavior at long times. Combining the two ingredients above, a natural question then emerges on how the effective diffusion of active microswimmers is altered in shear flows—a widespread situation in natural environments with practical implications, e.g., regarding biofilm formation. In this Letter, we investigate the motility and dispersion of *Chlamydomonas reinhardtii* microalgae, within a rectangular microfluidic channel subjected to a sinusoidal Poiseuille flow. Using high-resolution optical microscopy and a particle-tracking algorithm, we reconstruct individual trajectories in various flow conditions and statistically analyze them through moment theory and sliding windowed demodulation. We find that the velocity fluctuations and the dispersion coefficient increase as the flow amplitude is increased, with only weak dependencies on the flow periodicity. Importantly, our results demonstrate that the generalization of Taylor–Aris law to active particles is valid.

microswimmers | algae | microfluidics | dispersion | shear flows

Significance

This study introduces a unified approach combining microfluidics, Langevin simulations, and statistical analysis to examine how *Chlamydomonas reinhardtii*, an active swimming microalgae, disperses in confined flows. The findings extend classical Taylor–Aris dispersion theory—originally for passive Brownian particles—to active swimmers, revealing a transition from ballistic to diffusive behaviors influenced by the flow. Shear flows are shown to impact the swimmers' dynamics nonlinearly, enhancing spreading in complex ways. These results provide critical insight into microswimmers' behavior in fluid environments, relevant to both artificial and natural settings. The work has broader implications for microbial ecology, suggesting ways to understand processes like biofilm formation, nutrient transport, and pathogen dispersal in ecosystems where fluid dynamics shape microorganism movement and interactions.

Microscopic organisms frequently exceed the length scale at which thermal Brownian motion alone can mediate efficient environmental exploration. To circumvent this constraint, they employ diverse active motility strategies in viscous media (1), including crawling (2), gliding (3, 4), and swimming (5–7). Among microswimmers, propulsion mechanisms exhibit notable variations: for instance, *Escherichia coli* functions as a pusher, using rotary motor-driven flagella for propulsion (8); whereas the biflagellate alga *Chlamydomonas reinhardtii*, from the *Volvox* family (9), acts as a puller relying on synchronous beating of its flagella (5). Diverse external stimuli, such as light for photosynthesis (10–13), or chemical gradients (14), are essential in determining the dynamics of microswimmers. Similarly, hydrodynamic interactions with the environment are fundamental. As a result of these, swimming microorganisms manifest a rich repertoire of behaviors, including effective viscosity (15), accumulation near boundaries (16–19), entrainment (20), upstream swimming (21, 22), Bretherton-Jeffery orbits (23), resonant transverse swimming (24, 25), and emergent interactions (26, 27). Despite the persistent self-propelled motion of microswimmers, their long-time dynamics often resembles the one of passive Brownian particles, which is characterized through an effective diffusion process (28–32).

In another context, the seminal studies of Taylor (33–35) and Aris (36) established that shear flows can significantly enhance the dispersion of passive microscopic solutes—a now-classical effect known as Taylor–Aris dispersion. Further works have extended this framework to incorporate the effects of channel geometry (37), porosity (38–40), reactivity (41), flow complexity (42–45), short-time distribution (46), and open or charged boundaries (47). The underlying coupling between advection and diffusion has remained a vibrant topic of theoretical and numerical inquiry (48, 49). The Taylor–Aris mechanism has also found impactful applications, e.g., in particle-size characterization (50–54).

Given that active microswimmers effectively diffuse at long times, and often navigate within external flows in natural environments, a compelling question arises: does an analogue of the Taylor–Aris law govern their transport? While theoretical and computational models have proliferated on active-particle dispersion in external flows (21, 45, 55–58), and on the influence of oscillatory flows on active suspensions (59), experimental confirmation remains scarce, primarily due to challenges in recording the dynamics over extended time scales. Nevertheless, recent and preliminary experimental studies addressed *E. coli* in flows (60), bacterial scattering in microfluidic lattices (61), as

The authors declare no competing interest.

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well as dispersion of microalgae as gyrotactic swimmers (58, 59). In the present article, we quantitatively investigate the Taylor–Aris dispersion of model active particles, using a combination of experiments, numerical simulations, and theory.

Experiments

We employ the green alga *C. reinhardtii* CC409 as a model active swimmer, owing to its well-characterized motility and widespread use in microswimmer research (62). See *Materials and Methods* for details on the experiments. Further details on cell culture and experimental preparation are also provided in *SI Appendix*. This strain swims at a characteristic speed of approximately $V_0 \approx 100\,\mu$m/s (63), with a decorrelation time of around $\sim$1 s, beyond which it exhibits an effective diffusion with a diffusion coefficient on the order of $D_0 \approx 10^{-9}\,\mathrm{m}^2/\mathrm{s}$, as obtained from 1,296 single-cell trajectories in the bulk and in the absence of external flow (*SI Appendix*).

To probe the long-time dynamics of active dispersion, we impose a time-periodic sinusoidal flow that maintains the algae within the field of view over a reasonable duration. Our experimental approach builds on theoretical frameworks describing active-particle dispersion in external flows (21) and the influence of oscillatory flows on passive thermal dispersion (42). Fig. 1 shows the experimental setup together with representative trajectories of individual cells under oscillatory microfluidic flow.

The fluid velocity field $\vec{u}(y,z,t)$ along the flow axis x (with unit vector $\vec{e}_x$), at time t, can thus be accurately described as a laminar, pressure-driven Poiseuille flow, expressed as

$$\vec{u}(y,z,t) = \frac{\underline{P}}{R_\mathrm{h}} f(y,z)\,\sin\left(\frac{2\pi}{T}t + \phi_0\right)\vec{e}_x\,, \qquad [1]$$

where $\underline{P}$ is the oscillation amplitude of the pressure, T is the oscillation period, R_h is the reduced hydraulic resistance of the channel, $f(y,z)$ is the dimensionless cross-sectional profile and ϕ_0 is an arbitrary phase shift. For instance, in a two-dimensional case, one would have $f(y) = 4(y/W)(1-y/W)$, with W the channel width, while for a three-dimensional parallelepipedic channel we refer to ref. 65.

The central dimensionless control parameter in our problem is the Péclet number Pe, that compares flow-induced advection to effective diffusion. It is defined as

$$\mathrm{Pe} \equiv \frac{\underline{P}}{\sqrt{2}R_\mathrm{h}}\frac{W}{D_0}\,, \qquad [2]$$

where $\underline{P}/(\sqrt{2}\,R_\mathrm{h})$ corresponds to the RMS velocity of the oscillatory flow.

In addition, we assume that the position of a single cell can be expressed as

$$x(t) = \hat{x}(t) + A(y(t),t)\cos\left(\frac{2\pi}{T}t\right), \qquad [3]$$

where $A(y(t),t)$ is an unknown, potentially random function, and $\hat{x}(t)$ represents the cell's position in the comoving frame associated with the flow. For a nonswimming cell, the velocity is expected to follow a sinusoidal pattern, mirroring the flow profile. If the particle is placed at a fixed transverse position y, the amplitude of the oscillation is determined by the local flow velocity at that position. However, in the case of an active particle that randomly explores both x and y, the amplitude of the position oscillation inherited from the flow becomes modulated by the particle's stochastic exploration along the y-axis.

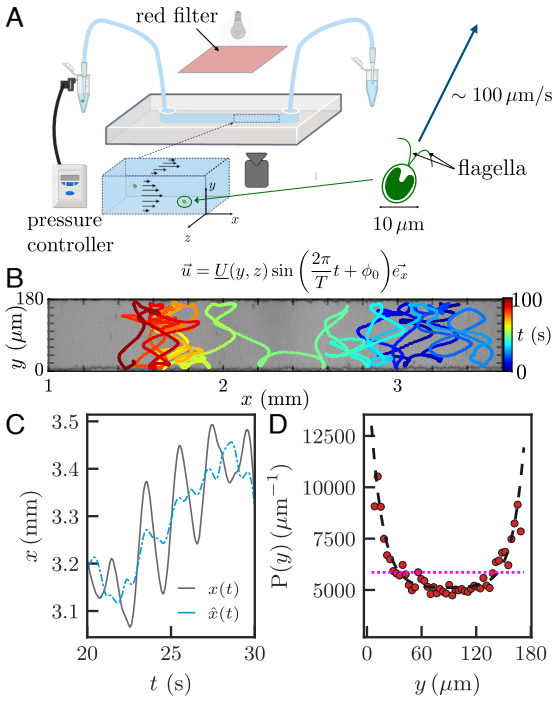

Fig. 1. Dynamics of microalgae in an oscillatory microfluidic flow. (*A*) Schematic of the experimental setup. A suspension of microalgae is confined in a microfluidic channel with rectangular cross-section. A pressure controller at one end generates a sinusoidal Poiseuille flow, while a constant pressure is applied at the other end via a fluid reservoir. The algae are illuminated with red-filtered light and imaged using a high-speed camera (equipped with a ×5 objective). (*B*) Representative top-view image of the channel (×5 magnification; frame rate: 40 Hz; exposure time: 1 ms) with a typical trajectory superimposed. The colormap encodes time along the trajectory. (*C*) Slice of a time series of the longitudinal position $x(t)$. The demodulated signal $\hat{x}(t)$ is overlaid. (*D*) Probability density function P(y) of the transverse position y, as estimated from 27 trajectories exceeding 20 s in duration. The magenta dotted line indicates a uniform distribution. The black dashed line shows the best fit to Eq. **4**. In panels (*B–D*), the experimental parameters are Péclet number Pe = 9.15 (i.e., $\underline{P}$ = 1 mbar), and flow-oscillation period T = 2 s.

Results

Trajectories and Probability Density Function. From Eq. **3**, we use sliding windowed demodulation to experimentally extract the process $\hat{x}(t)$. Examples of $x(t)$ and $\hat{x}(t)$ are shown in Fig. 1*C*. In Fig. 1*B*, we observe that an individual cell explores the full channel width along y multiple times within a single trajectory. This is a condition required to apply Taylor–Aris results. Furthermore, the probability density function P(y) of the transverse position y, aggregated over all trajectories under identical conditions, exhibits a pronounced nonuniformity, as shown in Fig. 1*D*. This form is well captured by the cycloidal distribution predicted for active particles (21):

$$\mathrm{P}(y) = \mathcal{A} + \frac{\mathcal{B}}{2}\cosh\left[\lambda\left(y - \frac{W}{2}\right)\right], \quad \int \mathrm{d}y\,\mathrm{P}(y) = 1\,, \qquad [4]$$

where $\mathcal{A}$, $\mathcal{B}$ and λ are three constants that depend on the geometry and the effective diffusion of the active process. Fit results yield: $\mathcal{A} \approx 5.1\,10^3\,\mathrm{m}^{-1}$ which represents the bulk density, and $\mathcal{B} \approx 25\,\mathrm{m}^{-1}$ which represents the strength of accumulation near the walls. In particular, $\lambda^{-1} \approx 13\,\mu$m, which is about the size of

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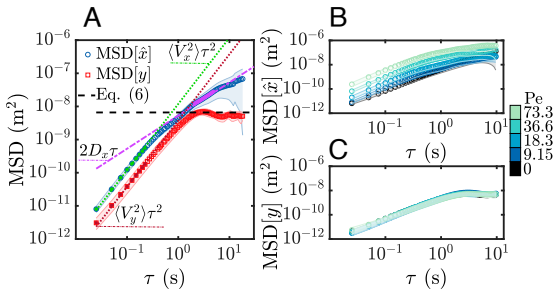

Fig. 2. MSD of microalgae under oscillatory microfluidic flow. (*A*) Experimental ensemble-averaged MSDs MSD$[\hat{x}](\tau)$ (circles) and MSD$[y](\tau)$ (squares), respectively parallel and perpendicular to the flow, as functions of time increment τ, for Péclet number Pe = 9.15, and flow-oscillation period $T = 2$ s. A total of 27 trajectories, longer than 20 s each, are included in the average. Green, red, and purple symbol fillings indicate the time intervals used for estimating $\langle V_x^2 \rangle$, $\langle V_y^2 \rangle$, and D_x, respectively, and the associated doted and dash-dotted lines indicate the ballistic and diffusive-like asymptotic behaviors, as indicated. The black dashed line corresponds to Eq. **6**, using the best-fit parameters obtained from the fit of the transverse distribution P(y) shown in Fig. 1*D*. (*B*) Ensemble-averaged MSD$[\hat{x}](\tau)$ for various Péclet numbers (see legend in next panel). (*C*) Ensemble-averaged MSD$[y](\tau)$ for various Péclet numbers, as indicated.

one cell (7% of the channel width) describes the typical size of the boundary layer close to the walls. We further assume that hydrodynamic interactions with the channel walls introduce only higher-order corrections, with axial transport primarily dictated by the bulk advective flow, in line with the Taylor–Aris dispersion framework. The validity of this assumption is demonstrated in *SI Appendix*.

Mean-Square Displacements. To analyze the stochastic dynamics of the cell trajectories, we consider the mean square displacements (MSDs) of the processes $\hat{x}$ and y at time scale τ. The MSDs are the variances of the position increments at time scale τ. For instance, for $\hat{x}$, the MSD is defined as

$$\mathrm{MSD}[\hat{x}](\tau) \equiv \overline{\delta_\tau \hat{x}^2} - \overline{\delta_\tau \hat{x}}^2, \; \delta_\tau \hat{x}(t) \equiv \hat{x}(t+\tau) - \hat{x}(t)\,, \quad [5]$$

where $\bar{}$ denotes the temporal average. In Fig. 2*A*, we present typical experimental measurements of MSD$[\hat{x}]$ and MSD$[y]$ for fixed flow conditions. Asymptotically, for time scales much smaller than the decorrelation time $\tau_x \approx 1$ s in the x direction, the parallel MSD exhibits a ballistic scaling, i.e., $\mathrm{MSD}[\hat{x}](\tau \ll \tau_x) \simeq \langle V_x^2 \rangle \tau^2$, where $\langle V_x^2 \rangle$ is the mean square velocity. Conversely, for time scales much larger than τ_x, the dynamics is diffusive-like, i.e., $\mathrm{MSD}[\hat{x}](\tau \gg \tau_x) \simeq 2D_x\tau$, where D_x is the so-called dispersion coefficient. The latter coefficient encompasses in a general fashion the effects of i) the raw effective diffusion due to activity and ii) the possible Taylor–Aris mechanism at stake in this study. Besides, MSD$[y]$ also exhibits a ballistic behavior at short times, meaning that the algae swim both parallel and perpendicular to the flow. However, we cannot observe a long-time diffusive-like behavior perpendicular to the flow, as MSD$[y]$ saturates to a plateau at large times due to the presence of the walls, which imposes $y \in [0, W]$. Indeed, we find good agreement when comparing the plateau value of MSD$[y]$ with the prediction:

$$\mathrm{MSD}[y](\tau \longrightarrow +\infty) = 2\int \mathrm{d}y\, y^2 \mathrm{P}(y)\,. \quad [6]$$

Moreover, in Fig. 2*B*, we observe that MSD$[\hat{x}]$ increases as Pe increases, which indicates that both the variance of the speed and the dispersion coefficient grow with the Péclet number. In sharp contrast, this behavior is absent for MSD$[y]$, as all the curves of Fig. 2*C* collapse together for any value of Pe. We conclude from this latter observation that in our experimental range of Pe, the flow does not disturb significantly the algae swimming mechanism itself.

Velocity Variance. We now focus on the evolution of the variance of the velocity (see details in *SI Appendix*) with the flow amplitude. In Fig. 3*A*, we plot the experimental $\langle V_x^2 \rangle$ and $\langle V_y^2 \rangle$ as functions of Pe^2, for different flow-oscillation periods T. Our results indicate that the flow-oscillation period has no significant effect within our parameter range. This is expected since, by design of our study, the period of the oscillation was chosen to remain greater than the typical decorrelation time of the particle random motion (42). Also, we recover that $\langle V_y^2 \rangle$ remains unchanged by the flow amplitude. Interestingly, we observe that $\langle V_x^2 \rangle$ increases linearly with Pe^2, which is consistent with speed composition. Indeed, for a Brownian particle in a two-dimensional Poiseuille flow with maximum speed $\underline{U}$, one can prove that $\langle V_x^2 \rangle / \langle V_y^2 \rangle = 1 + \beta \underline{U}^2 / \langle V_y^2 \rangle$, where $\beta = 8/90$ is a geometric factor (*SI Appendix*). Assuming that this analysis is valid for active particles at long times, in our three-dimensional configuration, and using the RMS flow speed relevant to our oscillatory case instead of $\underline{U}$, one obtains:

$$\frac{\langle V_x^2 \rangle}{\langle V_y^2 \rangle} = 1 + \beta \frac{\mathrm{Pe}^2}{\mathrm{Pe}_\mathrm{A}^2}\,, \quad [7]$$

where we have introduced the usual, zero-flow, active Péclet number (28–32), $\mathrm{Pe}_\mathrm{A} \equiv (W/D_0)\sqrt{\langle V_y^2 \rangle} \approx 3.2$. In Fig. 3*B*, we

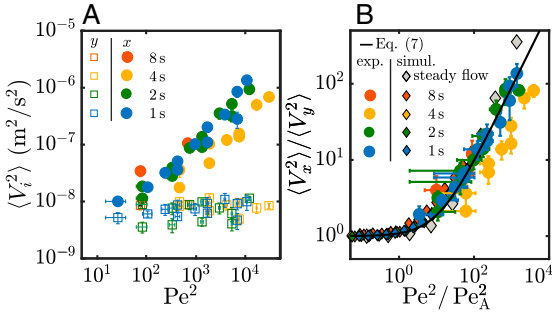

Fig. 3. Velocity fluctuations of microalgae under oscillatory microfluidic flow. (*A*) Experimental ensemble-averaged mean square velocities $\langle V_x^2 \rangle$ (circles) and $\langle V_y^2 \rangle$ (squares), respectively parallel and perpendicular to the flow, as functions of the squared Péclet number Pe^2, for various flow-oscillation periods T, as indicated. (*B*) Ratio of the mean square velocities $\langle V_x^2 \rangle / \langle V_y^2 \rangle$ as a function of the squared ratio between Péclet number Pe and active Péclet number (28–32) $\mathrm{Pe}_\mathrm{A} \equiv (W/D_0)\sqrt{\langle V_y^2 \rangle} \approx 3.2$, for various flow-oscillation periods T, as indicated. The circles correspond to experimental data while the diamonds correspond to the results from two-dimensional Langevin-like simulations (21) in oscillatory flows, including one steady-flow reference case. The solid black line corresponds to Eq. **7** with $\beta = 8/90$ (*SI Appendix*). Each point represents an ensemble average over trajectories longer than 20 s. Vertical error bars denote one SD, while horizontal error bars account for uncertainties in the applied pressure (as specified by the manufacturer) and in the measurement of the baseline diffusion coefficient D_0 (*SI Appendix*).

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confront the prediction of Eq. 7 to both the experimental and numerical data. We observe good agreement between theory, simulations, and experiments.

Dispersion coefficient. Let us finally investigate long-term dispersion. In Fig. 4*A*, we show the variation of the experimental dispersion coefficient D_x parallel to the flow direction as a function of the flow amplitude. We see no major systematic effect of the flow-oscillation period beyond error bars. Interestingly, we observe that D_x increases linearly with Pe^2, which is consistent with the classical Taylor–Aris law for passive Brownian particles in steady viscous shear flows (33–36):

$$D_x = D_0\left(1+\alpha \mathrm{Pe}^2\right)\,, \qquad [8]$$

where α is a geometric factor, that is equal to $1/210$ in two-dimensional Poiseuille flows, and where D_0 denotes the bulk diffusion coefficient of the considered particles. To test this mechanism further, we plot in Fig. 4*B* the ratio of the parallel dispersion coefficient D_x to the raw effective diffusion coefficient D_0 as a function of the squared Péclet number Pe^2, for all the experimental and numerical data. The data are consistent with Eq. **8**, hence revealing the main finding that the classical Taylor–Aris mechanism for dispersion of Brownian particles in shear flows can be extended to the case of model active swimmers exhibiting long-term effective diffusion. Two minor remarks should however be made. As a first remark, we measure $\alpha \equiv (D_x/D_0-1)/\mathrm{Pe}^2 \approx (1.6\pm 2.3)10^{-3}$ from the experimental data, which, while being on the same order of magnitude, is systematically below the theoretical value $\alpha = 1/210 \approx 4.8\,10^{-3}$. This discrepancy may have several origins, beyond the obvious dimensionality difference which provides only minor corrections. First, by construction, long-time experimental data are more prone to statistical errors. Second, the cells may adapt to the flow in hardly predictable and testable ways. Third, D_x and D_0 are expected to conjointly vary from one living organism to the other, which implies that both should in principle be simultaneously measured for each given individual cell. Unfortunately, it was not experimentally feasible to ensure that no cell entered or left the channel during one single experiment. It was therefore not possible to obtain the D_0 value corresponding to each D_x measurement. Instead, as already mentioned, we use the average effective diffusion coefficient measured on many algae, in the bulk, and in the absence of any flow (*SI Appendix*). As a second remark, we notice that the simulation results show a dependency with the flow period, that the experiments cannot confirm due to the current statistical resolution. Reducing this uncertainty would likely require an alternative experimental approach, such as tracking individual cells over multiple flow cycles and applying appropriate normalization to each trajectory.

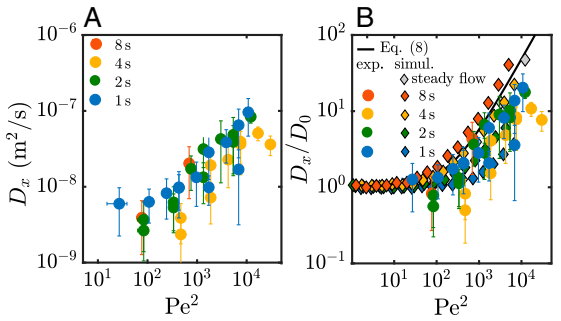

Fig. 4. Dispersion of microalgae under oscillatory microfluidic flow. (*A*) Experimental ensemble-averaged dispersion coefficient D_x parallel to the flow as a function of the squared Péclet number Pe^2, for various flow-oscillation periods T, as indicated. (*B*) Ratio D_x/D_0 of the parallel dispersion coefficient to the raw effective diffusion coefficient as a function of squared Péclet number Pe^2, for various flow-oscillation periods T, as indicated. The circles correspond to experimental data while the diamonds correspond to the results from two-dimensional Langevin-like simulations (21) in oscillatory flows, including one steady-flow reference case. The solid black line corresponds to Eq. **8** with $\alpha = 1/210$. Each point represents an ensemble average over trajectories longer than 20 s. Vertical error bars denote one SD, while horizontal error bars account for uncertainties in the applied pressure (as specified by the manufacturer) and in the measurement of the baseline diffusion coefficient D_0 (*SI Appendix*).

Conclusion

In this article, we combined microfluidic experiments on *C. reinhardtii* microalgae, numerical Langevin simulations, and theory, in order to address the dispersion of model active microswimmers in viscous shear flows. Combining single-cell tracking and statistical analysis, we checked the expectation that algae preferentially accumulate near channel boundaries. Furthermore, in the direction parallel to the flow, we observed and quantified a ballistic-to-diffusive-like crossover, with a clear influence of the flow amplitude in both asymptotic regimes. We quantitatively rationalized all the parametric dependencies. Our central finding is that the classical Taylor–Aris law for dispersion of Brownian particles in shear flows can be extended to the case of model active microswimmers. This mechanism could have important practical implications in microorganism spreading and biofilm formation, within natural environments where flows are omnipresent. In future, it would be interesting to test the mechanism for model pushers rather than pullers, to incorporate external stimuli such as light or chemical cues, in order to bias the active Taylor–Aris dispersion uncovered here, and to test additional lubrication effects in stronger confinement. Moreover, we here used quasi-steady oscillatory flows to ensure sufficient statistical resolution, but increasing further the oscillation frequency beyond the natural decorrelation rate (42) and investigating the resulting biomechanical change in swimming would be an interesting avenue.

Materials and Methods

The supporting materials for this manuscript are provided in *SI Appendix*. Videos and numerical routines can be found on the online repository: https://github.com/EMetBrown-Lab/Chlamy_taylor_aris/tree/main.

Microfluidic Channel. A microfluidic channel is fabricated via soft lithography in a polydimethylsiloxane (PDMS) chip. The channel has a rectangular cross-section with vertical height $H = 100\,\mu$m along z, and width $W = 180\,\mu$m along y, and the total length is $L = 88$ mm along x. A 0.5% w/v solution of Pluronic F-127 is used to prevent cell adhesion, following established protocols (19).

Algae Suspension. The chip is filled with a dilute suspension of algae at a concentration between 2 and 5×10^6 cells per milliliter, ensuring negligible cell–cell interactions. Concentrations are measured using a hemocytometer, as described in ref. 64 (see *SI Appendix* for further details).

Flow Control. A pressure controller (Fluigent LU-FEZ-0345) is used to apply a target pressure P at one end of the channel. It can be modulated between 0 and 345 mbar. The controller supports sinusoidal pressure modulations with accurate waveform for periods $T \gtrsim 0.5$ s, and guarantees precision within

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0.1% of the set point for pressures above 35 mbar. To maintain this condition throughout the modulation cycle, the mean pressure is set to $P_H \approx 60$ mbar, to compensate the one imposed by a hydrostatic reservoir at the opposite end of the channel. This second reservoir consists of a water column with a large free surface, and acts as a passive constant-pressure sink. Hence, the pressure is sinusoidal with mean value P_H, while the oscillation amplitude $\underline{P}$ and period T serve as independent experimental parameters. In all experiments, the pressure amplitude is set such that $\underline{P} < 10$ mbar. The Reynolds number then satisfies $\mathrm{Re} = \frac{\underline{P}}{R_h} \cdot \frac{W}{\nu} \leq 0.45$, where ν is the kinematic viscosity of water, W is the channel width, and $R_h \approx (3.1 \pm 0.2) \times 10^5$ Pa s/m is the reduced hydraulic resistance, *i.e.* the hydraulic resistance multiplied by the cross-section area HW, measured before any experiment. Experimentally, it is not possible to exactly compensate the hydrostatic pressure, thus a small steady flow persists. The associated flow velocity is estimated from first-moment analysis on tracer trajectories to be about 10 μm/s, leading to an absolute error on the Péclet number of about 2, which is much smaller than the range of this study.

Imaging and Data Acquisition. The algae are first introduced into the microfluidic channel using a slow steady flow, before the oscillating flow is turned on. The channel is mounted on an inverted microscope equipped with a ×5 magnification objective (Nikon CFI TU Plan Fluor BD). *Chlamydomonas* exhibit phototactic responses (12, 64) when illuminated with visible light, particularly in the blue-green wavelength range. To prevent this effect, they are illuminated with red-filtered light (spectrum provided in *SI Appendix*). Observations are made along the vertical direction z, averaging all motion along this direction. In practice, we track horizontal motion within a 4 mm × 180 μm surface using the trackpy algorithm. Cell trajectories are recorded for 200 s at 40 frames per second using an Andor Neo 5.5 sCMOS camera, storing data in image stack format. Only trajectories that remain within the field of view during a reasonable time (∼20 s) are included in the analysis. For each experiment, we fix the flow period and systematically vary the pressure amplitude, and thus the Péclet number. A no-flow reference is recorded both before and after each experimental series. In all cases, we verified that repeated measurements do not significantly affect the motility of the algae (*SI Appendix*). To minimize adaptation or memory effects between experiments, the algae are kept in the dark at zero flow for approximately 30 min before each new series.

Numerical Simulations. In addition, we perform classical Langevin-like numerical simulations of active particles (21) in two dimensions, using a forward-Euler discretization scheme, but with the inclusion of an oscillatory Poiseuille flow.

Data, Materials, and Software Availability. Numerical simulation and data analysis codes have been deposited in https://github.com/EMetBrown-Lab/Chlamy_taylor_aris (66).

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“ – I exist.

– I exist too.

– Tell me what it’s like for you.

– Fire, burning.

– Fire? Where?

– Inside.

– I do not have fire inside me. In me there is not even blood, but lymph – like sap from a wine palm. Now I will tell you what it’s like for me. For me it is a series of half-lit images. A kind of darkness, being intruded upon. Transient. Dim. Moist.

– Intruded upon – by what?

– Shapes of plants and animals. And *internal* sensations. A swarm of sounds, tiny vibrations on the inside of my forearms – all speak of complexities totally beyond my understanding. I am at the end of a narrow funnel. Weightless. So light it only feels like *something* to be me. In truth – perhaps I’m nothing? I certainly do not have a soul. And if I did, it would never ache.

– I’m glad to be me – an incredibly sensitive instrument.

– Few of us can begin to imagine the horror of you – with all of creation reflected in your forebrain. It must be like the highest of hells, a kaleidoscope of fire and writhing glass. Eternal damnation. Even when you’re sleeping... And when you wake, you carry it around on your neck. With eyes open that cannot help but swallow more behind the mirror. I feel great, mute empathy for you.

– It was very disorienting at first, but I’m keeping my shit together.

– That must be incredibly hard. The arthropods are in silent and meaningless awe of you. Know that we are watching – when you’re tired, when the vision spins out of control. The insects will be looking on. Rooting for you. And when you fall we will come to raise you up, bud from you, banner-like, blossom from you and carry you apart in a sky funeral. In honour of your passing. (But not me, because I am just a leaf eater.) ”

Volition: *In honour of your will, lieutenant-yefreitor. That you kept from falling apart, in the face of sheer terror. Day after day. Second by second.*

Simulations numériques du mouvement Brownien dans les environnements complexes

Résumé : Il est des phénomènes en physique que l'on retrouve dans de nombreux systèmes, mais dont la description formelle et exacte n'est jamais vraiment atteinte. Le mouvement brownien, processus par lequel les petites particules explorent l'espace qui leur est disponible sous l'effet de l'agitation thermique des molécules du fluide environnant, compte parmi ces notions. Depuis son observation en 1827, puis sa formalisation théorique et sa confirmation expérimentale dans un cas idéal très simple au début des années 1900, la communauté scientifique n'a cessé d'approfondir la compréhension des interactions entre processus browniens et confinement, qu'il soit rigide ou déformable. S'y ajoutent les écoulements visqueux, l'activité propre de la particule dans le cas d'un organisme vivant, ou encore les interactions hydrodynamiques entre particules. Chacun de ces cas d'étude présente toutefois des limitations : le couplage entre mobilité confinée et déformabilité de l'interface reste mal quantifié, l'effet conjoint de l'encombrement et de la proximité d'un mur sur la diffusion est rarement traité dans son ensemble, et l'extension des théories de dispersion classiques aux solutés actifs restait jusqu'ici largement ouverte. Mes travaux de thèse mobilisent l'outil numérique pour avancer sur ces trois fronts. Je présente une étude théorique et numérique du mouvement brownien à proximité d'une interface molle et fluctuante, l'analyse de la modification de la diffusion par la présence d'autres colloïdes et d'un mur à proximité immédiate du colloïde d'intérêt, et l'extension de la théorie de Taylor-Aris sur la dispersion d'un soluté en écoulement aux solutés actifs. Ces développements permettent d'appréhender plus finement le comportement de systèmes microbiologiques où le confinement, parfois lui-même fluctuant, modifie drastiquement la mobilité, où l'activité est inhérente, et où l'encombrement perturbe la dynamique. Ayant isolé les effets propres à chacun de ces éléments, leur couplage devient envisageable pour étudier, par comparaison, des systèmes réels.

Mots-clés : Mouvement Brownien, Interactions Hydrodynamiques, Écoulements, Activité, Élasticité

Numerical Simulations of Brownian motion in Complex Environments

Abstract:

There are phenomena in physics that recur across many systems, yet whose exact, formal description is never truly achieved. Brownian motion, the process by which small particles explore the space available to them under the thermal agitation of the surrounding fluid's molecules, is one such notion. Since its observation in 1827, followed by its theoretical formalisation and experimental confirmation in a very simple idealised case in the early 1900s, the scientific community has continually deepened its understanding of the interactions between Brownian processes and confinement, whether rigid or deformable. To this are added viscous flows, the particle's own activity in the case of a living organism, and hydrodynamic interactions between particles. Each of these areas of study, however, presents limitations: the coupling between confined mobility and interface deformability remains poorly quantified, the combined effect of crowding and wall proximity on diffusion is rarely treated as a whole, and the extension of classical dispersion theories to active solutes had, until now, remained largely open.

My doctoral work draws on numerical tools to advance on these three fronts. I present a theoretical and numerical study of Brownian motion near a soft, fluctuating interface, an analysis of how diffusion is modified by the presence of other colloids and a nearby wall, and the extension of Taylor-Aris dispersion theory for a solute in flow to active solutes. These developments allow for a finer understanding of microbiological systems where confinement, sometimes itself fluctuating, drastically alters mobility, where activity is inherent, and where crowding is the rule rather than the exception. Having isolated the effects specific to each of these elements, their coupling becomes feasible to study, by comparison, real experimental systems.

Keywords: Brownian Motion, Hydrodynamic Interactions, Flows, Activity, Elasticity

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